The Curious Case of Post-Earnings Announcement Drift Over the Past Century

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Abstract

We provide novel evidence that, unlike several accounting-based anomalies, the Post-Earnings Announcement Drift (PEAD) remains statistically and economically significant throughout the pre-publication era, extending back before the 1970s. Strikingly, PEAD follows an inverted U-shape trend: weak in the 1930s, rising through the mid-century, peaking in the 1970s–1980s, and declining thereafter. This trend challenges a typical mispricing anomaly interpretation of PEAD, which would imply stronger predictability prior to its discovery. Time-varying arbitrage efficacy and skill, as measured by the finance industry's size, wages, and educational composition, as well as changes in investor attention to earnings announcements better explain the results.

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1. Introduction

Over the past five decades, post-earnings announcement drift (PEAD) has emerged as the most significant challenge to the efficient markets paradigm. PEAD reflects the tendency for stock prices to adjust gradually, rather than immediately, to earnings surprises. Fama (1998) describes this anomaly as 'above suspicion,' stating: "The post-earnings announcement drift first reported by Ball and Brown (1968) has survived robustness checks, including extensions to more recent data (Bernard and Thomas, 1990; Chan et al., 1996)." This enduring stock return anomaly suggests that even the most basic accounting information, such as earnings, presents a significant challenge for the market to process, which hinders the timely adjustment of stock prices.

On the other hand, in a recent influential paper, Linnainmaa and Roberts (2018) offer compelling evidence that many accounting-based stock return anomalies lose their statistical significance when the data are analyzed before 1963, prior to their discovery in the academic literature. They conclude that these phenomena, which have been central to the return predictability literature, are not genuine anomalies but rather artifacts of data snooping. Their findings challenge the broader notion that markets systematically fail to process various types of accounting information.

Although Linnainmaa and Roberts (2018) do not directly examine post-earnings announcement drift (PEAD), their findings have important and untested implications for the extensive PEAD literature and the challenge it poses to market efficiency. Prior research shows that PEAD exhibits a pattern similar to many accounting-based anomalies that have been questioned for potentially reflecting data snooping bias. In particular, the PEAD effect is most pronounced during the original in-sample period but declines steadily afterward. This parallel

post-publication decline underscores the importance of analyzing the pre-publication period when evaluating the persistence and underlying sources of the PEAD anomaly.

Following this argument, in this paper, we examine PEAD over the past century, covering both the pre- and post-publication periods, to assess whether PEAD is also subject to data snooping bias—an issue raised for many other accounting-based anomalies. A major obstacle to such analysis has been the limited availability of historical earnings announcement data. Nearly all prior research on PEAD relies on Compustat, which begins in the 1970s, thereby restricting analysis to in-sample or post-publication periods. We meticulously hand-collect data on earnings announcements from the *Wall Street Journal* (WSJ) for the pre-publication period, 1934 to 1971, before the availability of Compustat. This process involves navigating archival records to assemble a unique dataset that enables us to analyze historical variation in PEAD over an extended period of 89 years from 1934 to 2022. By extending the data back several decades using historical records, our study is among the first to provide a comprehensive view of PEAD's behavior before and after its widespread recognition in the literature.

Figure 1 provides a first glance at our main findings by illustrating how the market's reaction to earnings announcements varies across 10-year periods extending from 1934 to 2022. For each quarterly sample of earnings announcements, we compute the 60-day post-earnings announcement drift (PEAD, CAR[+2,+61]), the initial 2-day return (CAR[0,+1]), and the total 62-day return (CAR[0,+61]) for a hedge portfolio based on standardized unexpected earnings (SUE). We then average these measures across all announcements during each 10-year period.

Panel A of Figure 1 plots the average PEAD (CAR[+2,+61]) for each 10-year period, while Panel B provides the analogous information for the initial reaction to earnings news, CAR[0,+1]. Panel C then plots the proportion of the total response (over days 0 to +61) that is

incorporated with a delay (CAR[+2,+61) / CAR[0,+61], in the **orange** bars) versus immediately (CAR[0,+1] / CAR[0,+61], in the **blue** bars). The **orange** bars in Panels A and C reveal that PEAD comprises a smaller proportion of the total response to earnings news in the early and late decades of our sample, but a larger portion in the middle decades. In contrast, the **blue** bars in Panels B and C show that the initial reaction comprises a larger proportion in the early and late decades of our sample, relative to the middle decades.

Figure 1 provides novel insights into the history of the market's reaction to earnings news over the past century. PEAD is statistically significant in every 10-year period since the 1930s, underscoring its robustness—even during the pre-discovery sample. This evidence stands in contrast to the main findings of Linnainmaa and Roberts (2018), who document substantial post-publication decay for many accounting-based anomalies. In contrast, our results suggest that PEAD has remained a persistent predictor of returns, particularly before the 1970s. Therefore, our findings strongly reject the argument that PEAD is a product of data snooping and establish that it is not subject to the same concerns that have been raised for other accounting-based anomalies.

Second, Figure 1 also brings to light further dramatic evidence that contrary to the typical pattern expected of mispricing-based anomalies—where return predictability is strongest before discovery and fades thereafter—PEAD follows an *inverted U-shaped trajectory*. It was relatively weak in the 1930s, grew in prominence from the 1940s through the 1980s, and has steadily diminished in more recent decades. The recent trend toward more timely reaction to earnings news since the 1980s (i.e., the right half of Panel A in Figure 1) is well-documented and widely acknowledged in the literature. What stands out in our findings is the strong upward trend in PEAD throughout the pre-publication period that extends from the 1930s through the 1970s (i.e.,

the left half of Panel A), before reversing after its discovery. This pattern challenges the view of PEAD as a manifestation of static market inefficiency, which would predict heightened return predictability in earlier periods when financial markets were less sophisticated, information processing capabilities were more limited. and arbitrageurs were unaware of the anomaly (McLean and Pontiff, 2016; Stambaugh, Yu, and Yuan, 2015). Instead, our findings suggest that PEAD became more pronounced as markets developed, peaking during the 1970s and 1980s—perhaps reflecting a transitional period of partial efficiency—before declining as investors became increasingly adept at incorporating earnings news into prices.

One potential concern with this analysis is that our hand-collected data from the WSJ may not adequately represent the CRSP universe over time. One might expect that this limitation was more of a concern in the earlier decades before WSJ coverage expanded closer to the 1970s. Such improving coverage could result in a more comprehensive sample as the 1970s approached, which could, in turn, lead to the increasing significance and robustness of PEAD that we find over this early period. However, contrary to this concern, our WSJ sample covers 57% of all CRSP firms in the 1930s when its coverage is lowest, and this coverage averages 76% of all CRSP firms from 1934 to 1971. For comparison, Linnainmaa and Roberts (2017) report that Compustat covers 59% of the CRSP universe in 1965, while our WSJ coverage averages 83% in the 1960s. In this light, the coverage of our WSJ sample compares favorably with Compustat coverage of CRSP during the same period, as well as the sample from Moody's and Compustat that is analyzed by Linnainmaa and Roberts (2018).

Another potential concern with the WSJ sample of earnings announcements is that the distribution of firms may be skewed toward larger companies due to reporting biases by the WSJ. We address this concern by comparing firm size for the WSJ sample to that in the CRSP

universe for every 10-year period, and we show that the distribution of firm size is similar in the WSJ and CRSP samples throughout our sample period. We conclude that our data provide comprehensive and balanced coverage of the distribution of firms in CRSP.

We further alleviate potential concerns about our sample selection by showing that the upward trend in PEAD from the 1930s to the 1970s continues to hold when we focus only on the smaller subset of firms that existed throughout the first four or five decades of our sample. In addition, we compare firms that existed over each pair of consecutive 10-year periods and find a similar upward trend in PEAD. Furthermore, our results generalize across different firm attributes. For example, while Farboodi, et al. (2022) find that the recent gains in price efficiency mainly come from large growth stocks, we find a similar U-shaped trend in PEAD since the 1930s for both small and large firms, value and growth stocks, and firms with low and high values of turnover, idiosyncratic volatility, or illiquidity. This U-shaped evolution of PEAD is also robust across the different industrial sectors in the economy.

One intriguing question that arises from our results is why PEAD is highly significant for all 10-year periods prior to the 1970s, unlike other accounting-based anomalies that Linnainmaa and Roberts (2018) identify as artifacts of data snooping. One obvious interpretation is that PEAD's return predictability is real and persistent. However, an alternative answer may lie in the short-lived return predictability that prevails for many anomalies. Bowles et al. (2024) find that anomaly returns are concentrated in the first month after information is released, but they decay soon thereafter. In contrast, Linnainmaa and Roberts (2018) make the conventional assumption that accounting data are only available six months after the fiscal year end. This lag in the timing of accounting data may weaken the power of the tests used by Linnainmaa and Roberts (2018) to detect short-term predictability, which could lead to insignificant results.

We investigate this anomaly timing issue by considering a particularly important and accessible anomaly, the predictability of the earnings-to-price (E/P) ratio. In our analysis, rather than skipping six months, we use our WSJ announcement dates to capture the first public release of earnings information. Contrary to Linnainmaa and Roberts (2018), we show that the E/P ratio significantly predicts future returns over the period, 1930 – 1970. This evidence suggests that anomaly timing may play a role in the insignificant results of Linnainmaa and Roberts (2018).

While anomaly timing might help to explain the significance of PEAD before the 1970s, it alone cannot account for the entire inverted U-shaped trend in PEAD over the past century. This surprising result calls for further analysis of the historical evolution in potential economic forces that might help to explain this nonlinear evolution of PEAD. We begin by discussing the historical trends of several economic factors that may help to explain the recent declining trend in PEAD since the 1980s, although they cannot account for the surprising earlier rise in PEAD from the 1930s through the 1970s.

First, we show that limits to arbitrage such as illiquidity, low turnover, and IVOL were much more severe in the 1930s, and have diminished steadily ever since (see also Jones, 2002). However, while the decline in these frictions since the 1980s is one likely cause of the recent decline in PEAD over this period, the fact that these same frictions have steadily declined ever since the 1930s is not compatible with the earlier rise in PEAD from the 1930s to the 1970s.

Second, McLean and Pontiff (2016) show that many predictable return patterns disappear after arbitrageurs learn about these anomalies through academic publications. In this light, one would expect a larger PEAD in the pre-Compustat era before much research was published about this anomaly. However, the smaller PEAD we find prior to the 1970s is at odds with this learning argument of McLean and Pontiff (2016).

A third potential explanation for the low PEAD in the 1930s is the possibility that corporate insiders may have been more active at the time in sharing and trading on private information prior to earnings announcements, before insider trading laws were first enforced in the 1960s. But contrary to this view, we show that pre-announcement returns do not comprise a larger portion of total earnings announcement returns in the early portion of our sample.

Fourth, we explore whether changes in the *earnings surprise gap*—the magnitude difference between positive and negative surprises—might explain the time variation in PEAD. As Huang (2022) shows for momentum, larger return spreads between winners and losers can amplify return predictability. Similarly, a widening earnings surprise gap over time, driven by evolving accounting standards and increasing complexity, could imply smaller reactions and weaker PEAD in the 1930s. However, our results remain robust even after controlling for changes in the surprise gap, suggesting it does not drive the inverted U-shaped pattern.

Next, we analyze structural trends and challenges faced by the finance industry that have profoundly affected the capacity of arbitrageurs over time. From the 1930s to the 1970s, there was enormous growth in the amount and complexity of information, but limited innovation in the information processing technology that is required for arbitrageurs to function effectively. In contrast, rapid technological progress since the 1970s has revolutionized the finance industry and enhanced the efficacy and skill of arbitrageurs to perform their price discovery function. We capture these trends by constructing three proxies for the efficacy and skill of arbitrageurs.

First, we measure the efficacy of arbitrageurs by the size of the trading sector relative to the number of public firms that require analysis. This proxy is motivated by the model of Martin and Nagel (2022), in which investors can forecast asset prices more effectively when the number of firm attributes (i.e., the complexity of information) is sufficiently small relative to the total

number of firms that require analysis. A higher ratio should be associated with greater arbitrage efficacy, and thus more timely adjustment of prices to earnings news.

We also construct two proxies for the skill of arbitrageurs, as the level of wages or education employed in the finance industry relative to the nonfarm private sector. Phillipon and Reshef (2012) establish that a U-shaped trend in financial deregulation over the past century led to a similar trend in the average skill level of employees in the financial sector, with a lag. While more severe financial regulation in the 1930s constrained the demand for skilled labor in finance for the next 50 years, deregulation in the 1980s reversed this trend to increase competition and the demand for skilled labor in finance. We conjecture that the inverted U-shaped trend in PEAD we document should be associated with these dramatic swings in the skill of arbitrageurs over time.

Consistent with this view, we find a robust relation between PEAD and all three proxies for arbitrage efficacy and skill. Our results suggest that financial deregulation has gradually influenced structural changes in the finance industry over the past century, which have in turn impacted these three aspects of the efficacy and skill of arbitrageurs. As a result, the inverted U-shaped trend in PEAD has evolved in tandem with a corresponding U-shaped trend in the relative size, wages, and education of the finance industry.

Finally, we provide further evidence to support our conjecture that the time varying arbitrage efficacy and skill of arbitrageurs cause the inverted U-shaped trend in PEAD. Arbitrage forces incorporate information into prices through their trading activity. Hence, elevated trading volume suggests that arbitrage forces are more active and attentive, facilitating the timely incorporation of earnings information into prices. In particular, arbitrageurs, with higher skills and efficacy, might have been more attentive to earnings both in the early decades, when earnings were one of the few reliable sources of financial information, and in more recent years,

as technology and data tools enabled renewed focus on earnings news. In contrast, the middle decades—marked by growing information complexity but limited processing capacity—may have diluted earnings salience and delayed price reactions. We explore this view by analyzing abnormal trading volume on the two days around the announcement, across the 10-year periods that span our sample. We find that abnormal volume around earnings releases follows a U-shaped trend that mirrors the inverted U-shaped path of PEAD over the past century. This evidence suggests that arbitrageurs have been more intact and attentive to earnings information and intensively trading in both the earlier and later years of the last century, relative to the middle.

2. Data and Variables

2.A. Wall Street Journal Data on Earnings Announcements for the Period, 1934 – 1971

The primary challenge to analyzing PEAD before 1970 is the lack of data on earnings announcements. Compustat coverage of earnings announcements does not begin until the second half of 1971, and this coverage is incomplete for 1971. We resolve this problem by hand-collecting earnings announcement data from the *Wall Street Journal* (WSJ) archives from 1934 through 1971 and then appending this sample with Compustat data after 1971.

In June of 1930, the WSJ began publishing earnings figures released by firms on the previous day in a table format.[1] We use the Proquest search engine to download these daily earnings report tables for the period between June 1930 and December 1971, from the digital archive containing pdf files of the WSJ. This search yields small samples for 1930 and 1931, but the coverage expands greatly in 1932. Since we require eight quarters of data to compute the firm's earnings surprise (SUE), our final sample begins in 1934 and extends through 2022.

[1] See Internet Appendix A.1. for details regarding the environment for information disclosure since the 1930s.

Figure 2 provides an example of this WSJ earnings report published on April 14, 1937. This example is indicative of the quality of the pdf files available, which varies considerably over the sample period, 1934 – 1971. In some cases, the Proquest search engine cannot detect the earnings report even though the table heading and keywords match. We manually check each daily WSJ with a missing earnings report to ensure that our coverage is complete.

For each earnings announcement, our primary variables from Figure 2 include the company name, earnings announcement date, period end date, and net income for the current quarter and the same quarter in the previous year. We identify the earnings announcement date (day 0) as the day that these earnings figures appear in the WSJ.[2] In these WSJ earnings reports, a symbol besides the earnings number points to a footnote if that number represents a net loss. We check all footnotes in each earnings report to ensure that losses are recorded properly. We exclude cases where we cannot read the company name or the earnings numbers in the pdf file. We also ensure the timeliness of announcements in our WSJ sample by deleting announcements that appear more than 90 calendar days after the quarter-end date provided in the WSJ earnings report. Finally, we collect the information from Figure 2 for all earnings announcements reported in the WSJ, and match this data with CRSP using the company name.

2.B. Coverage of Firms in WSJ Sample versus CRSP Universe over the Period, 1934 – 1971

In this subsection, we compare the coverage and size distribution of firms in our hand-collected sample of earnings announcements from the WSJ relative to all firms in the CRSP universe, over the years that comprise the WSJ sample, 1934 – 1971. One potential concern is

[2] Over the period, 1934 – 1971, these earnings announcement data typically appear in the WSJ on the day following their release by the firm. Thus, most investors at the time received earnings information on the day these data appear in the WSJ. Akbas, Ay, and Koch (2024) provide support for this choice of day 0, by showing that trading volume on the day that earnings annnouncements appear in the WSJ is dramatically higher than on the previous day. While it is unlikely that the date appearing in the WSJ archives is incorrect, we note that any potential error in this choice of day 0 would bias the results against our findings of a robust significant market response on day 0.

that the announcing firms in our WSJ sample may not be representative of the complete sample of firms covered by CRSP across these early years. Table 1 addresses this issue by presenting information about the number of unique firms and the distribution of firm size for our WSJ sample versus the broader CRSP universe, for each 10-year period from 1934 to 1971. The number of firms in our WSJ sample increases steadily from 630 over the first 10-year period (1934 – 1943) to 2,894 for the last 8 years (1964 – 1971), while the number of firms in CRSP increases from 895 to 3,099 over the same period. Thus, the WSJ coverage of firms as a proportion of the CRSP universe ranges from 70% for 1934 – 1943 to 93% for 1964 – 1971.

For comparison, Linnainmaa and Roberts (2018) hand-collect historical balance sheet data from Moody's manuals and complement this information with Compustat data since the 1920s. They report information about the coverage of firms in their hand-collected sample from Moody's and Compustat relative to the CRSP universe over five-year intervals from 1925 through 1965. Their hand-collected sample covers just 59% of the CRSP universe in 1965, and roughly 72% of the CRSP universe in the average year.[3] In comparison, Panel A of Table 1 shows that our WSJ sample of earnings announcements offers a more robust representation of the CRSP universe, with an average coverage of 81% across the four 10-year periods from 1934 to 1971 (or an average annual coverage of 76%).

Another potential concern is that the distribution of firms in our WSJ sample may be skewed toward larger firms due to potential reporting biases by the WSJ.[4] It is possible that the more efficient price response to earnings news that we document in the early portion of our sample could reflect a tendency for larger firms to have a lower PEAD and higher immediate

[3] We calculate the coverage from Linnainmaa and Roberts (2018) by dividing the number of firms in their Compustat and Moody's sample relative to the CRSP universe, using information from Table 1 of their paper.
[4] According to Frazzini and Lamont (2007), Compustat's low coverage of earnings announcements for small firms is due to the exclusion of the smallest firms by the WSJ, since the WSJ is one of the main sources for Compustat.

response following earnings news. In response to this concern, Panel B of Table 1 provides information about the distribution of firm size in the WSJ and CRSP samples, respectively, at the beginning of each 10-year period from 1934 to 1971, as well as in 1971. Throughout this period, the size of the average and median firms is similar for the CRSP and WSJ samples. For example, the market capitalization of the average firm in our WSJ sample (the CRSP universe) increases from $39.81 ($41.28) million in 1934 to $301.51 ($313.58) million in 1971. Likewise, the 25th and 75th percentiles of the firm size distribution are similar in the WSJ and CRSP samples across these 10-year periods. This discussion reveals that our WSJ data on earnings announcements offer comprehensive and balanced coverage of the CRSP universe when compared to other studies that rely on Compustat or historical accounting data over this early period.

2.C. Control Variables and Descriptive Statistics

All variables are defined in Table 2. First, the standardized unexpected earnings surprise (SUE) is defined as the difference between the EPS from the current quarter and the EPS four quarters earlier, divided by the standard deviation of these differences over the previous eight quarters, as in Foster, Olsen, and Shevlin (1984). Next, every quarter we sort all announcing stocks into quintiles (ranked 0 to 4) based on SUE. For each firm, we then divide this quintile rank by four to obtain the adjusted SUE rank (Adj_SUE), which ranges from 0 to 1. Note that a one-unit increase in Adj_SUE moves from the lowest SUE quintile to the highest SUE quintile.

Investors with access to the WSJ could compare the earnings performance of the announcing firm for the most recent quarter with the same figure from the corresponding quarter in the previous year. Since Compustat periodically updates its earnings data over time as firms restate their earnings, there is a discrepancy between the data actually seen by investors in the WSJ and the Compustat data used in many recent studies. This discrepancy might result in

understating or inflating the market's response to earnings news (see Livnat and Mendenhall, 2006). In contrast, from 1934 to 1971, we analyze the earnings figures that were actually reported in the WSJ as they became available to investors, and thus overcome this potential bias.

We obtain daily stock return data from CRSP. We use historical book-to-market data from Kenneth French's library and complement these data with Compustat annual data after 1950. The control variables for our main analysis include the adjusted rank of SUE, the lagged SUE rank, firm size in month *t-1*, the most recent publicly available book-to-market ratio,[5] the number of earnings announcements on the same day (Hirshleifer, Lim, and Teoh, 2009), the compounded daily returns and standard deviation across returns over the prior two weeks (i.e. days *t-11* to *t-2*), average daily stock illiquidity over month *t-1*, average daily turnover between days *t-11* and *t-2*, and the firm's market beta estimated over months *t-36* to *t-1* (from regressions of the firm's monthly stock return on the CRSP value-weighted market return). We also obtain the size and book-to-market cut-off values from Kenneth French's website.

In Table 3, we provide summary statistics for the main variables during each successive 10-year period of our sample. We exclude stocks with a price less than $1 at the end of the quarter before the earnings announcement, and stocks with a prior quarter-end market capitalization less than the 5th percentile cutoff by NYSE market capitalization, to ensure that our results are not driven by market microstructure biases. Table 3 indicates that the average firm size in this sample increases over time from $72 million in the first 10-year period (1934 – 1943) to $12.5 billion in the last nine years (2014 – 2022). Average liquidity just prior to the earnings announcement also rises steadily over the past century in terms of both share turnover and the

[5] Book equity data in Kenneth French's website are collected from Moody's Industrial Manuals, published every year in June. If the earnings announcement date is after June, we divide the current year's book-equity with the previous month's firm size. If the announcement is in or before June, we use the previous year's book-equity data.

Amihud (2002) illiquidity measure, while the average firm's book-to-market ratio declines. In contrast, the average firm's SUE, beta, and return volatility have no discernible trend over time.

3. The Market's Response to Earnings Announcements over the Past Century

3.A. Portfolio Approach

We first use a portfolio approach to examine post-earnings announcement drift (PEAD), as well as the initial response to earnings news. We begin by double-sorting all stocks each quarter into portfolios, first, above or below the median market capitalization and, second, based on the 30th and 70th percentile of book-to-market cutoffs from Kenneth French's data library. Then, for every earnings announcement, we measure PEAD (i.e., CAR[+2,+61]) by computing the cumulative abnormal return relative to the same matched size and book-to-market portfolio, over days +2 through +61 following the firm's earnings release on day 0 (i.e., CAR[+2,+61]). We similarly measure the initial response to earnings news on days 0 and +1 (i.e., CAR[0,+1]).

Next, each quarter we sort all earnings announcements into quintiles based on the firm's earnings surprise (SUE). For all announcements contained in each quarterly SUE quintile, we then average these two return measures across all announcements during every 10-year period over the years, 1934 – 2022. Our measures of PEAD and the initial response to earnings news for each 10-year period are the resulting mean CAR[+2,+61] and CAR[0,+1] for the hedge portfolio that is long stocks with high SUE and short stocks with low SUE every quarter.

The results are presented in Table 4. Column (1) reveals a significant PEAD (i.e., mean CAR[+2,+61]) for each 10-year period, which follows a clear *inverted U-shaped trend* over the past century. Specifically, the mean CAR[+2,+61] increases from 2.01% over the first 10-year period (1934 – 43) to 4.22% in the middle period (1974 – 83), before declining back toward 0.69% over the last nine years (2014 – 2022). Column (2) similarly indicates a significant initial

response to earnings news (i.e., mean CAR[0,+1]) for every 10-year period spanning the sample. However, this latter initial return measure follows a direct *U-shaped trend* in CAR[0,+1], which declines from 2.84% in the first 10-year period (1934 – 43) to 1.74% over 1974 – 83, before reversing back toward 3.88% over 2014 – 22. Together, this evidence leads to the clear U-shaped trend in the degree of PEAD-related price efficiency, plotted in Panel C of Figure 1.[6]

3.B. Regression Approach

In this subsection, we account for the possibility that the evidence in Table 4 may be due to firm characteristics, by repeating our analysis from Table 4 using a regression approach. In particular, we estimate the following regression model to analyze the market's response to earnings news for every 10-year period, while controlling for other firm attributes, as follows:

$$CAR[a,b]_{it} = \beta_0 + \beta_1 \, Adj_SUE_{it} + Controls + \delta_d + \varepsilon_{it}. \quad (1)$$

The dependent variable, $CAR[a,b]_{it}$, is the cumulative abnormal return for firm *i* during quarter *t*, measured over two time frames following the announcement on day 0: (i) the 60-day PEAD (CAR[+2,+61]), and (ii) the initial reaction to earnings news (CAR[0,+1]). The controls include lagged earnings surprise (Lagged SUE), number of announcements (# Ann), return over the prior two weeks (Ret [-11,-1]), book-to-market (BM), beta, size, return volatility (Ret_Vol), turnover, and illiquidity. All variables are defined in Table 2. In all regression analysis that follows, we include fixed effects for day of the week (δ_d), we adjust standard errors for heteroscedasticity, and we cluster standard errors by the earnings announcement day.

In Equation (1), the relation between the firm's adjusted earnings surprise (Adj_SUE) and CAR[+2,+61] or CAR[0,+1] is given by the partial derivative, $\partial CAR[a,b]_{it} / \partial Adj_SUE_{it} =$

[6] The two columns of Table 4 are plotted in Panels A and B of Figure 1, while the ratio of each column to the total announcement return (CAR[0,+61]) is plotted in Panel C of the same figure. Sixty days is the most common window used to measure PEAD in the literature (i.e., CAR[+2,+61]), since it ensures that the drift for one quarterly announcement is not confounded by the response to the next announcement.

β_1. This derivative shows that a one-unit increase in Adj_SUE from 0 to 1 (i.e., ranging from the bottom to the top SUE quintile) is associated with a change in the market's response to earnings news ($CAR[a,b]_{it}$) of β_1 percent, after controlling for firm attributes. Thus, β_1 is analogous to the return on a hedge portfolio that is long the top SUE quintile and short the bottom SUE quintile, over days *a* to *b* following the announcement, after controlling for firm attributes.[7]

In Panels A and B of Table 5, we present the results for CAR[+2,+61] and CAR[0,+1], respectively. The columns in each Panel provide the evidence for the different 10-year periods spanning the sample from 1934 to 2022. First consider the results for CAR[+2,+61] in Panel A. In the first row of Panel A, the coefficient of Adj_SUE (β_1) reveals a highly significant post-earnings announcement drift in the SUE hedge portfolio for every 10-year period of our sample, after controlling for other firm attributes. Furthermore, across the columns of Panel A, the magnitude of these coefficients (β_1) follows the same dramatic inverted U-shaped trend in PEAD over the past century that is documented in Figure 1 and Table 4.

Next, consider the results for CAR[0,+1] from the first row in Panel B of Table 5. Once again, the coefficient of Adj_SUE (β_1) indicates a highly significant immediate response to earnings news (i.e., CAR[0,+1]) for the SUE hedge portfolio in every 10-year period, after controlling for firm attributes. Furthermore, across columns of Panel B, the magnitude of these coefficients (β_1) follows a slight downward trend over the first half of the sample before 1983, which is followed by an upward trend after 1983. This behavior resembles the U-shaped trend in CAR[0,+1] presented in Figure 1 and Table 4. Overall, the evidence in Table 5 establishes that the dramatic U-shaped trend in PEAD-related efficiency over the past century is robust when we control firm attributes.

[7] See Akbas, Jiang, and Koch (2020) and Akbas, Ay, and Koch (2024) for other studies that use a similar regression specification in which the coefficient of an adjusted quintile rank variable is analogous to the hedge portfolio return.

4. The Timing of Matched Accounting Data to Stock Returns and the Significance of PEAD

An intriguing question that arises from our analysis is why we find a significant PEAD for each 10-year period from the 1930s to the 1980s, unlike many other accounting-based anomalies that Linnainmaa and Roberts (2018) identify as artifacts of data snooping (see also Martineau, 2021). Of course, an explanation could be that our findings are unique to the PEAD and other anomalies truly suffer from data snooping bias. However, an alternative explanation lies in how accounting data are matched to stock returns in the anomaly literature. According to Bowles, Reed, Ringgengerg, and Thornock (2024), anomaly returns are typically concentrated in the first month after information release dates, and they tend to decay soon thereafter. However, Linnainmaa and Roberts (2018) make the conventional assumption used in most academic research that accounting data are only available six months after the fiscal year end (Fama and French, 1993). This six-month delay in matching accounting data to stock returns may weaken the power of tests designed to detect short-term predictability, and may thus be one factor behind the insignificant results in Linnainmaa and Roberts (2018).

We investigate this issue by analyzing a related and prominent anomaly, the well-documented return predictability of the earnings-to-price (E/P) ratio. We focus on the E/P ratio because both earnings and price information are readily available to investors on the earnings announcement date published in the WSJ. According to Linainnmaa and Roberts (2018), the E/P ratio does not predict returns prior to the 1970s. In the spirit of Bowles, Reed, Ringgengerg, and Thornock (2024), we rely on the earnings announcement dates from our WSJ sample prior to the 1970s, and Compustat announcement dates thereafter, to capture the timing of the first public release of this earnings information. That is, we use information immediately available to

investors on the publication date, rather than assuming that earnings data are only available with a delay of six months after the end of the fiscal year, as in Linainnmaa and Roberts (2018).

First, each quarter we sort all announcing firms into quintiles based on the E/P ratio, calculated by dividing net income disclosed relative to the market capitalization of the announcing firm at the beginning of the quarter. We then compute the average return to each quintile portfolio during the first month of the next quarter and the following five months. Finally, we average the portfolio returns for each quintile portfolio across all quarters in every 10-year period since 1934.

The results are presented in Table 6. Contrary to the evidence reported in Linnainmaa and Roberts (2018), the high minus low E/P hedge portfolio return in month t+1 (i.e., the first month of the next quarter after the announcement) is significant for the 10-year periods extending from the 1930s until the 1990s. Furthermore, the magnitude of this E/P hedge portfolio return follows an inverted U-shaped trend over the past century, similar to the trend in PEAD from Figure 1 and Tables 4 and 5. Furthermore, when we use this portfolio approach to predict returns up to six months forward in the bottom rows of Table 6, the significance of the E/P hedge portfolio return declines. This evidence suggests that the six-month delay used to match accounting data to stock returns may play a crucial role in the findings of insignificant return predictability presented by Linnainmaa and Roberts (2018). We note that, while our results indicate significant return predictability for earnings-related anomalies prior to the 1970s, a more detailed examination of other anomalies is necessary to draw more generalized conclusions on other accounting-based anomalies.

5. Nonlinear Trend in the Reaction to Earnings Announcements: A Closer Examination

5.A. Divergent Linear Trend in CAR[+2,+61] or CAR[0,+1] before versus after 1983

The evidence provided in Figure 1, Table 4, and Table 5 indicates a robust nonlinear trend in both CAR[+2,+61] and CAR[0,+1] over the past century, which reverses direction around 1983. In this subsection, we expand Equation (1) to further examine whether the change in the direction of this trend around 1983 is statistically significant, as follows:

$$CAR[a,b]_{it} = \beta_0 + \beta_1 Adj_SUE_{it} + \beta_2 Trend_t + \beta_3 Adj_SUE_{it} * Trend_t + Controls + \delta_d + \varepsilon_{it}. \quad (2)$$

In this specification, we add a linear trend which indexes the quarters in our sample since 1934, as well as its interaction with Adj_SUE. Our main focus is on the coefficient of the interaction term, Adj_SUE * Trend (β_3), which reveals the nature of any trend in the coefficient of Adj_SUE (i.e., β_1, which is analogous to the SUE hedge portfolio return) during the period analyzed.

We investigate a possible divergent linear trend in CAR[+2,+61] or CAR[0,+1] before versus after 1983, by estimating Equation (1) over two periods, 1934 – 1983 versus 1984 – 2022. Panels A and B of Table 7 present this analysis for CAR[+2,+61] and CAR[0,+1], respectively. On the left (right) side of each panel, we provide the evidence before (after) 1983.

First, consider the results for PEAD (i.e., CAR[+2,+61]) in Panel A of Table 7. In the first row, the coefficient of Adj_SUE (β_1) indicates a highly significant PEAD (i.e., CAR[+2,+61]) for the SUE hedge portfolio during both subperiods, before and after 1983, while controlling for other firm attributes that have also been shown to predict returns. But our main focus is on the coefficient of the interaction term, Adj_SUE * Trend (β_3) for each subperiod. On the left side of Panel A, β_3 is significantly positive, indicating that PEAD (i.e. β_1 or CAR[+2,+61]) has an *increasing* trend during the first subperiod, 1934 – 1983. In contrast, the

right side of Panel A reveals a significant negative coefficient (β_3), indicating that PEAD has a *declining* trend during the second subperiod, 1984 – 2022, after controlling for firm attributes.

Panel B of Table 7 reveals the opposite result for the initial response to earnings news (CAR[0,+1]), with a significant negative coefficient of the interaction term, Adj_SUE * Trend (β_3), before 1983 versus a significant positive coefficient (β_3) after 1983. This outcome indicates a *declining* trend in CAR[0,+1]) from 1934 to 1983, versus an *increasing* trend in CAR[0,+1] after 1983. Together, Panels A and B of Table 7 corroborate our evidence in Figure 1 and Tables 4 and 5, indicating a dramatic V-shaped trend in PEAD-related efficiency over the past century.

5.B. Quadratic Trend in CAR[+2,+61] or CAR[0,+1] over the Entire Sample, 1934 - 2022

Next, we further investigate this non-linear trend in PEAD over the past century by using quadratic trend regression analysis to examine our entire sample of earnings announcements, from 1934 to 2022. We accomplish this task by revising Equation (2) to include both a quarterly trend and a trend-squared term, as well as their respective interactions with Adj_SUE, as follows:

$$CAR[a,b]_{it} = \beta_0 + \beta_1 Adj_SUE_{it} + \beta_2 Trend_{it} + \beta_3 Adj_SUE_{it} \times Trend_t + \beta_4 Trend^2_{it} + \beta_5 Adj_SUE_{it} \times Trend_t^2 + Controls + \delta_d + \varepsilon_{it}. \quad (3)$$

In this specification, we are most interested in the coefficient of the interaction term between Adj_SUE and $Trend^2$ (β_5). The inverted U-shaped trend for PEAD (i.e., CAR[+2,+61]) shown in Figure 1 and Tables 4, 5, and 7 implies a negative coefficient for Adj_SUE x $Trend^2$ (β_5), while the U-shaped trend for CAR[0,+1] implies a positive coefficient, β_5.

The three columns on the left side of Table 8 present the results for Equation (3) that pertain to CAR[+2,+61], while the right side provides analogous evidence for CAR[0,+1]. The first column on each side of Table 8 excludes the controls, while the second column includes these controls, and the third column adds interaction terms between each control variable and

both Trend and Trend^2 (these interaction terms are not reported, for brevity). Across all three specifications, the coefficient of interest (β_5) is significantly negative for CAR[+2,+61] and significantly positive for CAR[0,+1], as expected. This evidence provides additional corroboration for the U-shaped trend in PEAD-related efficiency over the past century.

5.C. Robustness Tests

This section explores the robustness of our results to a number of different subsamples, as well as alternative modeling specifications.

5.C.1. Alternative Sampling Decisions

We first consider alternative samples that either impose or relax certain limitations on the data. Our results are robust to the following screens:

i) When we include all stocks;
ii) When we only keep NYSE firms in our sample;
iii) When we only keep NYSE firms and do not exclude firms with prices less than $1, or firms below the fifth percentile NYSE size cutoff;
iv) When we only keep NYSE and AMEX firms;
v) When we only keep firms that existed throughout each of the first four or five 10-year periods of our sample;
vi) When we compare announcements by the firms that existed over each consecutive pair of 10-year periods of our sample.

The evidence for the first four screens above appears in Table A.1. and Figure A.1. in the internet appendix. These tests show that our main results are not driven by the large increase in the number of firms due to the expansion of CRSP to include AMEX firms in 1962 or NASDAQ firms in 1973. The fifth screen above forces the composition of our sample to remain stable over the first four or five decades. Table A.2. shows that the increasing (declining) trend in CAR[+2,+61] (CAR[0,+1]) remains when we only keep firms that exist throughout the first four or five 10-year periods between 1934 and 1973. Figure A.2. shows a similar trend in the proportion of the total response to earnings news that is realized from days +2 to +61 (i.e.

CAR[2,61] / CAR[0,61]). The final screen above compares announcements made by the firms that existed throughout each consecutive pair of 10-year periods. Table A.3. and Figure A.3. again reveal a similar trend in PEAD-related efficiency over each consecutive decade up to 1983.

5.C.2. Alternative Modeling Decisions

We next examine the robustness of our results to alternative modelling decisions. Our evidence is robust to the following decisions:

i) When we use decile portfolios instead of quintile portfolios;
ii) When we use market-adjusted cumulative abnormal returns instead of size and book-to-market adjusted returns**;**
iii) When we use market-adjusted returns with decile portfolios instead of quintile portfolios;
iv) When we use a wider window that includes day -1 (i.e., CAR[-1,+1] / CAR[-1,+61]).

The above four screens are presented in Table A.4., and reveal a similar U-shaped pattern in PEAD-related efficiency when we employ these alternative modeling decisions.

5.D. Portfolio Approach, Subsets of Firms with Different Attributes

Recent work on information trends in capital markets suggests that, since the 1970s, the trends toward greater production of information and price informativeness have been mainly concentrated among large growth stocks, but they do not apply to the entire market.[8] Motivated by this work, we investigate whether our main findings are confined to certain subsets of firms or industries. In Figure A.5. of Appendix A.2., we reproduce the analysis in Figure 1 by plotting the mean: (i) CAR[+2,+61], (ii) CAR[0,+1], and (iii) CAR[+2,+61] / CAR[0,+61] for terciles of stocks each quarter with small versus large values of firm size or book-to-market based on NYSE cutoffs, averaged across all announcements in each 10-year period.

[8] Bai, Philippon, and Savov (2016) show that prices are more informative about long term cash flows for large growth firms, while Farboodi, Matray, Veldkamp, and Venkateswaran (2022) show that data about these firms is growing in abundance and importance.

First, consider the results for small versus large stocks in Panels A – C of Figure A.5. Panels A and B indicate that, although CAR[+2,+61] and CAR[0,+1] are substantially larger for small firms, both small and large stocks have a similar inverted U-shaped trend in CAR[+2,+61] along with a U-shaped trend in CAR[0,+1]. Combining the evidence in Panels A and B, Panel C reveals the same U-shaped trend in the proportion of total announcement returns that is realized over days +2 to +61 for both small and large stocks, across the 10-year periods in our sample.

Second, consider the analogous comparison for value versus growth firms in Panels D – F of Figure A.5. This evidence indicates that value and growth firms display the same nonlinear evolution in CAR[+2,+61], CAR[0,+1], and CAR[+2,+61] / CAR[0,+61] over the past century.

5.E. Regression Approach, Subsets of Firms with Different Attributes

Next, we conduct formal regression analysis of the nonlinear evolution in the market's underreaction to earnings news, for subsets of firms with high versus low values of the following five firm attributes: size, book-to-market, return volatility, illiquidity, and turnover. Each quarter we first independently sort stocks into terciles based on every firm attribute. We then re-estimate Equation (3) for the top or bottom tercile of announcements by each firm attribute, respectively.

The results are provided in Panels A and B of Table 9 for CAR[+2,+61] and CAR[0,+1], respectively. Similar to the estimation of Equation (3) in Table 8, we focus on the coefficient of the interaction between Adj_SUE and the trend squared term (β_5), which reveals whether CAR[+2,+61] or CAR[0,+1] follows a nonlinear trend over our sample period. This coefficient (β_5) is again significantly negative across all ten columns of Panel A in Table 9, while it is significantly positive across all ten columns of Panel B.

Figure A.5 and Table 9 establish that PEAD (CAR[+2,+61]) and CAR[0,+1] each reveal a similar nonlinear trend over the past century for both small and large firms, as well as firms

with either high or low values of book-to-market, return volatility, illiquidity, and turnover. Hence, in contrast to the recent literature on information trends in capital markets, the nonlinear trend in PEAD over the last century is fairly homogeneous across subsets of stocks with different attributes, and is thus likely to be driven by economic forces that affect all stocks similarly.[9]

5.F. Regression Approach, Subsets of Firms in Different Industries

Finally, we investigate whether our main results are driven by subsets of stocks in any specific industrial sector. The twelve columns in Panels A and B of Table 10 present the results from re-estimating Equation (3) for CAR[+2,+61] and CAR[0,+1], respectively, for the subsets of firms in the twelve industry portfolios provided by Kenneth French (i.e., the 12 Fama and French industries). Now, the coefficient of the interaction term, Adj_SUE x Trend (β_5), is significantly negative for nine of the twelve columns of Panel A in Table 10, and significantly positive across all twelve columns in Panel B. Once again, this analysis confirms that CAR[+2,+61] and CAR[0,+1] each follow a similar nonlinear trend over the past century across these industry portfolios, providing further corroboration of our other robustness tests.

6. Potential Factors Behind the Rise in PEAD Before the 1970s: The Usual Suspects

We now explore several alternative economic mechanisms to account for the nonlinear trend in PEAD over the past century. The well-documented recent decline in PEAD since the 1980s (i.e., the right half of Panel A in Figure 1), is commonly explained by declining market frictions, advances in information processing technology, and growing investor sophistication.[10]

[9] In Table A.5. of Internet Appendix 2, we also reproduce our results from Table 5 across subsets of firms with high versus low values of the same five firm attributes: size, book-to-market, illiquidity, return volatility, and turnover.

[10] For recent studies of PEAD, see Ali, Chen, Yao, and Yu (2020), Beaver, McNichols, and Wang (2020), He and Narayanamoorthy (2020), Hirshleifer, Peng, and Wang (2021), Hirshleifer and Sheng (2021), Li and Lytvynenko (2021), Li, Nekrasov, and Teoh (2020), Martineau (2022), Meursault, Liang, Routledge, and Scanlon (2022), and Thomas, Zhang, and Zhu (2021). See also Ayers, Li, and Yeung (2011), Baker, Litov, Wachter, and Wurgler (2010), Barber and Odean (2000), Bartov (1992), Campbell, Ramadorai, and Schwartz (2009), DellaVigna and Pollet (2009), Hirshleifer, Lim, and Teoh (2009), Hirshleifer, Myers, Myers, and Teoh (2008), and Hirshleifer and Teoh (2003).

In contrast, the opposite trend toward an increasing PEAD from the 1930s to the 1970s (i.e., the left half of the Panel A in Figure 1) is surprising for several reasons. We begin by discussing the historical trends of several economic factors that cannot account for the surprising rise in PEAD from the 1930s to the 1970s, although they can help to explain the decline in PEAD since the 1980s. Later, we will turn our attention to the historical behavior of several other economic factors that help to explain both the rise in PEAD before the 1970s and its decline thereafter.

6.A. Declining Trend in Limits to Arbitrage from the 1930s to the 1970s

The asset pricing literature overwhelmingly argues that investor underreaction to new information is due to trading frictions such as illiquidity, transaction costs, and information costs, as well as information uncertainty and a lack of investor sophistication.[11] In this light, we first examine the evolution of trading frictions such as illiquidity, return volatility, and share turnover across 10-year periods that span our sample period. Table 3 documents that the 1930s witnessed greater illiquidity and return volatility, along with less share turnover, when compared with the subsequent 10-year periods. Furthermore, trading frictions associated with turnover and illiquidity have declined steadily ever since the 1930s. However, while these declining limits to arbitrage over the past century help to explain the recent fall in PEAD since the 1980s, they cannot account for the surprising rise in PEAD from the 1930s to the 1970s.

6.B. Learning from the Publication of Ball and Brown (1968)

It is commonly argued that the recent proliferation of research on PEAD is at least partly responsible for greater attention being focused on this trading opportunity since the 1980s, and thus the subsequent decline in PEAD. This argument is consistent with McLean and Pontiff

[11] For example, see Chordia, Goyal, Sadka, Sadka, and Shivakumar (2009), Chordia, Subrahmanyam, and Tong (2014), and Ng, Tjomme, and Rodrigo (2008) for evidence regarding transaction costs, and Bartov, Radhakrishnan, and Krinsky (2000) and Vega (2006) for evidence regarding investor sophistication.

(2016), who contend that many predictable return patterns disappear after investors learn about them through academic publications. According to this argument, arbitrageurs should be less efficient at updating prices in the pre-Compustat era before much research was published about these anomalies. However, the greater efficiency embodied in the smaller level of PEAD that we document in the 1930s, along with the increasing magnitude of PEAD from the 1930s to the 1970s, are inconsistent with this learning argument of McLean and Pontiff (2016).

6.C. Insider Trading and Information Leakage Before Earnings Announcements Prior to 1961

An alternative potential explanation for the surprisingly high degree of price efficiency (i.e., the low PEAD) in the 1930s is the possibility that corporate insiders were more active at the time in trading on and sharing private information before earnings announcements. While the Securities and Exchange Act of 1934 instituted a prohibition against illegal insider trading, these insider trading laws were not enforced in the U.S. prior to 1961. Bhattacharya and Daouk (2002) and Bhattacharya (2023) argue that it is the enforcement and not merely the existence of illegal insider trading laws that matters for price efficiency. In this light, we might expect that greater pre-announcement information leakage could explain the more timely adjustment of stock prices around these events (i.e., the lower PEAD) that we find in the early portion of our sample.

We compute the pre-announcement cumulative abnormal returns over the 10 or 20 days before earnings are released (i.e., CAR[-11,-2] or CAR[-21,-2]) for the SUE hedge portfolio each quarter, and we average these returns across all announcements during each 10-year period. In Figure 3, we plot the ratio of these average pre-announcement returns to total announcement returns (i.e., CAR[-11,-2] / CAR[-11,+61] and CAR[-21,-2] / CAR[-21,+61]), across the 10-year periods from 1934 to 2022. This figure reveals that pre-announcement returns do not comprise a greater proportion of total announcement returns prior to 1961, before illegal insider trading laws

were first enforced in the U.S. This evidence suggests that insider trading and information leakage ahead of earnings announcements were not responsible for the surprisingly high degree of price efficiency (i.e., the low PEAD) that characterizes the early portion of our sample.

6.D. Historical Evolution of Corporate Finance Risk-Taking Activities

Historical trends in corporate policies toward risk-taking, such as engagement in M&A and IPO activities or exposure to leverage and credit risk, may also help to explain the nonlinear path of PEAD over the past century. For example, higher IPO intensity makes it more difficult to value firms due to the greater complexity of information that must be processed. Likewise, more corporate defaults intensify uncertainty and make it more challenging to price or hedge stocks.

However, the intensity of corporate defaults was high in the early 1940s, lower from the 1950s to the 1990s, and then again subsequently higher. Likewise, M&A activity was low in the 1970s before spiking in the 1980s, while IPO intensity also spiked after the 1990s (Philippon and Reshef, 2012). This historical behavior is inconsistent with the inverted U-shaped trend in PEAD that we find because the relatively low M&A activity, IPO intensity, and credit risk during the 1970s should imply a higher (not lower) PEAD at the time.

6.E. The Earnings Surprise Gap

Huang (2022) finds that the momentum gap, or the difference in returns between past winners versus losers during the portfolio formation period, helps to explain the return predictability of the momentum strategy. In a similar vein, it is possible that the magnitude of positive or negative earnings surprises in the first half of our sample might be smaller than those after the 1970s, due to changes in accounting standards and the complexity of accounting rules. As a result, the earnings surprise gap, or the difference between positive and negative earnings

surprises, might be smaller in the early portion of our sample, which could result in both a smaller initial reaction to earnings news and less post-announcement drift over this early period.

We test this possibility by expanding the quadratic trend regression in Equation (3) to also control for the magnitude of the earnings surprise gap in the current or previous quarter. Our measure of the earnings surprise gap is the difference between the 5th and 95th percentiles of SUE each quarter. We control both the earnings surprise gap and its interaction with the Adj_SUE. The results are provided in Table 11, and reaffirm the robustness of the nonlinear trends in both PEAD and CAR[0,+1], after controlling for the earnings surprise gap.

6.F. The Volatility of Earnings Surprises

Historical trends in earnings volatility might also conceivably influence the market's response to earnings news over time. We test this possibility by expanding our quadratic trend regression model in Equation (3) to also control for the volatility of earnings surprises each quarter, as well as its interaction with Adj_SUE. Our measure of earnings surprise volatility (Surprise_Vol) is the standard deviation across all earnings surprises (SUE) during the quarter. The results of this analysis are presented in Table 12, and again establish that the inverted U-shaped trend in PEAD and the U-shaped trend in CAR[0,+1] continue to persist after accounting for earnings surprise volatility. This evidence indicates that earnings volatility does not explain the nonlinear trend in either PEAD or the initial response to earnings news that we document.

7. Potential Economic Explanations for the Inverted U-Shaped Trend in PEAD

Our results thus far establish a robust inverted U-shaped trend in PEAD over the past century. We show that this pattern is not an artifact of data snooping and further demonstrate that traditional mispricing explanations based on market frictions cannot fully account for our findings. However, we have not yet explored the underlying economic mechanisms that may be

driving these surprising results. In this section, we propose and evaluate potential explanations that could help us better understand the dynamics behind the documented pattern.

7.A. Time Variation in Arbitrage Capacity

In financial markets, the degree of price efficiency evolves in tandem with the arbitrageurs' capacity to perform the necessary arbitrage activity to push prices toward fundamental values. We argue that this arbitrage efficacy should vary over time, depending on the capacity of the finance industry to support the work of arbitrageurs. Accordingly, we propose that time variation in the magnitude of PEAD should be closely linked to structural trends and challenges that affect the ability of the finance industry to support the efforts of arbitrageurs.

Over the past century, the efficacy of arbitrageurs has varied substantially in response to the dynamic interplay between the growth in financial information versus the capacity of arbitrageurs to process this information. Throughout this period, there has been enormous growth in the amount and complexity of information available for investors to scrutinize. For example, this period witnessed expanding regulation that compels firms to disclose ever more information with greater complexity.[12] Furthermore, the introduction of Nasdaq firms in the 1960s amplified the challenges faced by investors to analyze the growing number of earnings announcements. Over the first half of our sample, from the 1930s to the 1970s, the capacity of arbitrageurs to process information may not have kept pace with the enormous growth in this information, which could help to explain the increasing PEAD that we document over this period.

In contrast, the recent period since the 1970s witnessed extraordinary technological innovation, including the arrival of the computer age, the internet, big data analytics, and AI. This innovation in information processing technology revolutionized the ability of arbitrageurs to

[12] See Internet Appendix A.1. for a discussion of the history of expanding U.S. regulation on corporate disclosure.

effectively analyze the continuing growth in the amount and complexity of information. As a result, the increasing capacity of arbitrageurs since the 1980s may have overtaken the continuing growth in this information, which could help to explain the declining PEAD over this period.

We focus on two aspects of this dynamic interplay between the growth in financial information versus the capacity of arbitrageurs to process this information that has altered the efficacy of arbitrageurs over the past century, to help explain the U-shaped trend in PEAD. First, we propose that *arbitrage efficacy* can be effectively proxied by the relative size of the trading sector, defined as the market value of the trading sector scaled by the total number of publicly traded firms that must be analyzed (labeled as Relative_Size). This proxy is motivated by the model of Martin and Nagel (2022), in which investors can forecast asset prices more efficiently when the number of firm attributes (i.e., the complexity of information) is sufficiently small relative to the total number of firms that require analysis.

Second, we follow Philippon and Reshef (2012) to proxy *arbitrage skill* as the average annual level of wages or education in the financial sector, relative to that in the nonfarm private sector (i.e., Relative_Wage or Relative_Education). We obtain data on sector-based compensation, employment, and education from the National Income and Product Accounts at www.bea.gov. We calculate the average wage by dividing total compensation by total employment in the industry. The relative wage in the financial sector is then defined as the average wage in the finance industry scaled by the average wage in the nonfarm private sector. Similarly, our measure of relative education in the finance industry is calculated as the difference between the share of workers in the financial sector with more than a high school education versus the corresponding share in the non-financial private sector. We update the relative education data provided on Thomas Philippon's website up to 2022 by using the March Current

Population Survey. We calculate the proportion of educated workers in each sector using individual-level education attainment data, adjusted by sampling weights and hours worked.[13]

We next analyze the association between the timeliness of the market's response to earnings news (i.e., PEAD and CAR[0.+1]) and each proxy for arbitrage efficacy or arbitrage skill (i.e., Relative_Size, Relative_Wage, or Relative_Education). In this context, we expect that a larger value for each proxy should enhance the timeliness of price adjustment around earnings announcements (i.e., reduce PEAD and increase CAR[0,+1]).[14]

We begin this analysis by estimating three expanded versions of Equation (1) that include each proxy for arbitrage efficacy (i.e., Size_Per_Firm) or arbitrage skill (i.e., Relative_Wage or Relative_Education), as well as the proxy's interaction with Adj_SUE. We also control for different aspects of the aggregate stock market and macroeconomic conditions that may also affect investors' response to earnings news, as well as interactions of these conditions with Adj_SUE. Controls include cumulative market returns (Aggregate Returns), daily market return volatility (Aggregate Volatility), Amihud illiquidity (Aggregate Illiquidity), monthly stock return volatility (Aggregate Disagreement), the Price/Dividend Ratio, Inflation, Industrial Production, the Money Supply, NBER Recessions, and the Treasury Bill Rate. All variables are defined in Table 2.

The three columns on the left side of Table 13 present the relation between PEAD (i.e., CAR[+2,+61]) and each proxy for arbitrage efficacy or skill, while the three columns on the right provide the analogous relation between each proxy and CAR[0,+1]. Once again, the key variable is the interaction between Adj_SUE and each proxy (i.e., Relative_Size, Relative_Wage, or

[13] https://pages.stern.nyu.edu/~tphilipp/research.htm

[14] Pastor, Stambaugh, and Taylor (2015) find that the ability of actively managed funds to outperform passive benchmarks declines as the size of the active mutual fund industry increases. This evidence is also consistent with the conjecture that markets become more efficient as the relative size of the finance industry grows.

Relative_Education). In each case, the coefficient of this interaction term reflects the impact of the proxy on the timeliness of the market's response to earnings news (i.e., PEAD or CAR[0,+1]). In this context, we expect that a higher level of arbitrage efficacy or skill should decrease PEAD and increase CAR[0,+1] after controlling for market conditions.

The results in Table 13 uniformly support these expectations. In columns (1) and (4), the coefficients of the interaction between Adj_SUE and Relative_Size imply that greater arbitrage efficacy is associated with lower PEAD and a higher immediate response to earnings news (i.e., for PEAD, β_3 = -0.059, t-ratio = -2.98, for CAR[0,+1], β_3 = 0.041, t-ratio = 6.18). Columns (2) and (3) similarly reveal a significant *negative* coefficient on the interaction term involving each proxy (i.e., for wage, β_5 = -0.032, t-ratio = -5.14; and for education, β_7 = -0.257, t-ratio = -3.59) when the dependent variable is PEAD. In contrast, columns (5) and (6) both reveal a significant *positive* coefficient on the interaction when the dependent variable is CAR[0,+1] (i.e., for wage, β_5 = 0.018, t-ratio = 8.27; and for education, β_7 = 0.102, t-ratio = 4.43). Together, the evidence in Table 14 indicates a more timely response to earnings news (i.e., smaller PEAD and a larger immediate response) when the finance industry exhibits greater arbitrage efficacy or skill.

7.B. Time Variation in the Capacity of Arbitrageurs: The Role of Financial Deregulation

In the previous section, we demonstrate that time variation in arbitrage efficacy and skill plays a central role in explaining our results. We argue that it is the evolving internal dynamics of the financial industry that shape arbitrageurs' capacity to incorporate earnings information into stock prices. However, one potential concern is that broader economic conditions—such as financial crises or other omitted macroeconomic factors—might simultaneously influence both the magnitude of PEAD and the efficacy or skill of arbitrageurs. If so, our results could be driven by an omitted variable rather than by such shifts in the structure of the finance industry.

We address this concern by focusing on financial deregulation, which has been well-documented as a primary driver of structural changes within the finance industry. According to Philippon and Reshef (2012), deregulation plays a pivotal role in shaping the wages and educational composition of the finance labor force, with its effects unfolding gradually over a lag of approximately five years. While deregulation encourages innovation and allows firms to better utilize skilled labor, a more heavily regulated financial environment imposes binding constraints on firm organization, limiting the deployment of human capital. Crucially, Philippon and Reshef (2012) show that adverse macroeconomic events—such as financial crises, bank failures, or high unemployment—do not predict wage or education trends in finance, whereas deregulation does. Additionally, regulation directly influences the size of the financial sector, as more (less) strict rules reduce (expand) its capacity to grow and allocate resources to functions like arbitrage.

Based on this argument, we posit that deregulation in the financial sector exerts a gradual effect on our three proxies for arbitrage efficacy and skill, which in turn shape the market's response to earnings announcements. The slow-moving nature of this impact—manifesting over a five-year horizon— allows us to isolate the influence of internal structural changes within the financial industry due to deregulation from those due to transitory macroeconomic shocks. In contrast to the persistent and delayed effects of deregulation on the financial sector—and, ultimately, on PEAD—macroeconomic conditions are more likely to have temporary short-lived effects, making them less likely to be the primary drivers of our main findings.

In Figure 4, we first plot the evolution of the financial deregulation index compiled by Phillipon and Reshef (2012) over the past century, along with relative wages and education in the finance industry. The passage of Glass-Steagall in 1933 marked an end to the unregulated laissez-faire era of the 1920s. This figure indicates that the increasingly regulated financial

environment of the 1930s resulted in a gradual exodus of skilled labor from the finance industry (revealed by declines in relative wages and education) with a lag of roughly five years, which then prevailed until the 1980s. The subsequent return to extensive financial deregulation in the 1980s then reversed this trend back toward greater demand for highly skilled labor in the financial sector. Overall, Figure 4 illustrates that the timing and pattern of financial deregulation closely mirror that of PEAD, suggesting a strong potential impact of deregulation on our results.

Next, we test this conjecture by decomposing the total variation in the relative size, wages, or education of the financial sector into two components: (i) the variation attributable to financial deregulation lagged by five years, and (ii) the residual variation unrelated to past deregulation. Specifically, for each year, we estimate the fitted and residual values of our three proxies for arbitrageur efficacy and skill using the following time-series regression models:

$$\text{Relative_Size}_T = \beta_0 + \beta_1 \text{Deregulation}_{T-5} + \varepsilon_{iT}\,, \tag{9}$$

$$\text{Relative_Wage}_T = \beta_0 + \beta_1 \text{Deregulation}_{T-5} + \varepsilon_{iT}\,, \tag{10}$$

$$\text{Relative_Education}_T = \beta_0 + \beta_1 \text{Deregulation}_{T-5} + \varepsilon_{iT}\,. \tag{11}$$

We then estimate an expanded version of Equation (5) that relates our measures of the market reaction to earnings announcements (i.e., PEAD or CAR[0,+1]) to the resulting fitted and/or residual values of these three proxies, separately. In our analysis, the fitted values capture the component of time variation in arbitrage skill or efficacy that is attributable to structural changes in the financial sector, specifically those induced by deregulation. To the extent that the results in the previous section are driven by the gradual impact of deregulation on the internal dynamics of the finance industry, we expect the main explanatory power to come from the fitted components of each of the three proxies, rather than from their residual counterparts.

Panels A and B of Table 14 present this analysis pertaining to PEAD and CAR[0,+1], respectively. In columns (1), (3) and (5) of each Panel, we replace the actual values of each proxy for arbitrage efficacy or skill from the analysis of Table 14 with the fitted values from Equations (9), (10), and (11), as well as the other controls.[15] Columns (2), (4), and (6) then append this analysis to also include the residuals from Equations (9), (10), and (11), to further explore whether the residual variation in each proxy that is not attributable to lagged financial deregulation is also associated with PEAD or CAR[0,+1].

First, consider the evidence in Panel A of Table 14, which documents the relation between the delayed response to earnings news (i.e., PEAD) and the *predicted* components of the relative size, wages, or education in the financial sector. In all six columns of Panel A, the coefficients of the interaction terms involving Adj_SUE and the *predicted* component of each proxy (i.e., Adj_SUE x Pred_Size , Adj_SUE x Pred_Wage, or Adj_SUE x Pred_Educ) are significantly negative. This negative relation indicates that higher levels of each proxy for arbitrage efficacy or skill that are predicted by lagged deregulation are associated with more timely adjustment to earnings news (i.e., smaller PEAD), as expected. In addition, in columns (2), (4), and (6), the coefficients of the interactions involving the *residual* component of each proxy (i.e., Adj_SUE x Resid_Size , Adj_SUE x Resid_Wage, or Adj_SUE x Resid_Educ) are either insignificant or the wrong sign. Furthermore, the coefficients of the former interactions involving the *predicted* components of these three proxies remain robust when we append the model to include the *residual* components, as well as their interactions with Adj_SUE.

Second, consider the corresponding analysis that pertains to the immediate response to earnings news (i.e., CAR[0,+1]), provided in Panel B of Table 14. In all six columns of Panel B,

[15] This approach is analogous to two stage least squares estimation of the analysis in Table 14, which relies on lagged financial deregulation as an instrument for each proxy for the efficacy or skill of arbitrageurs.

this analysis reveals significant positive coefficients for the same three interactions involving the *predicted* component of each proxy (i.e., Adj_SUE x Pred_Size , Adj_SUE x Pred_Wage, and Adj_SUE x Pred_Educ). Once again, this evidence indicates that higher levels of each proxy for arbitrage efficacy or skill that are predicted by lagged deregulation are associated with more timely price adjustment to earnings news (now, a larger CAR[0,+1]), as expected. Furthermore, the coefficients of the analogous interaction terms involving the residual variation in each proxy (i.e., Adj_SUE x Resid_Size , Adj_SUE x Resid_Wage, or Adj_SUE x Resid_Educ) are again either insignificant or the wrong sign.

Together, this evidence implies that markets are more efficient in pricing earnings information during periods when lagged values of the financial deregulation index predict that more resources with greater skill will be subsequently devoted to arbitrage activity. Our results suggest that, rather than being driven by episodic macroeconomic shocks such as financial crises that may impede market efficiency and raise endogeneity concerns, the slow-moving effects of financial deregulation represent a primary force that is responsible for enhancing or restraining the capacity of arbitrageurs. Over time, these consequential institutional changes in the finance industry then drive the speed and completeness with which information is incorporated into prices around earnings announcements.

7.C. Trading Activity Around Earnings Announcements

Price efficiency in financial markets is achieved through the trading activity of arbitrageurs. The literature on post-earnings announcement drift (PEAD) suggests that investor attention plays a crucial role in the delayed market response to earnings news (Hirshleifer, Lim, and Teoh, 2009). Stocks that attract more attention tend to exhibit higher trading volume around earnings announcements, stronger immediate price responses, and lower subsequent drift. Hence,

elevated trading volume suggests that arbitrage forces are more active and attentive, facilitating the timely incorporation of earnings information into prices.

In context of arbitrage capacity we discussed above, in the 1930s, earnings announcements likely commanded significant investor attention due to the scarcity of alternative aspects of compelled financial disclosure at the time. With fewer informational substitutes available to investors in the 1930s, these announcements stood out as highly visible key events, accelerating the assimilation of earnings news into prices. Since the 1930s, regulation compelling more financial disclosure has evolved to generate a growing body of financial information with greater complexity.[16] By the 1970s, the introduction of Compustat and the growth of data vendors dramatically expanded investors' access to financial information. However, with the limited computational tools and analytical capacity at the time, investors likely began to diversify their focus across a broader set of accounting metrics, potentially diluting the salience of earnings announcements. In more recent decades, advances in information technology and data processing capacity may have restored the ability of arbitrageurs to refocus on the centrality of earnings announcements, drawing investor attention back to these events. Accordingly, the combination of heightened investor attention and improved arbitrage capabilities in both the earlier and more recent portions of the past century may have contributed to more timely price adjustment, helping to explain the observed inverted U-shaped trend in PEAD.

We assess whether arbitrageurs devoted greater attention to earnings announcements in the earlier and later periods of the last century, by analyzing the historical trend in abnormal volume on the two days around earnings announcements (AVOL[0,+1]). This measure has been

[16] See Appendix A.1. for a discussion of the evolution of compelled financial disclosure since the 1930s.

widely used as a proxy for investor attention around earnings announcements.[17] We measure abnormal volume as the difference between average log dollar volume on days 0 and +1 versus average log dollar volume over days -60 to -11 prior to the announcement. Every quarter, we compute the firm's AVOL[0,+1] for the subset of stocks in the top and bottom quintile portfolios sorted by SUE. We then compute the mean abnormal volume for these subsets of announcing firms with a large absolute SUE, averaged across all quarters during each 10-year period.

The results are plotted in Figure 5 and show that abnormal volume around earnings announcements follows a distinct U-shaped trend over the last century that mirrors the inverted U-shaped path of PEAD. This evidence indicates that investors were more attentive to earnings news during both the earlier and later years of our sample, relative to the middle years. That is, they traded more actively around earnings announcements near the beginning and end of the past century relative to the middle, which could help to further explain the more timely adjustment of prices to earnings news (i.e., the lower PEAD) we document at these times.

Next, we analyze the impact of abnormal volume on our findings by estimating an expanded version of Equation (1) that includes our measure of abnormal volume, Avg_AVOL, along with its interaction with Adj_SUE. We calculate Avg_AVOL as the quarterly average of AVOL[0,+1] over all announcements. In this light, increased investor attention proxied by higher abnormal volume should be associated with lower PEAD. The results are presented in Table 15. The coefficients of the interaction terms between Adj_SUE and Avg_AVOL in columns (1) and (2) imply a significantly more timely response to earnings news, in the form of lower PEAD and higher immediate response (i.e., for PEAD, β_3 = -0.024, t-ratio = -7.22, for CAR[0,+1], β_3 =

[17] Bai, Philippon, and Savov (2016) show that greater volume facilitates more efficient incorporation of information into prices. See also Akbas (2016), Atiase and Bamber (1994), Bamber (1987), Berkman et al. (2009), Garfinkel and Sokobin (2006), Gervais, Kaniel, and Mingelgrin (2001), and Hirshleifer, Lim, and Teoh (2009), for more evidence regarding abnormal trading volume around earnings announcements.

0.011, t-ratio = 8.63), when abnormal volume around the announcement is greater. Together, the evidence in Figure 5 and Table 15 collectively suggests that investors respond to earnings news more promptly during periods when greater investor attention is paid to these events.

8. Conclusion

Our study sheds new light on the historical evolution of post-earnings announcement drift (PEAD), challenging conventional views on its origins and persistence. By hand-collecting earnings announcement data from the *Wall Street Journal* for the pre-Compustat period (1934–1971) and analyzing an extended sample through 2022, we establish that PEAD has been a robust and enduring anomaly for nearly a century. Contrary to the findings of Linnainmaa and Roberts (2018) for other accounting-based anomalies, PEAD does not appear to be an artifact of data snooping, as it remains statistically significant even before its formal discovery in the academic literature.

A key insight from our analysis is the timing of PEAD's strength over the past century. Rather than following the typical pattern of a mispricing anomaly—strongest before discovery and fading after—PEAD shows an inverted U-shape. It was weaker in the 1930s, grew from the 1940s through the 1980s, and has declined in recent decades. This challenges the view that PEAD reflects static inefficiency, which would imply it was most pronounced early on when markets were less developed and arbitrageurs unaware. Instead, our findings suggest PEAD intensified as markets evolved, peaking in the 1970s–1980s, then declining as participants grew more efficient at processing earnings news.

Additionally, we document a historical link between the finance industry's structural evolution and the magnitude of PEAD. We introduce two novel proxies—arbitrage efficacy (the ratio of finance industry size to public firms) and arbitrage skill (relative wages and education in

the finance sector)—and find that these metrics align closely with PEAD's rise and fall. The increasing complexity of financial markets from the 1930s to the 1980s may have hindered timely price adjustment to earnings news, whereas the post-1980s expansion of arbitrage capacity and the influx of highly skilled financial professionals improved market efficiency, reducing PEAD.

We propose that the evolution of PEAD reflects shifts in investor attention, disclosure practices, and arbitrage capacity. In the 1930s, limited public disclosure meant that earnings announcements were among the most significant corporate events, likely drawing heightened investor attention and facilitating a more immediate price response. As financial disclosures expanded with the introduction of Compustat in the 1970s, investors were faced with a greater volume of accounting information, potentially diluting their focus on earnings announcements and allowing PEAD to grow stronger. However, as financial technology and information processing capabilities advanced in the post-1980s era, the market's ability to incorporate earnings news in a timely manner improved, contributing to the decline in PEAD.

Our findings suggest that PEAD is not a static market anomaly but a phenomenon that evolves with changes in investor behavior, financial disclosure regimes, and arbitrage activity. The fact that PEAD grew stronger over time before its discovery—and then weakened post-1980s—contradicts the traditional mispricing narrative and highlights the importance of analyzing an anomaly's full historical trajectory. Future research should further investigate whether other anomalies exhibit similar nonlinear trends and explore the broader implications of evolving financial market efficiency.

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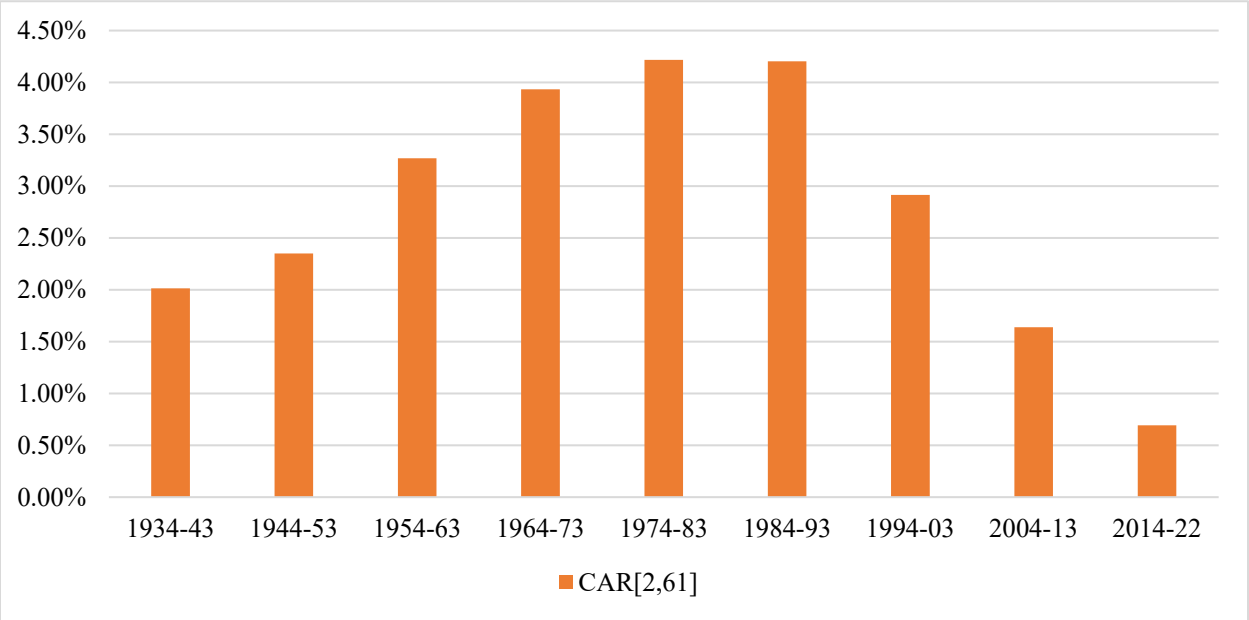
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Figure 1. Market's Response to Earnings News over 10-Year Periods from 1934 to 2022

Panels A and B plot the magnitudes of PEAD (CAR[+2.+61], in the **orange** bars) and the initial announcement return (CAR[0,+1], in the **blue** bars), respectively, for the SUE hedge portfolio that is long the quintile of earnings announcements each quarter with high standardized unexpected earnings (SUE) and short the quintile with low SUE, averaged across all announcements during every 10-year period. The 95% significance level for each average return measure is indicated by a hash mark on the **orange** and **blue** bars in Panels A and B, respectively. Panel C then plots the proportion of the total response to earnings news (over days 0 to +61) that is incorporated with a delay (CAR[+2,+61) / CAR[0,+61], in the **orange** bars) versus immediately (CAR[0,+1] / CAR[0,+61], in the **blue** bars).

Panel A. Magnitude of Post-Earnings Announcement Drift (CAR [2,61])



Panel B. Magnitude of Immediate Response to Earnings Announcements (CAR[0,1])

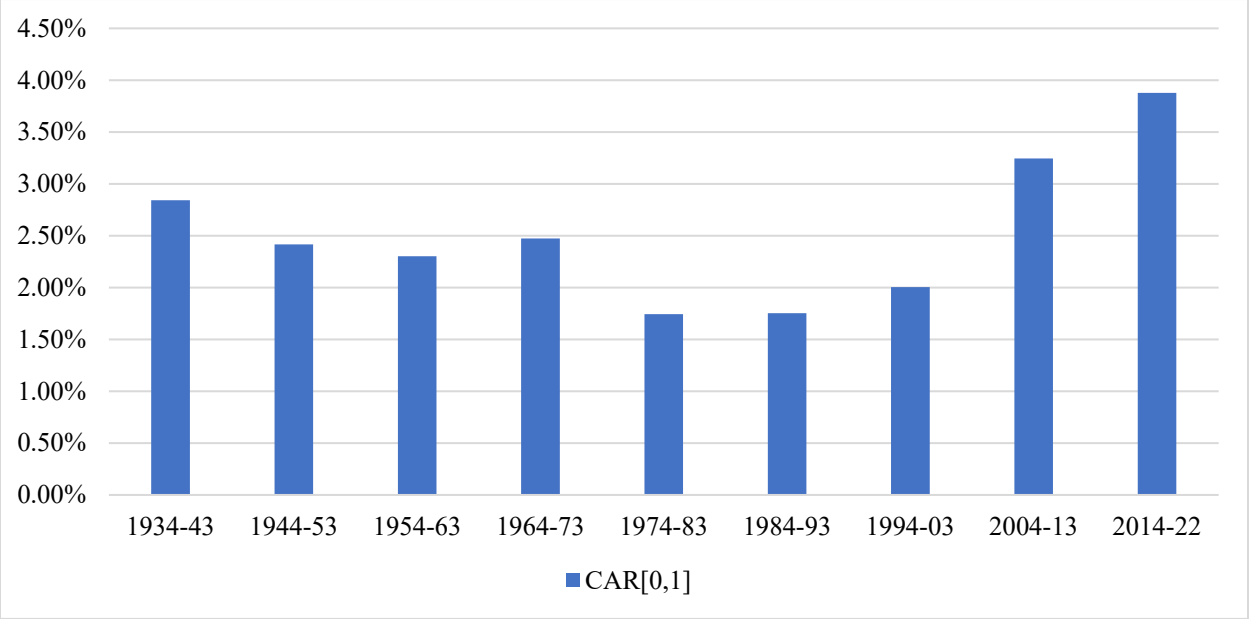


Figure 1, continued

Panel C. Proportion of Total Announcement Return Realized on Days 2 to 61, or 0 and 1, (CAR[2,61] / CAR[0,61]) or (CAR[0,1] / CAR[0,61]), averaged over Each 10-Year Period

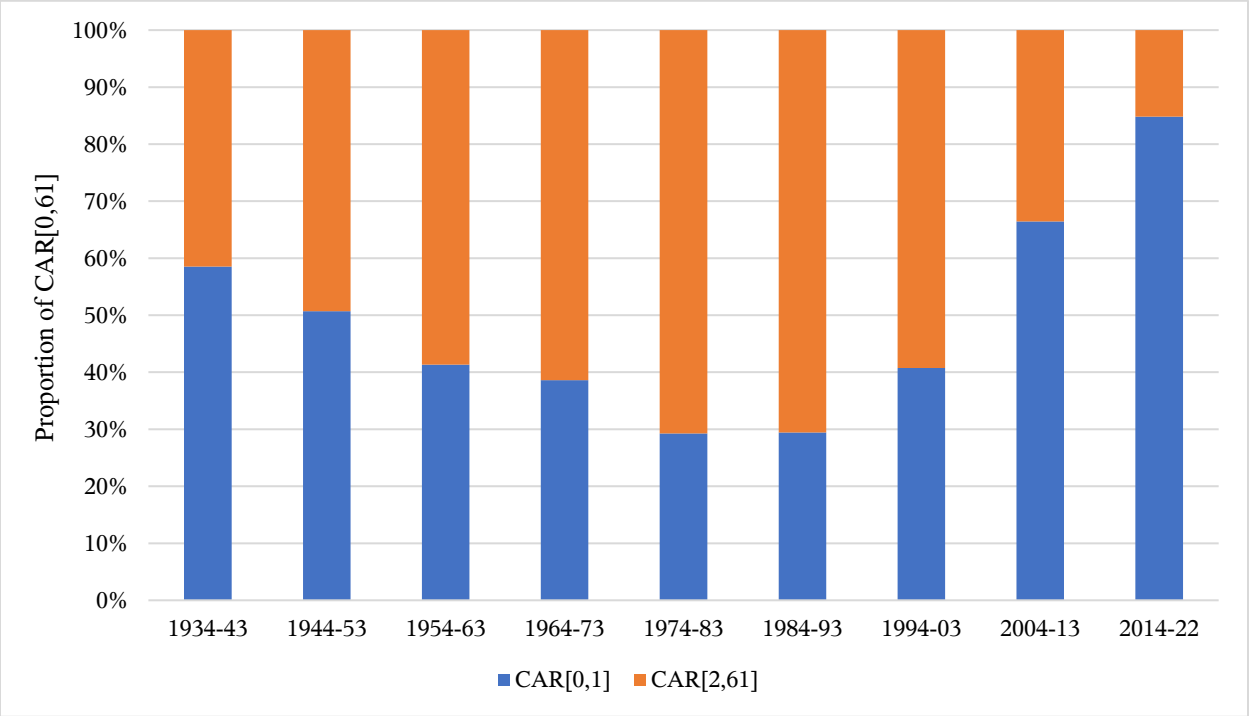


Figure 2. Current Earnings Report from the April 14, 1937 issue of the *Wall Street Journal*

Current Earnings Reports

Following is Dow-Jones summary of earnings reported by various companies, giving the net income after federal taxes, common share earnings for the period indicated and annual dividend rate on common stocks. When profit is before federal taxes, no calculation of share earnings is made. If stock is listed on the New York Stock Exchange the tape symbol is given in parentheses after name of company:

Tuesday, April 13, 1937:

Company:	Annual dividends	Net income 1937	Net income 1936	Common share earnings 1937	Common share earnings 1936	Surplus after dividends 1937	Surplus after dividends 1936
Allen Industries, Inc.:							
March 31 quarter	§1.00	$177,496	$162,415	h$.70	h$ 66	*.....	*.. ..
Barker Bros. Corp. (BKR):							
March 31 quarter	f...	173,694	67,814	h.72	h.15	.....	.....
Bridgeport Machine Co.:							
March 31 quarter	1 00	‡175,897	‡77,878	....	....	.....	.....
Cleveland-Cliffs Iron Co.:							
March 31 quarter	f...	108,499	†72,180	....	....	.....	.....
Collins & Aikman Corp. (CK):							
Year, February 27	w.50	4,991,659	3,974,687	8.15	6.28	$1,775,997	$2,973,911
General Finance Corp.:							
February 27 quarter	f...	138,742	83,587	h.16	h.12	.....	.....
Goldblatt Bros., Inc.:							
x Thirteen months, January 31	2.40	1,947,414	1,114,226	h3.07	h1.86	.....	.....
Hoe & Co., R.:							
Six months, March 31	f...	182,695	†43,518	....	....	.....	.....
International Great Northern R. R.:							
Two months, February 28	....	†494,916	†479,412	....	....	.....	.....
Kimberly-Clark Corp. (KMB):							
March 31 quarter	1.00	t451,561	t204,282	.93	.42	.....	.....
Twelve months, March 31	1.00	t1,105,763	t536,668	2.27	1.10	.....	.....
Nash-Kelvinator Corp. (NSK):							
February 28 quarter	1.00	709,563	65,731	....	....	.....	.....
Pacific Finance Corp. of Calif.:							
March 31 quarter	§1.80	392,759	412,463	.70	.79	.....	.....
Parker Rust Proof Co. (PRK):							
March 31 quarter	§1.50	329,850	250,393	.76	.58	.....	.....
Pittsburgh & West Virginia Ry. (PW):							
Two months, February 28	f...	101,234	55,463	.33	.18	.....	.....
Transue & Williams Steel Forging Corp. (TU):							
March 31 quarter	w.15	‡78,349.	‡33,515	....	....	.....	.....
		1936	1935	1936	1935	1936	1935
American Pneumatic Service Co.:							
Year, December 31	f...	†$139,458	†$210,247	....	....	.....	.....
Chicago & North Western Rwy. System (NW):							
Year, December 31	f...	†9,388,038	†11,448,630	....	....	.....	.....
Cleveland Automatic Machine Co.:							
Year, December 31	f...	31,259	†13,800	p$1.84	....	.....	.....
Continental Diamond Fibre Co. (CDF):							
Year, December 31	k$2.00	596,060	167,677	1.30	$.37	d$317,340	d$311,738
Crowell Publishing Co.:							
Year, December 31	3.00	2,923,118	2,399,539	3.81	3.12	612,826	1,026,931
Imperial Oil, Ltd.:							
Year, December 31	k1.25	25,628,285	25,239,850	.95	.93	d8,078,062	d8,467,427
Kansas Gas & Electric Co.:							
Twelve months, December 31	....	1,287,971	1,062,791	....	....	.....	.....
Kildun Mining Corp.:							
Year, December 31	f...	†90,702	†17,061	....	....	.....	.....
Massachusetts Utilities Assoc.:							
Year, December 31	f...	1,470,741	1,441,865	....	....	.....	.....
Montana Power Co.:							
Twelve months, December 31	....	3,306,096	2,608,406	....	....	.....	.....
Nebraska Power Co.:							
Twelve months, December 31	....	1,825,026	2,048,966	....	....	.....	.....
Smith (L. C.) & Corona Typewriters, Inc. (SLT):							
December-31 quarter	§1.00	279,141	.....	1.36	....	.....	.....

*Not available. †Net loss. ‡Profit before federal income taxes. §Plus extras. d Deficit. f No common dividend. h On shares outstanding at close of respective periods. k Paid in year 1936. p On preferred stock. t Surplus available for common stock, after preferred dividends. w Last dividend declared; period not announced by company. x Thirteen months ended January 31, 1937, compared with year ended December 31, 1935.

Figure 3. Pre-Announcement Returns as a Proportion of Total Announcement Returns over 10-year Periods from 1934 to 2022

This figure plots the proportion of the total 73-day (or 83-day) response to earnings news that occurs over days -11 to +61 (or days -21 to +61) around the earnings announcement on day 0, which is realized prior to the earnings release over days -11 to -2 (or days -21 to -2), for each 10-year period from 1934 to 2022. We first compute the mean 10-day (or 20-day) pre-announcement return, CAR[-11,-2] (or CAR[-21,-2]) and the total 73-day (or 83-day) total announcement return, CAR[-11,-+61] (or CAR[-21,+61]) for the hedge portfolio based on SUE each quarter. We then average these return measures across all quarters during every 10-year period. The **blue** line tracks the proportion of the total response to earnings news that is incorporated during the 10-day or 20-day pre-announcement interval (i.e., CAR[-11,-2] / CAR[-11,+61] or CAR[-21,-2] / CAR[-21,+61]), across the 10-year periods that span our sample.

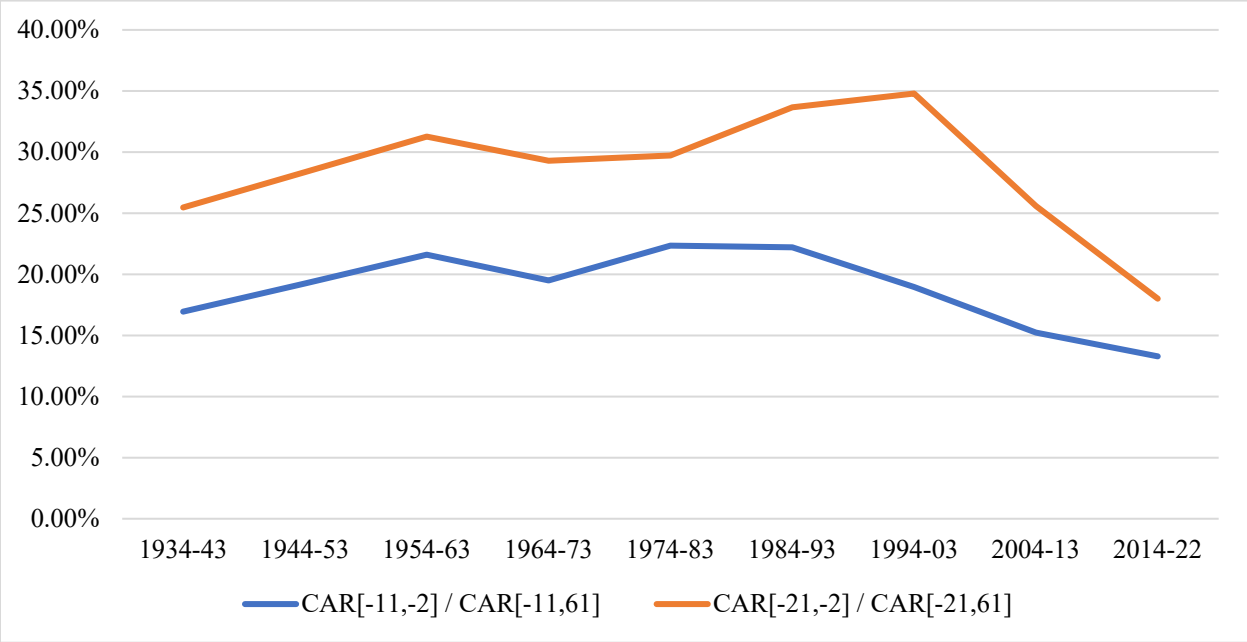


Figure 4. Wages, Education, and Deregulation in the Finance Industry from 1929 to 2022

This figure tracks the historical evolution in the level of wages (**blue**), education (**orange**), and deregulation (**black**) in the financial sector, relative to other industries, from 1929 to 2022. We rely on the three time series developed in Philippon and Reshef (2012), until 2006. We then extend these measures from 2006 to 2022, by following their definitions. For example, in 2010 we adjust the financial deregulation index downward to reflect the introduction of the Dodd-Frank Act, which created stricter regulations for systemically important banks. The relative wage is defined as the ratio of the average wage in the finance industry scaled by the average wage in the nonfarm private sector. The measure of relative education is the difference between the share of workers with more than a high school education in the finance industry versus the analogous share in the nonfarm private sector.

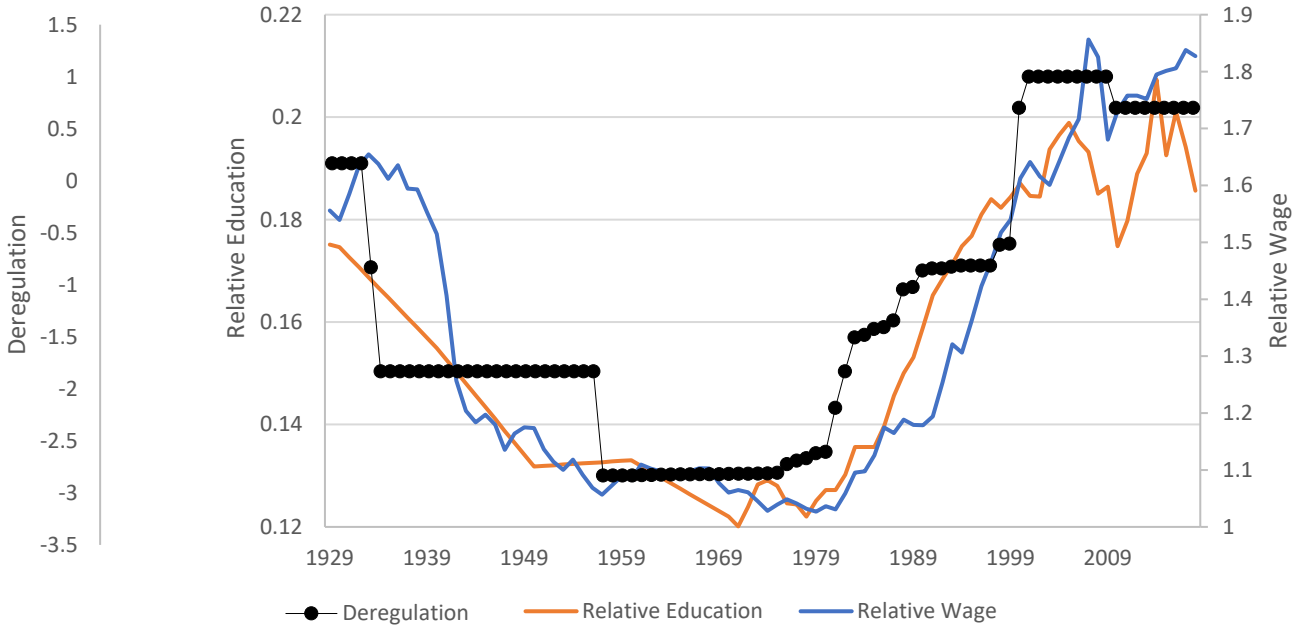


Figure 5. Abnormal Volume Around Earnings Announcements over 10-Year Periods from 1934 to 2022

This figure documents evolution in the average abnormal trading volume on the two days around earnings announcements (AVOL[0,1]), for subsets of stocks each quarter with a large absolute earnings surprise, averaged across all such earnings announcements during each 10-year period between 1934 and 2022. For each earnings announcement, abnormal trading volume on days 0 and +1 is the difference between the average log dollar volume in the announcing firm's stock on days 0 and +1, versus the average log dollar volume over days -60 to -11 prior to the announcement (on day 0). Every quarter, we first compute every firm's AVOL[0,1] for the subset of stocks in the top or bottom quintile portfolios based on SUE. We then plot the mean daily abnormal volume for these subsets of announcements with a large absolute earnings surprise, averaged across all quarters during each 10-year period.

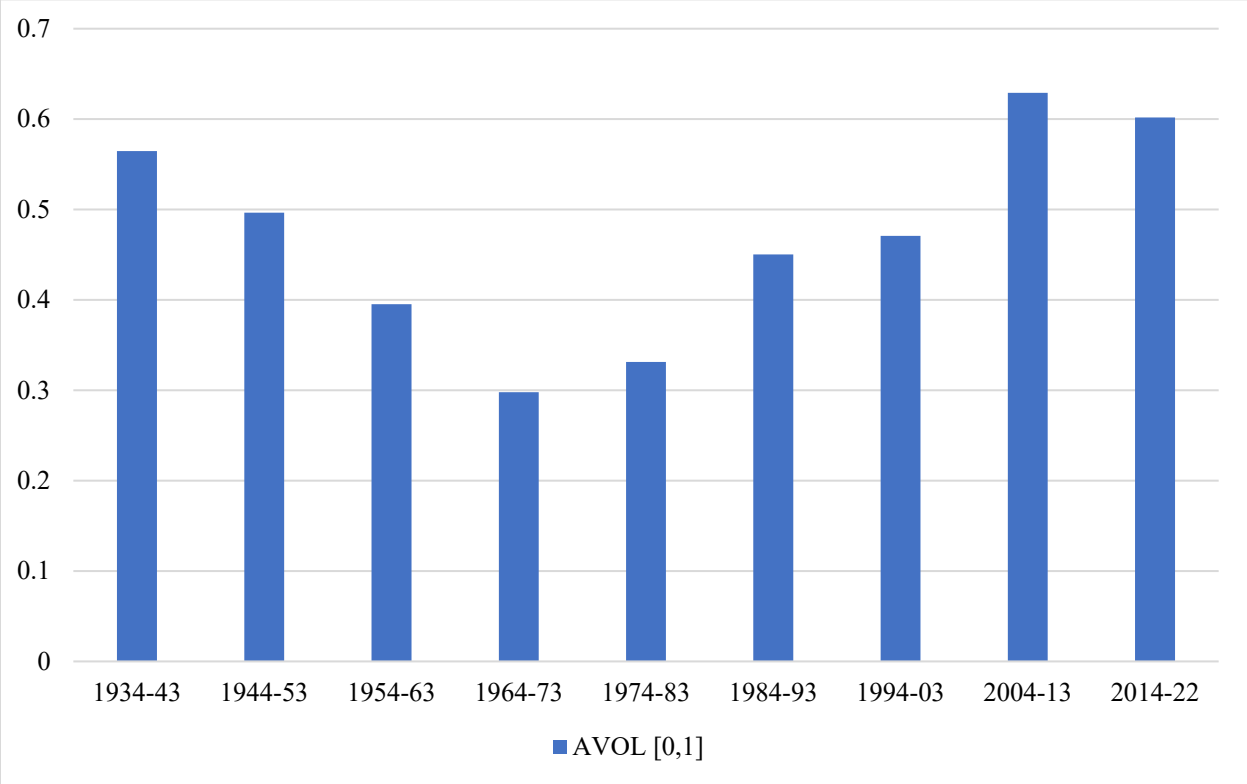


Table 1. Coverage and Size Distribution of Firms in the WSJ Sample of Earnings Announcements versus the CRSP Universe

Panel A reports the number of unique firms in our hand-collected sample of earnings announcements from the Wall Street Journal (WSJ) and the CRSP universe, respectively, over each ten-year period from 1934 to 1971. Panel B reports descriptive statistics that reflect the distribution of firm size (i.e., market capitalization) for the firms in the WSJ Sample and the CRSP universe, respectively, at the beginning of each 10-year period and the end of the sample.

Panel A. Number of Uniques Firms in the WSJ Sample versus the CRSP Universe

	Year			
	1934-43	**1944-53**	**1954-63**	**1964-71**
Number of Firms				
CRSP Universe	895	1,081	2,264	3,099
WSJ Sample	630	897	1,793	2,894
WSJ Coverage	70%	83%	79%	93%

Panel B. Size Distribution of Firms in the WSJ Sample versus the CRSP Universe

	Year				
	1934	**1944**	**1954**	**1964**	**1971**
Average Size					
CRSP Universe	39.81	56.52	151.78	232.34	301.51
WSJ Sample	41.28	58.60	162.19	267.22	313.58
25th Percentile					
CRSP Universe	2.19	6.92	13.91	8.88	15.49
WSJ Sample	2.19	7.00	14.76	11.14	15.90
Median Size					
CRSP Universe	7.05	15.32	38.42	30.41	47.85
WSJ Sample	7.41	14.40	40.59	38.00	50.28
75th Percentile					
CRSP Universe	29.53	38.57	113.48	114.55	185.27
WSJ Sample	29.40	36.83	113.48	133.13	193.25

Table 2. Variable Definitions

	Dependent Variables
CAR [a, b]$_{i,t}$	Cumulative abnormal return for stock *i* from day *d* = *a* to *b*, following the earnings announcement in quarter *t*. Abnormal returns are calculated as the difference between compounded daily returns for stock *i* (Ret_{id}) and the size and book-to-market matched portfolio ($MatchRet_{md}$) between days *a* and *b*. Each stock is matched to one of six portfolios based on median market capitalization and 30th and 70th percentile of book-to-market based on cutoffs provided by Kenneth French. $\prod_{d=a}^{d=b}(1 + Ret(i,d)) - \prod_{d=a}^{d=b}(1 + MatchRet(m, d))$
	Independent Variables
SUE	Standardized unexpected earnings (SUE) is based on the definition of Foster, Olsen, and Shevlin (1984). SUE = ($EPS_{i,q} - EPS_{i,q-4}$) / $\sigma_{q-8,q-1}$ where $EPS_{i,q}$ and $EPS_{i,q-4}$ are firm *i*'s earnings per share in quarters *q* and *q-4*, and $\sigma_{q-8,q-1}$ is the standard deviation of $EPS_{i,q} - EPS_{i,q-4}$ over the past eight quarters.
Adj_SUE	Adjusted Rank of SUE, constructed by sorting the cross-section of earnings announcements each quarter into quintiles (0 – 4), and then dividing each quintile rank by 4. The adjusted rank ranges from 0 for the lowest SUE quintile to +1 for the top SUE quintile.
# Ann	The logarithm of the number of same-day announcements.
Ret [-11, -2]	The cumulative return for stock *i* over days *t-11* to *t-2*.
Ret_Vol	The standard deviation of returns for stock *i* over days *t-11* to *t-2*.
Illiquidity	The average daily Amihud (2002) illiquidity measure, computed as the ratio of the daily absolute return to the dollar trading volume for stock *i* in month *t-1*, multiplied by 10^4.
Trend	The quarterly time trend divided by 1000.
$Trend^2$	The quarterly squared time trend divided by 1000.
Turnover	The logarithm of average daily turnover for stock *i* over days *t-11* to *t-2*.
BM	The logarithm of the book-to-market ratio for firm *i* for the most recent period prior to day *t*.
Size	The logarithm of the market capitalization for stock *i* in the month prior to day *t*.
Beta	The market beta of stock *i*, estimated by regressing monthly returns for stock *i* against the CRSP value-weighted market index over months *t-36* to *t-1*.
AVOL [0, +1]	For each stock, abnormal trading volume on days 0 and +1 is the difference between average log dollar volume on days 0 and +1 versus average log dollar volume over days -60 to -11 prior to the announcement (on day 0). Every quarter, we compute the firm's AVOL[0,+1] for the subsets of stocks in the top and bottom quintile portfolios by SUE. We then plot the mean abnormal volume for these subsets of firms with a large absolute earnings surprise, averaged across all quarters during each 10-year period.
Aggregate Returns	The cumulative monthly market returns averaged during *quarter q*.
Aggregate Volatility	The daily market return volatility averaged during *quarter q*.
Aggregate Illiquidity	Amihud (2002) illiquidity averaged during *quarter q*.

Table 2., continued

	Independent Variables Contd.
Aggregate Disagreement	The mean monthly stock return volatility averaged during *quarter q*.
Price Dividend Ratio	The stock market price-dividend ratio measured at the end of month *t-1* averaged during *quarter q.*
Inflation	Change in the natural log of the CPI from month *t-13* to *t-1* averaged during *quarter q.*
Industrial Production	Change in the natural log of industrial production from month *t-13* to *t-1* averaged during *quarter q.*
Money Supply	Change in the natural log of the money supply (M1) from month *t-13* to *t-1* averaged during *quarter q.*
NBER Recession	NBER recession indicator averaged during *quarter q.*
Treasury Bill Rate	Three-month treasury bill rate averaged during *quarter q*.
Deregulation	The financial deregulation index from 1934 to 2007 is based on Philippon and Reshef (2012) and consists of the following four components: i) the proportion of the population living in states that have removed bank branching restrictions, ii) the separation of commercial and investment banking through the Glass-Steagall Act, iii) interest rate ceilings, and iv) the separation of banks from insurance companies. The deregulation index is given by deregulation = (i) – (ii) – (iii) – (iv) We extend this measure beyond 2007 through 2022. We retain the value of 1.0 from 2007 until 2009 and reduce this value to 0.7 through 2022, to reflect tighter regulation imposed by Dodd-Frank Act.
Relative Wage	The ratio of the average wage in the finance sector to the average wage in the nonfarm private sector, as in Philippon and Reshef (2012, p. 1557).
Relative Education	The difference between the shares of educated workers in the finance sector and the nonfarm private sector, as in Philippon and Reshef (2012, p. 1556).
Relative Size	The ratio of the market capitalization of the trading sector scaled by the total number of unique firms in CRSP

Table 3. Summary Statistics for Main Variables over 10-Year Periods since 1934

This Table provides summary statistics for the variables in our main analysis over each 10-year period since 1934. All variables are defined in Table I. The sample period covers January 1934 through December 2022.

	Period	SUE	Size	Beta	BM	Ret[11,02]	Ret_Vol	Turnover	Illiquidity
Mean	1934-43	0.183	71.75	1.405	1.897	0.006	0.027	0.001	0.153
	1944-53	0.254	110.49	1.251	1.255	0.003	0.016	0.001	0.025
	1954-63	0.087	349.14	1.007	0.995	0.005	0.015	0.001	0.008
	1964-73	0.494	612.60	1.289	0.808	0.003	0.019	0.001	0.004
	1974-83	0.427	602.38	1.209	1.164	0.01	0.021	0.002	0.007
	1984-93	0.027	1209.52	1.122	0.789	0.005	0.023	0.003	0.007
	1994-03	0.121	3277.95	1.015	0.633	0.008	0.029	0.006	0.002
	2004-13	0.107	5757.21	1.287	0.654	0.003	0.024	0.009	0.001
	2014-22	0.200	12508.86	1.235	0.626	0.005	0.021	0.009	0.001

	Period	SUE	Size	Beta	BM	Ret[11,02]	Ret_Vol	Turnover	Illiquidity
Median	1934-43	0.178	11.71	1.354	1.118	0.000	0.022	0.000	0.029
	1944-53	0.089	23.73	1.209	0.942	0.000	0.014	0.001	0.010
	1954-63	0.112	77.71	0.991	0.767	0.000	0.013	0.001	0.003
	1964-73	0.340	148.54	1.225	0.666	0.000	0.016	0.001	0.001
	1974-83	0.309	155.43	1.157	1.045	0.002	0.018	0.001	0.001
	1984-93	0.136	227.66	1.091	0.665	0.000	0.019	0.002	0.000
	1994-03	0.153	415.90	0.875	0.497	0.000	0.023	0.003	0.000
	2004-13	0.136	989.52	1.143	0.494	0.004	0.019	0.007	0.000
	2014-22	0.115	1873.35	1.158	0.447	0.005	0.017	0.006	0.000

	Period	SUE	Size	Beta	BM	Ret[11,02]	Ret_Vol	Turnover	Illiquidity
Std	1934-43	2.056	262.69	0.629	3.755	0.087	0.023	0.002	0.550
	1944-53	1.774	399.86	0.531	1.619	0.052	0.009	0.001	0.069
	1954-63	6.501	1388.73	0.464	1.092	0.052	0.009	0.001	0.020
	1964-73	15.124	2216.47	0.556	0.668	0.072	0.011	0.002	0.010
	1974-83	2.594	2064.53	0.556	0.727	0.078	0.012	0.002	0.024
	1984-93	4.338	3888.37	0.577	0.766	0.084	0.017	0.004	0.049
	1994-03	3.892	15139.85	0.864	0.671	0.106	0.021	0.009	0.010
	2004-13	3.590	20821.4	0.867	1.049	0.086	0.018	0.010	0.044
	2014-22	8.102	59265.65	0.761	1.335	0.082	0.016	0.015	0.075

Table 4. Evolution of PEAD and the Initial Response to Earnings News from 1934 to 2022

Column (1) presents the mean cumulative abnormal return over days +2 through +61 (i.e., PEAD or CAR[+2,+61]), while column (2) provides the initial announcement return over days 0 and +1 (i.e., CAR[0,+1]) following the earnings announcement on day 0, for the hedge portfolio that is long the top SUE quintile and short the bottom SUE quintile sorted every quarter. We report the mean CARs averaged across the sample of all earnings announcements during each 10-year period spanning the sample from 1934 to 2022.

Period	(1) CAR [2, 61]	t Value	(2) CAR [0, 1]	t Value
1934-43	2.01%	3.40	2.84%	16.76
1944-53	2.35%	6.68	2.42%	23.33
1954-63	3.27%	11.41	2.30%	27.51
1964-73	3.93%	14.11	2.47%	29.81
1974-83	4.22%	19.11	1.74%	29.73
1984-93	4.20%	21.14	1.75%	28.25
1994-03	2.92%	10.99	2.00%	25.37
2004-13	1.64%	6.77	3.25%	34.47
2014-22	0.69%	2.16	3.88%	29.50

Table 5. Evolution of PEAD and the Initial Response to Earnings News from 1934 to 2022, Controlling for Firm Characteristics

Panels A and B present the results from estimating Equation (1) for CAR[2,61] and CAR[0,1], respectively, over each 10-year period beginning in the year 1934. The dependent variable (CAR[+2,+61] or CAR[0,+1]) is measured by the cumulative abnormal return for stock *i* compounded over days a to b, following the earnings announcement (on day 0) for firm *i* during quarter *t*. The coefficient of Adj_SUE (β1) is analogous to the return on a hedge portfolio that is long the top SUE quintile and short the bottom SUE quintile each quarter, after controlling for firm attributes. All models include fixed effects for day of the week. Standard errors are adjusted for heteroscedasticity and clustered by the day of the earnings announcement. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Post-Earnings Announcement Drift: CAR [2, 61]

Variables		CAR [2, 61]								
		1934-43	1944-53	1954-63	1964-73	1974-83	1984-93	1994-03	2004-13	2014-22
Adj_SUE	**(β_1)**	**0.019*** **	**0.025*** **	**0.034*** **	**0.044*** **	**0.045*** **	**0.044*** **	**0.035*** **	**0.021*** **	**0.016*** **
		(3.33)	**(7.47)**	**(12.76)**	**(16.21)**	**(20.52)**	**(22.83)**	**(14.50)**	**(10.44)**	**(5.47)**
Lagged SUE	(β_2)	-0.000	-0.001***	-0.000	-0.001***	-0.001***	-0.000	-0.001*	-0.000*	-0.000
		(-0.40)	(-2.82)	(-0.81)	(-2.65)	(-3.64)	(-1.60)	(-1.91)	(-1.89)	(-0.05)
#Ann	(β_3)	-0.001	0.002	0.001	-0.001	-0.000	0.000	0.002	0.002**	-0.000
		(-0.29)	(1.31)	(1.26)	(-0.75)	(-0.22)	(0.07)	(1.32)	(2.15)	(-0.13)
Ret [-11, -2]	(β_4)	0.001	-0.022	0.050**	0.014	-0.042***	-0.007	-0.002	-0.051**	-0.023
		(0.03)	(-0.88)	(2.20)	(0.78)	(-3.47)	(-0.59)	(-0.14)	(-2.22)	(-0.79)
BM	(β_5)	0.004	0.002	0.004*	0.002	0.003**	0.002	0.003*	0.008***	0.005***
		(1.62)	(0.99)	(1.89)	(0.98)	(2.32)	(1.39)	(1.82)	(4.24)	(3.56)
Beta	(β_6)	0.000	0.012***	-0.002	-0.008***	0.000	0.001	-0.000	-0.001	0.001
		(0.06)	(3.25)	(-0.70)	(-2.75)	(0.00)	(0.76)	(-0.23)	(-0.65)	(0.26)
Size	(β_7)	0.001	-0.000	-0.001*	-0.002***	-0.001	-0.001***	-0.003***	0.001	0.000
		(0.65)	(-0.52)	(-1.87)	(-2.76)	(-1.40)	(-2.97)	(-4.16)	(1.14)	(0.09)
Ret_Vol	(β_8)	0.143	-0.206	-0.337**	0.147	-0.033	-0.123*	-0.141*	0.606***	0.386***
		(1.25)	(-1.11)	(-2.13)	(1.19)	(-0.38)	(-1.80)	(-1.74)	(4.03)	(2.67)
Turnover	(β_9)	0.000	-0.001	0.001	-0.002	0.002***	0.002***	0.006***	0.000	0.004**
		(0.94)	(-1.23)	(0.83)	(-1.20)	(2.68)	(3.97)	(6.51)	(0.25)	(1.98)
Illiquidity	(β_{10})	-0.006	0.040*	0.006	-0.219*	-0.006	-0.014	-0.108	-0.013	-0.016***
		(-1.26)	(1.86)	(0.09)	(-1.90)	(-0.16)	(-0.81)	(-1.34)	(-0.96)	(-3.25)
Constant	(β_0)	-0.013	-0.023*	0.008	0.003	0.014	0.011	0.056***	-0.028**	0.006
		(-0.78)	(-1.89)	(0.75)	(0.16)	(1.19)	(1.20)	(4.35)	(-2.18)	(0.36)
Observations		7,169	8,869	13,546	25,749	55,435	81,388	109,845	93,426	67,428
Adjusted R-squared		0.002	0.011	0.014	0.013	0.009	0.008	0.003	0.006	0.002

Table 5, continued

Panel B. Initial Announcement Return: CAR [0, 1]

Variables		CAR [0, 1]								
		1934-43	1944-53	1954-63	1964-73	1974-83	1984-93	1994-03	2004-13	2014-22
Adj_SUE	**(β_1)**	**0.027*** **	**0.025*** **	**0.023*** **	**0.029*** **	**0.021*** **	**0.022*** **	**0.025*** **	**0.040*** **	**0.044*** **
		(17.43)	**(25.35)**	**(28.65)**	**(35.04)**	**(34.47)**	**(35.11)**	**(32.76)**	**(44.94)**	**(40.39)**
Lagged SUE	(β_2)	0.000***	-0.000	-0.000	-0.000***	-0.000***	-0.001***	-0.001***	-0.002***	-0.002***
		(2.67)	(-0.23)	(-0.13)	(-5.43)	(-6.53)	(-10.32)	(-11.04)	(-20.61)	(-15.35)
#Ann	(β_3)	0.001	-0.000	-0.000	-0.000	-0.001***	-0.000	-0.001*	-0.001	-0.000
		(1.00)	(-1.58)	(-1.42)	(-1.31)	(-3.18)	(-0.94)	(-1.66)	(-1.60)	(-0.18)
Ret [-11, -2]	(β_4)	-0.053***	-0.031***	-0.032***	-0.036***	-0.040***	-0.030***	-0.039***	-0.012**	-0.011*
		(-4.66)	(-4.22)	(-4.71)	(-7.47)	(-7.11)	(-6.11)	(-10.20)	(-2.11)	(-1.73)
BM	(β_5)	0.000	0.001	0.001*	0.004***	0.002***	0.002***	0.004***	0.002***	0.002***
		(0.57)	(1.28)	(1.90)	(8.53)	(7.25)	(7.66)	(9.16)	(4.94)	(4.04)
Beta	(β_6)	0.002	0.001	-0.001	-0.000	-0.000	-0.000	-0.002***	-0.000	-0.000
		(1.32)	(1.29)	(-1.12)	(-0.60)	(-0.16)	(-0.72)	(-3.76)	(-1.01)	(-0.19)
Size	(β_7)	-0.001*	0.000	0.001***	0.000	-0.000	0.000	0.001***	-0.000	-0.001***
		(-1.65)	(0.57)	(3.19)	(1.09)	(-0.24)	(0.43)	(4.82)	(-1.34)	(-3.01)
Ret_Vol	(β_8)	-0.013	0.074	-0.051	-0.098***	0.034	0.009	-0.008	-0.070**	0.038
		(-0.38)	(1.56)	(-1.02)	(-2.66)	(1.18)	(0.41)	(-0.34)	(-2.04)	(1.08)
Turnover	(β_9)	-0.000***	-0.000	-0.000	0.000	-0.000**	0.000	0.001***	0.000	-0.000
		(-2.90)	(-1.31)	(-0.69)	(1.01)	(-2.08)	(0.64)	(3.31)	(0.08)	(-0.50)
Illiquidity	(β_{10})	0.003	0.000	0.051**	0.112***	0.016	0.027**	0.089***	-0.002	-0.004
		(1.44)	(0.01)	(2.09)	(3.37)	(1.10)	(2.29)	(3.10)	(-0.19)	(-1.44)
Constant	(β_0)	-0.016***	-0.015***	-0.021***	-0.008**	-0.009***	-0.006**	-0.008**	-0.003	-0.003
		(-3.23)	(-5.03)	(-6.90)	(-2.15)	(-3.05)	(-2.17)	(-2.15)	(-0.66)	(-0.48)
Observations		7,169	8,869	13,546	25,749	55,435	81,388	109,845	93,426	67,428
Adjusted R-squared		0.066	0.094	0.079	0.065	0.034	0.023	0.017	0.027	0.031

Table 6. Matching Accounting Data to Stock Returns, and the Earnings/Price Anomaly

This table reports the average monthly returns to portfolios formed based on the earnings to price (E/P) ratio, using the date that net income is disclosed for each earnings announcement in our sample (i.e., day 0 from the WSJ sample of earnings announcements prior to 1971, and the Compustat sample of earnings announcements thereafter). First, each quarter we sort all announcing firms into quintiles based on the E/P ratio, calculated by dividing the quarterly net income disclosed relative to the market capitalization of the disclosing firm at the beginning of the quarter. We then compute the average return to each quintile portfolio during the first six months period starting the next quarter. Finally, we average these portfolio returns for each quintile portfolio across all quarters during every 10-year period since 1934. The t-statistics for the mean portfolio returns (in parenthesis) are based on Newey-West robust standard errors. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

E / P Quintile	**Portfolio Returns**								
	1934-43	1944-53	1954-63	1964-73	1974-83	1984-93	1994-03	2004-13	2014-22
Low	3.04	1.31	0.89	0.65	1.53	-0.17	1.64	0.86	0.87
	(2.00)	(1.55)	(1.40)	(0.77)	(1.37)	(-0.18)	(1.24)	(0.71)	(0.61)
2	1.27	1.23	0.90	0.86	1.70	0.20	0.73	0.52	1.67
	(1.04)	(2.20)	(1.81)	(1.02)	(1.70)	(0.25)	(0.88)	(0.61)	(1.93)
3	2.35	1.74	1.36	1.14	2.10	0.62	1.08	0.76	1.28
	(2.31)	(2.74)	(2.98)	(1.43)	(2.03)	(0.82)	(1.79)	(1.00)	(1.90)
4	2.56	2.16	1.74	1.88	2.49	1.00	1.54	0.85	1.34
	(2.72)	(3.22)	(3.71)	(2.14)	(2.55)	(1.29)	(2.90)	(1.01)	(1.89)
High	4.68	3.13	2.59	2.78	3.59	1.24	2.48	1.48	2.25
	(3.58)	(3.86)	(4.90)	(2.68)	(3.19)	(1.38)	(3.79)	(1.68)	(2.56)
Month t+1 (High – Low)	**1.64***	**1.82*****	**1.70*****	**2.13*****	**2.06*****	**1.41*****	**0.84**	**0.62**	**1.38**
	(1.77)	**(5.16)**	**(6.53)**	**(4.42)**	**(4.84)**	**(4.06)**	**(1.04)**	**(0.85)**	**(1.59)**
Month t+2 (High – Low)	1.38*	0.67*	1.03***	-0.26	0.56	1.04***	-0.28	0.27	-0.31
	(1.99)	(1.75)	(5.37)	(-0.91)	(1.26)	(3.93)	(-0.24)	(0.68)	(-0.39)
Month t+3 (High – Low)	1.29	1.32***	0.91***	0.79***	1.18***	1.39***	1.74*	0.08	-0.96
	(1.22)	(2.91)	(3.44)	(3.08)	(4.21)	(3.60)	(1.83)	(0.21)	(-0.91)
Month t+4 (High – Low)	0.22	0.68**	0.96***	1.01**	1.03**	1.25***	0.06	0.82*	2.08**
	(0.29)	(2.53)	(2.92)	(2.49)	(2.63)	(3.60)	(0.07)	(1.90)	(2.70)
Month t+5 (High – Low)	0.04	0.45*	0.60*	-0.08	0.57	0.50**	-0.30	0.39	-0.44
	(0.05)	(1.71)	(1.83)	(-0.21)	(1.27)	(2.24)	(-0.25)	(1.18)	(-0.69)
Month t+6 (High – Low)	0.66	0.85**	0.23	0.80**	0.77**	1.20***	1.63*	-0.26	0.10
	(0.78)	(2.32)	(1.21)	(2.61)	(2.30)	(3.10)	(1.73)	(-0.83)	(0.20)

Table 7. The Divergent Trend in CAR[+2,+61] and CAR[0,+1] before versus after 1983

Panels A and B present the results from estimating Equation (2) for CAR[+2,+61] and CAR[0,+1], respectively, over two subperiods, 1934 – 1983 and 1984 – 2022. The dependent variable (CAR[+2,+61] or CAR[0,+1]) is measured by the cumulative abnormal return for stock i compounded over days a to b, following the earnings announcement (on day 0) for firm i during quarter t. The coefficient of Adj_SUE (β_1) is analogous to the return on a hedge portfolio that is long the top SUE quintile and short the bottom SUE quintile each quarter. The variable, Trend, indexes the period of the observation across quarters. For ease of exposition of the coefficient magnitudes, we divide 'Trend' by 1000. In columns (1) and (4), we estimate Equation (2) with only the trend terms without the controls, while the models in columns (2) and (5) add the controls. Columns (3) and (6) also add the interactions of all the controls with 'Trend' (the coefficients of these additional interaction terms are omitted here for brevity). Tabhe 2 defines all variables. All models include fixed effects for day of the week. Standard errors are adjusted for heteroscedasticity and clustered by the day of the earnings announcement. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Divergent Trend in CAR[2, 61]: 1934 – 1983 versus 1984 – 2022

Variables		1934 – 1983			1984 – 2022		
		(1)	(2)	(3)	(4)	(5)	(6)
Adj_SUE	(β_1)	0.018***	0.020***	0.018***	0.046***	0.051***	0.049***
		(4.92)	(5.42)	(4.76)	(20.96)	(22.21)	(21.52)
Trend	(β_2)	-0.029*	-0.037*	0.122	0.165***	0.170***	-0.278*
		(-1.73)	(-1.96)	(1.17)	(8.34)	(7.96)	(-1.78)
Adj_SUE x Trend	**(β_3)**	**0.127*** **	**0.146*** **	**0.156*** **	**-0.262*** **	**-0.272*** **	**-0.245*** **
		(5.00)	**(5.75)**	**(5.74)**	**(-9.53)**	**(-9.76)**	**(-9.03)**
Constant		-0.013***	0.002	-0.015	-0.025***	0.006	0.041***
		(-5.08)	(0.28)	(-1.19)	(-12.54)	(0.85)	(3.51)
Controls		N	Y	Y	N	Y	Y
Trend x Controls		N	N	Y	N	N	Y
Observations		110,768	110,768	110,768	352,087	352,087	352,087
Adjusted R^2		0.009	0.009	0.010	0.003	0.004	0.004

Panel B. Divergent Trend in CAR[0, 1]: 1934 – 1983 versus 1984 – 2022

Variables		1934 – 1983			1984 – 2022		
		(1)	(2)	(3)	(4)	(5)	(6)
Adj_SUE	(β_1)	0.029***	0.031***	0.029***	0.012***	0.019***	0.016***
		(29.66)	(31.20)	(26.97)	(18.34)	(27.48)	(21.76)
Trend	(β_2)	0.025***	0.033***	0.061**	-0.088***	-0.072***	0.007
		(5.92)	(6.68)	(2.23)	(-14.85)	(-11.40)	(0.16)
Adj_SUE x Trend	**(β_3)**	**-0.060*** **	**-0.052*** **	**-0.034*** **	**0.183*** **	**0.165*** **	**0.205*** **
		(-8.82)	**(-7.63)**	**(-4.48)**	**(20.56)**	**(18.42)**	**(21.08)**
Constant		-0.015***	-0.017***	-0.021***	-0.007***	0.000	-0.005
		(-22.63)	(-11.49)	(-5.92)	(-13.24)	(0.03)	(-1.54)
Controls		N	Y	Y	N	Y	Y
Trend x Controls		N	N	Y	N	N	Y
Observations		110,768	110,768	110,768	352,087	352,087	352,087
Adjusted R^2		0.039	0.048	0.049	0.018	0.023	0.024

Table 8. Nonlinear Trend in CAR[+2,+61] and CAR[0,+1] over the Entire Sample

This Table presents the results from estimating Equation (3). The dependent variable is either PEAD (CAR[+2,+61]) or the initial response to earnings news (CAR[0,+1]), measured by the cumulative abnormal return for stock *i* compounded over days a to b, following the earnings announcement of firm *i* during quarter *t*. The coefficient of Adj_SUE (β_1) is analogous to the return on a hedge portfolio that is long the top SUE quintile and short the bottom SUE quintile each quarter. The variable, Trend, indexes the period of the observation each quarter. For ease of exposition of the coefficients, we divide 'Trend' and 'Trend^2' by 1000. Across the models (columns) estimated, we include different subsets of control variables. In columns (1) and (4), we estimate Equation (2) with only the trend terms without the controls, while columns (2) and (5) add the control variables. Columns (3) and (6) also add the interactions of all the control variables with Trend and Trend^2 (the coefficients of these interaction terms are omitted here for brevity). Table 2 defines all variables. All models include fixed effects for day of the week. Standard errors are adjusted for heteroscedasticity and clustered by the day of the announcement. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Variables		CAR [2, 61]			CAR [0, 1]		
		(1)	(2)	(3)	(4)	(5)	(6)
Adj_SUE	(β_1)	0.009*	0.009*	0.009**	0.037***	0.039***	0.035***
		(1.89)	(1.91)	(2.01)	(27.03)	(28.74)	(24.57)
Trend	(β_2)	-0.200***	-0.249***	0.615***	0.103***	0.098***	0.065
		(-5.79)	(-6.80)	(3.24)	(10.46)	(9.31)	(1.12)
Adj_SUE x Trend	(β_3)	0.372***	0.415***	0.398***	-0.233***	-0.195***	-0.200***
		(7.36)	(8.18)	(7.76)	(-15.15)	(-12.62)	(-12.09)
Trend^2	(β_4)	0.0006***	0.0007***	-0.0014***	-0.0003***	-0.0003***	-0.0001
		(6.88)	(7.75)	(-2.64)	(-12.45)	(-10.30)	(-0.37)
Adj_SUE x Trend^2 (β_5)		**-0.0011*** **	**-0.0012*** **	**-0.0011*** **	**0.0007*** **	**0.0006*** **	**0.0007*** **
		(-8.69)	**(-9.39)**	**(-9.05)**	**(17.68)**	**(15.08)**	**(15.76)**
Lagged SUE			-0.000***	-0.000		-0.001***	0.000
			(-3.81)	(-0.54)		(-29.10)	(1.47)
#Ann			0.001	0.001		-0.000	0.000
			(1.20)	(0.45)		(-1.46)	(0.07)
Ret [-11, -2]			-0.019**	-0.007		-0.029***	-0.035***
			(-2.33)	(-0.23)		(-13.05)	(-3.25)
BM			0.005***	0.006**		0.002***	-0.002**
			(6.75)	(2.13)		(13.41)	(-2.17)
Beta			-0.000	0.009**		-0.001***	0.004***
			(-0.04)	(2.04)		(-3.04)	(3.52)
Size			-0.001***	0.002		0.000	-0.001***
			(-2.98)	(1.49)		(0.08)	(-3.62)
Ret_Vol			0.153***	0.270*		-0.006	0.013
			(2.99)	(1.86)		(-0.42)	(0.29)
Turnover			0.002***	-0.000		-0.000	-0.001***
			(6.61)	(-0.70)		(-1.23)	(-3.96)
Illiquidity			-0.010***	-0.007		0.004**	-0.001
			(-2.67)	(-1.15)		(2.54)	(-0.29)
Constant		-0.004	0.020***	-0.045***	-0.018***	-0.014***	-0.018***
		(-1.15)	(3.71)	(-2.94)	(-20.31)	(-9.03)	(-3.91)
Trend x Controls		N	N	Y	N	N	Y
Trend^2 x Controls		N	N	Y	N	N	Y
Observations		462,855	462,855	462,855	462,855	462,855	462,855
Adjusted R^2		0.003	0.004	0.005	0.020	0.025	0.026

Table 9. Nonlinear Trend in CAR[+2,+61] and CAR[0,+1] for Subsamples of Stocks Based on Firm Attributes

Panels A and B present results from estimating the pooled regression model in Equation (2) applied to CAR[+2,+61] and CAR[0,+1], respectively, for subsets of stocks based on firm size (small versus large based on NYSE terciles), book-to-market (i.e., value versus growth), ivol, illiquidity, and turnover. These subsets of stocks are selected each quarter as the tercile portfolios with low versus high values of each firm attribute. The dependent variable is the cumulative abnormal return compounded over days 0 and +1, or days +2 to +61, following the earnings announcement of firm *i* in quarter *t*. The coefficient of Adj_SUE (β_1) is analogous to the return on a hedge portfolio that is long the highest SUE quintile and short the lowest SUE quintile each quarter. The variable, Trend, indexes the quarterly observations. For ease of exposition, we divide "Trend" by 1000 and "Trend^2" by 1000. For each model (column), we include all control variables from the model in Table V, as well as fixed effects for day of the week. Table I provides variable definitions. Standard errors are adjusted for heteroscedasticity and clustered by the day of the announcement. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Nonlinear Trend in PEAD, CAR[2,+61], for Subsamples Based on Firm Attributes

Variables		Small	Large	Value	Growth	High RetVol	Low RetVol	Illiquid	Liquid	High Turnover	Low Turnover
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Adj_SUE	(β_1)	-0.012	0.027***	0.006	0.021***	-0.004	0.018***	-0.009	0.027***	0.001	0.011
		(-1.13)	(5.00)	(0.63)	(3.07)	(-0.34)	(3.84)	(-0.98)	(4.53)	(0.14)	(1.28)
Trend	(β_2)	-0.288***	-0.001	-0.162**	-0.147**	-0.495***	-0.118***	-0.268***	-0.049	-0.201***	-0.322***
		(-4.10)	(-0.03)	(-2.57)	(-2.53)	(-5.17)	(-3.28)	(-3.67)	(-0.99)	(-3.55)	(-5.35)
Adj_SUExTrend	(β_3)	0.715***	0.052	0.371***	0.343***	0.673***	0.251***	0.584***	0.142**	0.590***	0.333***
		(6.47)	(0.89)	(3.74)	(4.60)	(5.07)	(5.09)	(5.16)	(2.28)	(7.23)	(3.62)
Trend^2	(β_4)	0.001***	0.000**	0.001***	0.001***	0.001***	0.000***	0.001***	0.000***	0.001***	0.001***
		(5.75)	(2.40)	(3.31)	(3.83)	(5.74)	(4.48)	(2.94)	(3.57)	(5.55)	(5.23)
Adj_SUExTrend^2	**(β_5)**	**-0.0018*** **	**-0.0002** **	**-0.0009*** **	**-0.0011*** **	**-0.0018*** **	**-0.0008*** **	**-0.0012*** **	**-0.0006*** **	**-0.0016*** **	**-0.0010*** **
		(-6.87)	**(-2.43)**	**(-3.83)**	**(-6.04)**	**(-5.20)**	**(-7.16)**	**(-4.17)**	**(-4.12)**	**(-8.13)**	**(-4.35)**
Constant		0.075***	0.017**	-0.012	0.031***	0.057***	0.010*	0.032**	-0.008	0.041***	-0.012
		(5.25)	(2.04)	(-1.35)	(3.68)	(3.92)	(1.79)	(2.39)	(-0.88)	(4.87)	(-1.41)
Controls		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Adjusted R^2		0.006	0.004	0.008	0.004	0.005	0.006	0.008	0.003	0.006	0.004

Table 9., continued

Panel B. Nonlinear Trend in the Initial Response to Earnings News, CAR[0,+1], for Subsamples Based on Firm Attributes

Variables		Small	Large	Value	Growth	High RetVol	Low RetVol	Illiquid	Liquid	High Turnover	Low Turnover
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Adj_SUE	(β_1)	0.053***	0.029***	0.047***	0.030***	0.039***	0.035***	0.041***	0.032***	0.059***	0.019***
		(16.27)	(17.72)	(17.20)	(14.46)	(12.27)	(21.61)	(13.26)	(17.89)	(19.49)	(11.95)
Trend	(β_2)	0.142***	0.090***	0.147***	0.038**	0.041	0.118***	0.093***	0.114***	0.157***	0.032***
		(6.60)	(6.99)	(7.80)	(2.12)	(1.62)	(9.56)	(4.15)	(7.91)	(6.12)	(2.92)
Adj_SUExTrend	(β_3)	-0.270***	-0.154***	-0.253***	-0.106***	-0.136***	-0.200***	-0.136***	-0.175***	-0.223***	-0.130***
		(-8.16)	(-8.15)	(-8.59)	(-4.25)	(-3.65)	(-11.51)	(-3.74)	(-8.67)	(-6.25)	(-7.88)
$Trend^2$	(β_4)	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***
		(-7.34)	(-6.65)	(-8.04)	(-3.50)	(-2.59)	(-10.16)	(-3.69)	(-8.05)	(-6.11)	(-4.56)
$Adj_SUExTrend^2$	**(β_5)**	**0.0008*** **	**0.0004*** **	**0.0007*** **	**0.0004*** **	**0.0005*** **	**0.0006*** **	**0.0005*** **	**0.0005*** **	**0.0008*** **	**0.0004*** **
		(10.49)	**(7.48)**	**(9.86)**	**(5.90)**	**(5.32)**	**(12.85)**	**(4.79)**	**(9.22)**	**(8.80)**	**(9.24)**
Constant		-0.016***	-0.010***	-0.019***	-0.009***	-0.023***	-0.007***	-0.008**	-0.015***	0.003	-0.020***
		(-4.26)	(-3.60)	(-6.71)	(-3.59)	(-5.78)	(-4.04)	(-2.01)	(-5.50)	(0.81)	(-11.37)
Controls		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Adjusted R^2		0.035	0.013	0.029	0.021	0.024	0.027	0.041	0.013	0.044	0.023

Table 10. Nonlinear Trend in CAR[+2,+61] and CAR[0,+1] for Different Industries

Panels A and B present results from estimating the pooled regression model in Equation (2) applied to CAR[+2,+61] and CAR[0,+1], respectively, for subsets of stocks included in each of the twelve industries based on the definitions provided in Kenneth French's website. The dependent variable is the cumulative abnormal return compounded over days 0 and +1, or days +2 to + 61, following the earnings announcement of firm *i* in quarter *t*. The coefficient of Adj_SUE (β_1) is analogous to the return on a hedge portfolio that is long the highest SUE quintile and short the lowest SUE quintile each quarter. The variable, Trend, indexes the quarterly observations. For ease of exposition, we divide "Trend" by 1000 and "Trend^2" by 1000. For each model, we include all controls from the model in Table V, as well as fixed effects for day of the week. Table I provides variable definitions. Standard errors are adjusted for heteroscedasticity and clustered by day of the announcement. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Nonlinear Trend in PEAD, CAR[+2,+61], for Fama-French 12 (FF12) Industries

Variables		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Adj_SUE	(β_1)	0.022**	0.011	0.005	0.009	0.012	0.022	-0.046	-0.002	0.007	0.046*	-0.039	0.014
		(2.23)	(0.62)	(0.67)	(0.53)	(0.74)	(1.31)	(-1.20)	(-0.14)	(0.28)	(1.93)	(-1.43)	(1.00)
Trend	(β_2)	-0.128*	-0.218*	-0.356***	0.045	-0.208*	-0.221*	-0.217	-0.351***	-0.542***	0.109	-0.085	-0.279***
		(-1.72)	(-1.77)	(-5.99)	(0.30)	(-1.74)	(-1.81)	(-0.91)	(-3.26)	(-4.39)	(0.50)	(-0.53)	(-2.77)
Adj_SUExTrend	(β_3)	0.266**	0.504***	0.470***	0.269	0.264	0.408**	0.896**	0.387***	0.595***	0.122	0.736***	0.399***
		(2.41)	(2.66)	(5.69)	(1.21)	(1.46)	(2.44)	(2.43)	(2.64)	(2.68)	(0.48)	(3.35)	(2.76)
Trend^2	(β_4)	0.000*	0.001**	0.001***	-0.000	0.001*	0.001***	0.000	0.001***	0.002***	0.000	0.000	0.001***
		(1.71)	(2.45)	(7.02)	(-0.24)	(1.93)	(2.96)	(0.92)	(3.47)	(5.44)	(0.12)	(0.53)	(3.41)
Adj_SUExTrend^2	**(β_5)**	**-0.0008*** **	**-0.0017*** **	**-0.0014*** **	**-0.0006**	**-0.0007**	**-0.0014*** **	**-0.0022*** **	**-0.0011*** **	**-0.0017*** **	**-0.0007**	**-0.0017*** **	**-0.0012*** **
		(-2.79)	**(-3.56)**	**(-6.59)**	**(-0.94)**	**(-1.55)**	**(-3.56)**	**(-2.64)**	**(-3.15)**	**(-3.32)**	**(-1.15)**	**(-3.88)**	**(-3.39)**
Controls		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Constant		-0.026**	0.039**	0.014	0.044**	0.054***	0.028	0.052	0.016	0.021	-0.022	-0.019	0.026*
		(-2.30)	(2.01)	(1.49)	(2.27)	(2.68)	(1.58)	(1.64)	(0.79)	(1.15)	(-0.72)	(-0.90)	(1.92)
Adjusted R^2		0.009	0.009	0.007	0.006	0.007	0.004	0.004	0.009	0.007	0.004	0.007	0.005

Table 10., continued

Panel B. Nonlinear Trend in the Initial Response to Earnings News, CAR[0,+1], for Fama-French 12 (FF12) Industries

Variables		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Adj_SUE	(β_1)	0.040***	0.052***	0.040***	0.038***	0.037***	0.043***	0.046***	0.020***	0.051***	0.035***	0.018*	0.023***
		(11.37)	(10.19)	(17.23)	(9.01)	(8.45)	(7.97)	(4.01)	(4.08)	(8.80)	(4.59)	(1.88)	(4.70)
Trend	(β_2)	0.117***	0.181***	0.082***	0.105***	0.110***	0.079**	0.329***	0.048*	0.143***	0.073	0.046	0.052
		(4.04)	(4.47)	(4.19)	(3.18)	(2.78)	(2.00)	(3.88)	(1.67)	(3.63)	(1.37)	(0.80)	(1.54)
Adj_SUExTrend	(β_3)	-0.219***	-0.329***	-0.171***	-0.230***	-0.222***	-0.171***	-0.474***	-0.124***	-0.229***	-0.166**	-0.071	-0.041
		(-5.22)	(-5.29)	(-5.97)	(-4.49)	(-3.87)	(-3.00)	(-3.84)	(-2.67)	(-3.91)	(-2.09)	(-0.90)	(-0.83)
$Trend^2$	(β_4)	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***	-0.001***	-0.000**	-0.000***	-0.000**	-0.000	-0.000**
		(-4.58)	(-4.22)	(-3.39)	(-3.47)	(-2.61)	(-2.87)	(-4.84)	(-2.45)	(-4.01)	(-2.19)	(-1.29)	(-2.14)
$Adj_SUExTrend^2$	**(β_5)**	**0.0007*** **	**0.0010*** **	**0.0005*** **	**0.0006*** **	**0.0007*** **	**0.0006*** **	**0.0014*** **	**0.0003*** **	**0.0007*** **	**0.0006*** **	**0.0003** **	**0.0003** **
		(6.04)	**(5.73)**	**(6.38)**	**(4.30)**	**(4.17)**	**(4.12)**	**(4.79)**	**(3.26)**	**(4.84)**	**(3.15)**	**(1.97)**	**(2.50)**
Controls		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Constant		-0.020***	-0.023***	-0.016***	-0.023***	-0.012*	-0.021***	-0.014	-0.011	-0.021***	-0.011	-0.002	-0.007*
		(-4.79)	(-3.35)	(-4.86)	(-4.59)	(-1.89)	(-3.45)	(-1.31)	(-1.55)	(-3.75)	(-1.44)	(-0.21)	(-1.65)
Adjusted R^2		0.035	0.038	0.033	0.019	0.028	0.023	0.026	0.027	0.032	0.025	0.023	0.023

Table 11. Nonlinear Trend in CAR[+2,+61] and CAR[0,+1]: Controlling for Earnings Surprise Gap

This Table presents results from estimating an expanded version of Equation (3) that controls for the earning surprise gap, Surprise_Gap, defined as the difference between the 95th and 5th percentiles of SUE each quarter over all announcements, and its interaction with the Adj_SUE. Lag_Surprise_Gap is previous quarter's Surprise_Gap. The dependent variable is either CAR[+2,+61] or CAR[0,+1]. For brevity, we only provide results for the relevant variables and interaction terms. We also include the other controls in Equation (1), as well as fixed effects for day-of-the-week. The sample period covers January 1934 through December 2022. Standard errors are adjusted for heteroscedasticity and clustered by the day of the earnings announcement. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Variables		CAR [2, 61]		CAR [0, 1]	
		(1)	(2)	(3)	(4)
Adj_SUE	(β_1)	0.024***	0.006	0.037***	0.036***
		(4.01)	(1.25)	(22.70)	(21.38)
Trend	(β_2)	-0.358***	-0.235***	0.096***	0.098***
		(-7.99)	(-6.40)	(8.22)	(9.25)
Adj_SUE x Trend	(β_3)	0.522***	0.404***	-0.207***	-0.212***
		(8.90)	(7.82)	(-11.92)	(-13.20)
$Trend^2$	(β_4)	0.001***	0.001***	-0.000***	-0.000***
		(8.53)	(7.44)	(-9.46)	(-10.30)
Adj_SUE x $Trend^2$	**(β_5)**	**-0.001*** **	**-0.001*** **	**0.001*** **	**0.001*** **
		(-9.87)	**(-9.10)**	**(14.41)**	**(15.52)**
Surprise_Gap	(β_6)	0.005***		0.000	
		(4.55)		(0.31)	
Adj_SUE x Surprise_Gap	**(β_7)**	**-0.005*** **		**0.001**	
		(-4.01)		**(1.64)**	
Lag_Surprise_Gap	(β_8)		-0.001**		0.000
			(-2.32)		(0.48)
Adj_SUE x Lag_Surprise_Gap	**(β_9)**		**0.001**		**0.001*** **
			(1.19)		**(3.64)**
Constant		0.006	0.023***	-0.014***	-0.014***
		(0.96)	(4.06)	(-8.49)	(-8.54)
Controls		Y	Y	Y	Y
Observations		462,855	462,855	462,855	462,855
Adjusted R^2		0.005	0.004	0.025	0.025

Table 12. Nonlinear Trend in CAR[+2,+61] and CAR[0,+1]: Controlling for Earnings Surprise Volatility

This Table presents results from estimating an expanded version of Equation (3) that controls for the earning surprise volatility, Surprise_Vol, defined as the standard deviation of SUE each quarter over all announcements, and its interaction with the Adj_SUE. Lag_Surprise_Vol is previous quarter's Surprise_Vol. The dependent variable is either CAR[+2,+61] or CAR[0,+1]. For brevity, we only provide results for the relevant variables and interaction terms. We also include the other controls in Equation (1), as well as fixed effects for day-of-the-week. The sample period covers January 1934 through December 2022. Standard errors are adjusted for heteroscedasticity and clustered by the day of the earnings announcement. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Variables		CAR [2, 61]		CAR [0, 1]	
		(1)	(2)	(3)	(4)
Adj_SUE	(β_1)	0.009*	0.006	0.039***	0.039***
		(1.96)	(1.33)	(28.80)	(28.70)
Trend	(β_2)	-0.250***	-0.245***	0.098***	0.098***
		(-6.81)	(-6.84)	(9.27)	(9.32)
Adj_SUE x Trend	**(β_3)**	**0.415*** **	**0.408*** **	**-0.195*** **	**-0.195*** **
		(8.18)	**(8.15)**	**(-12.62)**	**(-12.59)**
$Trend^2$	(β_4)	0.001***	0.001***	-0.0003***	-0.0003***
		(7.72)	(8.19)	(-10.33)	(-10.38)
Adj_SUE x $Trend^2$	**(β_5)**	**-0.001*** **	**-0.001*** **	**0.001*** **	**0.001*** **
		(-9.37)	**(-9.71)**	**(15.11)**	**(15.10)**
Surprise_Vol	(β_6)	0.0003**		0.0001**	
		(2.13)		(2.26)	
Adj_SUE x Surprise_Vol	**(β_7)**	**-0.0001**		**-0.0001** **	
		(-0.62)		**(-2.12)**	
Lag_Surprise_Vol	(β_8)		-0.002***		0.000
			(-6.99)		(1.17)
Adj_SUE x Lag_Surprise_Vol	**(β_9)**		**0.002*** **		**0.000**
			(4.99)		**(0.10)**
Constant		0.019***	0.022***	-0.014***	-0.014***
		(3.57)	(4.12)	(-9.18)	(-9.09)
Controls		Y	Y	Y	Y
Observations		462,855	462,855	462,855	462,855
Adjusted R^2		0.004	0.004	0.025	0.025

Table 13. Relation between the Response to Earnings News (CAR[2,61] or CAR[0,1]) and Arbitrage Efficacy or Skill

This Table presents results from estimating an expanded version of Equation (1) that controls for the arbitrage efficacy and skill of the finance industry, separately. Our measure of arbitrage efficacy, Size_Per_Firm, is defined as the ratio of the market value of the trading sector and the number of publicly traded firms in CRSP from 1934 to 2022. Every quarter, we take the average market value of the firms in the trading sector and divide it by the number of public firms that exist in CRSP during the same quarter. Our measures of arbitrage skill are relative wages and relative education in the financial sector. We rely on the measures of relative wages and education in the finance industry from Philippon and Reshef (2012), extended over our entire sample period, 1934 through 2022. The measure of relative wage is the ratio of the average wage in the finance industry to the average wage in the nonfarm private sector; the relative education level is the difference between the share of educated workers in the finance industry versus the nonfarm private sector. We also control for the stock market and macroeconomic conditions, including Aggregate Returns, Volatility, Illiquidity, and Disagreement, Price Dividend Ratio, Inflation, Industrial Production, Money Supply, NBER Recessions, and Treasury Bill Rate, as well as the ten respective interaction terms between each variable and the adjusted rank of SUE (e.g., Adj_SUE x Aggregate Volatility, etc.). The dependent variable is either CAR[+2,+61] or CAR[0,+1]. For brevity, we only provide results for the relevant variables and interaction terms. We also include the other controls in Equation (1), as well as fixed effects for day-of-the-week. The sample period covers January 1934 through December 2022. Standard errors are adjusted for heteroscedasticity and clustered by the day of the announcement. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Variables		CAR [2, 61]			CAR [0, 1]		
		(1)	(2)	(3)	(4)	(5)	(6)
Adj_SUE	(β_1)	0.060***	0.058***	0.056***	0.059***	0.010***	0.016***
		(8.12)	(8.56)	(6.29)	(7.79)	(4.51)	(5.58)
Relative_Size	(β_2)	0.017			-0.021***		
		(1.10)			(-4.48)		
Adj_SUE x Relative_Size	**(β_3)**	**-0.059*** **			**0.041*** **		
		(-2.98)			**(6.18)**		
Relative_Wage	(β_4)		0.001			-0.008***	
			(0.18)			(-5.17)	
Adj_SUE x Relative_Wage	**(β_5)**		**-0.032*** **			**0.018*** **	
			(-5.14)			**(8.27)**	
Relative_Education	(β_6)			0.075			-0.046***
				(1.46)			(-2.85)
Adj_SUE x Relative_Education	**(β_7)**			**-0.257*** **			**0.102*** **
				(-3.59)			**(4.43)**
Constant		0.007	0.005	-0.003	-0.012***	-0.002	-0.005**
		(1.35)	(0.73)	(-0.36)	(-7.00)	(-1.27)	(-2.21)
Stock Market & Macro Conditions		Y	Y	Y	Y	Y	Y
Other Controls		Y	Y	Y	Y	Y	Y
Observations		462,855	462,855	462,855	462,855	462,855	462,855
Adjusted R^2		0.027	0.006	0.005	0.025	0.025	0.025

Table 14. Response to Earnings News (CAR[2,61] or CAR[0,1]) and the Predicted versus Residual Variation in Size, Wages or Education in the Financial Sector

This Table relates the market response to earnings news (i.e., CAR[+2,+61] or CAR[0,+1]) to the fitted level of size, wages, or education in the finance industry that is predicted by the level of financial deregulation five years earlier, versus the residual variation in the same three proxies for arbitrage efficacy or skill. In Panels A and B, we estimate an expanded version of Equation (1) that relates CAR[+2,+61] or CAR[0,+1] to: (a) the resulting fitted values of size, wages, or education in the financial sector, and (b) both the fitted and residual values of size, wages, or education in the financial sector, as well as their respective interactions with Adj_SUE. We control for the same stock market and macroeconomic conditions analyzed in Table 14. We also include the interaction term between each control variable and the adjusted rank of SUE (e.g., Adj_SUE x Aggregate Volatility, etc.). We include fixed effects for day-of-the-week. Standard errors are adjusted for heteroscedasticity and clustered by the day of the earnings announcement.

Panel A. PEAD (CAR[2,61]) and the Fitted versus Residual Level of Size, Wages, or Education in the Financial Sector

Variables		CAR [2, 61]					
		(1)	(2)	(3)	(4)	(5)	(6)
Adj_SUE	(β_1)	0.017***	0.017***	0.059***	0.060***	0.093**	0.094***
		(3.08)	(3.11)	(7.79)	(8.12)	(8.23)	(8.27)
Pred_Size	(β_2)	0.024	0.023				
		(0.95)	(0.91)				
Adj_SUE x Pred_Size	**(β_3)**	**-0.180*** **	**-0.179*** **				
		(-5.40)	**(-5.39)**				
Res_Size	(β_4)		0.011				
			(0.59)				
Adj_SUE x Res_Size	**(β_5)**		**-0.015**				
			(-0.63)				
Pred_Wage	(β_6)			-0.002	0.002		
				(-0.40)	(0.32)		
Adj_SUE x Pred_Wage	**(β_7)**			**-0.034*** **	**-0.035*** **		
				(-5.04)	**(-5.10)**		
Resid_Wage	(β_8)				-0.004		
					(-0.35)		
Adj_SUE x Resid_Wage	**(β_9)**				**-0.015**		
					(-0.87)		
Pred_Education	(β_{10})					0.145	0.149**
						(2.16)	(2.22)
Adj_SUE x Pred_Education	**(β_{11})**					**-0.539*** **	**-0.547*** **
						(-6.02)	**(-6.10)**
Resid_Education	(β_{12})						-0.080
							(-0.90)
Adj_SUE x Resid_Education	**(β_{13})**						**0.256**
							(2.04)
Constant		0.007	0.007	0.004	0.004	-0.013	-0.013
		(1.21)	(1.22)	(0.52)	(0.58)	(-1.36)	(-1.41)
Market & Macro Conditions & Controls		Y	Y	Y	Y	Y	Y
Observations		462,855	462,855	462,855	462,855	462,855	462,855
Adjusted R^2		0.006	0.006	0.006	0.006	0.005	0.006

Table 14, continued

Panel B. CAR[0,1]) and the Fitted versus Residual Level of Size, Wages, or Education in the Financial Sector

Variables		CAR [0, 1]					
		(1)	(2)	(3)	(4)	(5)	(6)
Adj_SUE	(β_1)	0.033***	0.033***	0.004	0.005**	-0.008	-0.009**
		(20.11)	(19.97)	(1.58)	(2.02)	(-2.18)	(-2.40)
Pred_Size	(β_2)	-0.055***	-0.054***				
		(-6.89)	(-6.78)				
Adj_SUE x Pred_Size	**(β_3)**	**0.119*** **	**0.117*** **				
		(10.35)	**(10.24)**				
Res_Size	(β_4)		-0.008				
			(-1.37)				
Adj_SUE x Res_Size	**(β_5)**		**0.013**				
			(1.65)				
Pred_Wage	(β_6)			-0.010***	-0.010***		
				(-6.27)	(-6.24)		
Adj_SUE x Pred_Wage	**(β_7)**			**0.024*** **	**0.023*** **		
				(10.18)	**(10.08)**		
Resid_Wage	(β_8)				0.007		
					(1.81)		
Adj_SUE x Resid_Wage	**(β_9)**				**-0.017*** **		
					(-2.93)		
Pred_Education	(β_{10})					-0.133***	-0.136***
						(-6.42)	(-6.58)
Adj_SUE x Pred_Education	**(β_{11})**					**0.283*** **	**0.289*** **
						(9.46)	**(9.66)**
Resid_Education	(β_{12})						0.118***
							(4.00)
Adj_SUE x Resid_Education	**(β_{13})**						**-0.231*** **
							(-5.23)
Constant		-0.013***	-0.013***	0.0001	-0.0003	0.007**	0.007
		(-7.22)	(-7.20)	(0.04)	(-0.14)	(2.39)	(2.55)
Market & Macro Conditions & Controls		Y	Y	Y	Y	Y	Y
Observations		462,855	462,855	462,855	462,855	462,855	462,855
Adjusted R^2		0.025	0.025	0.025	0.025	0.025	0.025

Table 15. Relation between the Response to Earnings News (CAR[2,61] or CAR[0,1]) and Abnormal Trading Volume

This Table presents results from estimating an expanded version of Equation (1) that controls for the average quarterly abnormal volume over all announcements, and its interaction with the Adj_SUE. Avg_AVOL is calculated as the quarterly average of AVOL [0, +1] as defined in Table II. The dependent variable is either CAR[+2,+61] or CAR[0,+1]. For brevity, we only provide results for the relevant variables and interaction terms. We also include the other controls in Equation (1), as well as fixed effects for day-of-the-week. The sample period covers January 1934 through December 2022. Standard errors are adjusted for heteroscedasticity and clustered by the day of the earnings announcement. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Variables		**CAR [2, 61]** (1)	**CAR [0, 1]** (2)
Adj_SUE	(β_1)	0.044***	0.025***
		(23.03)	(37.23)
Avg_AVOL	(β_2)	0.014***	-0.002***
		(5.77)	(-3.01)
Adj_SUE x Avg_AVOL	**(β_3)**	**-0.024*** **	**0.011*** **
		(-7.22)	**(8.63)**
Constant		-0.003	-0.009***
		(-0.54)	(-6.35)
Controls		Y	Y
Observations		462,855	462,855
Adjusted R^2		0.004	0.024

Appendix A.1. The Environment for Financial News and Corporate Disclosure Regulation

Appendix A.1.A. The Environment for Disclosure of Financial News from 1900 to the 1930s

At the beginning of the 1900s, financial news was being disseminated fairly efficiently, with the help of technological developments such as the telegraph, telephone, and stock ticker, as well as financial newspapers and newswires. These developments contributed to help open the world of trading to the general public. The available technology also allowed investors to submit and execute orders within a relatively short period of time, typically within the same day.[18]

Competition among newswires helped to enhance the timely reporting of financial news.[19] The *Wall Street Journal* (WSJ) has traditionally been the premier newspaper reporting financial news. The *New York Times* and the *Tribune* have long been the other popular daily newspapers in the New York district. However, while the latter two outlets provide financial news, they are not regarded as reliable or comprehensive in their coverage compared to the WSJ. Baker, Bloom, Davis, and Sammon (2019, p. 6) emphasize that the WSJ has "the most thorough coverage of financial news and has the most complete and consistent archive back to 1900."

Appendix.A.1.B. The Evolution of Regulation on Corporate Disclosure Since 1900

The history of U.S. regulations on compelled corporate disclosure has been shaped by legal, economic, and social factors over the past century. Prior to 1900, the U.S. did not have a formal regulatory framework regarding corporate disclosure. This environment spawned varied practices by firms and substantial fraud over this early period. From 1906 to the 1930s, the rise of corporations witnessed the beginnings of disclosure regulations aimed at protecting investors.

The New York Stock Exchange (NYSE) had been leading efforts to improve disclosure practices long before the legislators acted in the 1930s. As early as 1895, the NYSE required all

[18] See New York Stock Exchange (1936) for details about the inner workings of the exchange during the 1930s.
[19] https://archive.org/details/0474_Nations_Market_Place_01_11_58_00

listed companies to submit annual reports that included balance sheet and income statement data, and distribute these to stockholders.[20] The NYSE similarly mandated the filing of quarterly reports in 1926.[21] By 1929, all stocks listed on the NYSE reported their net income regularly.[22] Furthermore, the NYSE's mandate also compelled firms to immediately share any value-relevant information with the major newspapers in New York that publish financial news.[23]

The crash of 1929 and the Great Depression led to major legislative reforms with the Securities Act of 1933 and the Securities Exchange Act of 1934. Both acts placed a new emphasis on corporate disclosure to help prevent financial manipulation and provide investors with the information required to make informed decisions. For example, the 1934 act required public firms to file detailed balance sheet and income statements annually within 120 days after the fiscal year-end (10K's), as well as less detailed reports semiannually within 45 days after the first half of the fiscal year (9K's), and a current report within ten days of the end of any month when a significant event occurred (8K's).[24]

Since the 1930s these regulations have expanded to place greater emphasis on 'full and fair disclosure.' In the 1960s and 1970s, the disclosure framework addressed issues like insider trading and accounting irregularities. In 1975, rules were introduced concerning forward-looking financial statements, which allowed companies to provide investors with forecasts under certain conditions. In the 1980s and 1990s, the growth of technological innovations and the rise of the internet transformed how companies disseminated information, and regulations adapted to address these emerging forms of communication. In 1996, the SEC proposed rules to allow

[20] Sobel, R. (1968, p. 177).
[21] See Reilly (2004, p. 11). Karmel (2001) states that companies agreed to disclose reports quarterly in 1932, while Mahoney (1997) reports that the NYSE began pushing companies to report quarterly in the 1920s.
[22] Benston (1969).
[23] Zarb and Kerekes (1970, p. 38 and p. 99).
[24] See Benston (1973) and Zarb and Kerekes (1970, p. 38).

electronic filing and dissemination of financial information to enhance accessibility (e.g., through EDGAR). These developments were followed by Regulation Fair Disclosure in 2000.

In 2001 – 2002, high-profile corporate scandals emerged (e.g., Enron and WorldCom), which prompted significant regulatory reforms. These reforms included the Sarbanes-Oxley Act of 2002 which introduced stringent reforms, including enhanced financial disclosure requirements, the establishment of the Public Company Accounting Oversight Board (PCAOB), and more severe penalties for corporate fraud. Following the liquidity crisis of 2008, the regulators passed Dodd-Frank which expanded disclosure requirements further and addressed issues like executive compensation, financial stability, and transparency in corporate governance.

Since 2010, the regulators have placed new emphasis on social issues such as conflict minerals and disclosures related to environmental, social, and governance (ESG) issues. Both regulators and investors have recently demanded more comprehensive disclosures related to sustainability, ethics, and corporate social responsibility (CSR), and the SEC has sought to integrate ESG disclosure into the broader corporate disclosure framework. Internationally, organizations like the International Financial Reporting Standards (IFRS) Foundation are developing frameworks for sustainability reporting, which are influencing domestic practices. In addition, the SEC has adopted regulations to address challenges posed by digital assets, cybersecurity risks, and the growing importance of non-financial disclosures.

In summary, the evolution of regulations on compelled corporate disclosure reflects an ongoing balancing act among efforts to protect investors, promote market integrity, and accommodate corporate innovation. This regulatory landscape has evolved as societal expectations evolve, and it will continue to adapt in response to both emerging risks and

opportunities. Current trends suggest a more integrated approach to financial and non-financial disclosure going forward, emphasizing transparency and accountability in corporate governance.

Internet Appendix A.2. CRSP Coverage, Summary Statistics, and Robustness Tests

Appendix Table A.1. presents our main results across different sample choices.

Appendix Table A.2. presents the average initial response to earnings news and PEAD for the subsets of stocks that exist during the first four or five 10-year periods.

Appendix Table A.3. presents the average initial response to earnings news and PEAD for the subsets of stocks that exist over successive 10-year periods.

Appendix Table A.4. provides our main results when we use alternative definitions of abnormal returns or portfolio formations or announcement windows.

Appendix Table A.5. presents the nonlinear trend in initial response to earnings news and PEAD for the subsets of stocks based on firm attributes over each 10-year periods since 1934.

Figure A.1. presents the evolution in PEAD across different samples used in Table A.I.

Figure A.2. plots the proportion of the total 62-day announcement return realized on days 0 and +1 for the subsets of stocks that exist during the first four or five 10-year periods.

Figure A.3. plots the proportion of the total 62-day announcement return realized on days 0 and +1 for the subsets of stocks that exist over both successive 10-year periods.

Figure A.4. presents the evolution in PEAD when we use alternative definitions of abnormal returns or portfolio formations or announcement windows, as in Table A.IV.

Figure A.5. plots the proportion of the total 62-day announcement return realized on days 0 and +1, as well as the mean CAR[0,+1] and CAR[+2,+61], for terciles of stocks each quarter with small versus large values of firm size or book-to-market, averaged across all quarters during each 10-year period.

Table A.1. Nonlinear Trend in PEAD and CAR[0,+1]: Alternative Samples

This table presents the results from estimating Equation (2). The dependent variable is either PEAD (CAR[+2,+61]) or the initial response to earnings news (CAR[0,+1]), measured by the cumulative abnormal return for stock *i* compounded over days a to b, following the earnings announcement of firm *i* during quarter *t*. The coefficient of Adj_SUE (β_1) is analogous to the return on a hedge portfolio that is long the top SUE quintile and short the bottom SUE quintile. The variable, Trend, indexes the period of the observation each quarter. For ease of exposition of the coefficients, we divide "Trend" and "Trend^2" by 1000. All models include the control variables in Table VIII, and add the interactions of all the control variables with Trend and Trend^2 (the coefficients of the controls and the interaction terms are omitted for brevity). In models (1) and (5), we include all stocks. In models (2) and (6), we only keep NYSE firms in our sample. In models (3) and (7), we only keep NYSE firms and do not exclude firms with prices less than $1, or firms below the fifth percentile NYSE size cutoff. In models (4) and (8), we only keep NYSE and AMEX firms. Table I provides variable definitions. All models include fixed effects for day of the week. Standard errors are adjusted for heteroscedasticity and clustered by the day of the announcement. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Variables		CAR [2, 61]				CAR [0, 1]			
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Adj_SUE	(β_1)	0.001	0.015***	0.016***	0.012**	0.024***	0.038***	0.039***	0.037***
		(0.10)	(3.18)	(3.16)	(2.54)	(13.99)	(27.18)	(23.45)	(26.67)
Trend	(β_2)	-0.137	0.353	0.200	0.474**	0.261***	-0.013	0.008	-0.004
		(-0.45)	(1.61)	(0.92)	(2.22)	(4.18)	(-0.21)	(0.12)	(-0.07)
Adj_SUE x Trend	(β_3)	0.535***	0.297***	0.289***	0.362***	-0.014	-0.234***	-0.237***	-0.208***
		(9.51)	(5.77)	(5.33)	(7.14)	(-0.77)	(-14.14)	(-12.73)	(-12.82)
Trend^2	(β_4)	0.001	-0.001	-0.000	-0.001*	-0.001***	0.000	0.000	0.000
		(1.03)	(-1.28)	(-0.59)	(-1.77)	(-4.57)	(1.04)	(0.60)	(0.81)
Adj_SUE x Trend^2 (β_5)		**-0.0014*** **	**-0.0010*** **	**-0.0009*** **	**-0.0011*** **	**0.0003*** **	**0.0007*** **	**0.0007*** **	**0.0006*** **
		(-10.06)	**(-7.13)**	**(-6.52)**	**(-8.53)**	**(5.87)**	**(15.50)**	**(14.53)**	**(14.18)**
Constant		-0.017	-0.030*	-0.022	-0.037**	-0.026***	-0.012***	-0.012**	-0.013***
		(-0.88)	(-1.94)	(-1.39)	(-2.42)	(-4.80)	(-2.68)	(-2.35)	(-2.87)
Other Controls		Y	Y	Y	Y	Y	Y	Y	Y
Trend x Controls		Y	Y	Y	Y	Y	Y	Y	Y
Trend^2 x Controls		Y	Y	Y	Y	Y	Y	Y	Y
Observations		618,246	271,908	283,709	300,035	618,246	271,908	283,709	300,035
Adjusted R^2		0.006	0.008	0.008	0.007	0.030	0.025	0.026	0.026

Table A.2. Average PEAD and CAR[0,+1]: Holding Stocks that Exist over Each of the First Four or Five 10-Year Periods

This Table presents the mean cumulative abnormal returns on days +2 through and +61 (CAR[+2,+61] and on days 0 and +1 CAR[0,+1]) following earnings announcements on day 0, for the hedge portfolio that is long the top SUE quintile and short the bottom SUE quintile. Panels A and B report these average returns for firms that continuously exist during each of the first four and five 10-year periods, respectively. For each 10-year period, we report the mean CARs across the sample of earnings announcements.

Panel A. Firms that exist in each of the first four 10-year periods between 1934 and 1973

Period	CAR [2, 61]	t Value	CAR [0, 1]	t Value
1934-43	1.94%	3.02	2.85%	14.63
1944-53	2.18%	5.5	2.46%	19.02
1954-63	3.76%	8.45	2.27%	17.46
1964-73	3.56%	7.98	1.97%	14.58

Panel B. Firms that exist in each of the first five 10-year periods between 1934 and 1983

Period	CAR [2, 61]	t Value	CAR [0, 1]	t Value
1934-43	2.21%	2.98	2.45%	11.16
1944-53	2.19%	4.74	2.23%	15.78
1954-63	3.66%	6.73	2.13%	14.22
1964-73	3.37%	7.06	1.93%	13.03
1974-83	3.03%	5.59	1.37%	10.69

Table A.3. Average PEAD and CAR[0,+1]: Holding Same Stocks over Successive Pairs of 10-Year Periods

This Table presents the mean cumulative abnormal returns on days +2 through and +61 (CAR[+2,+61] and on days 0 and +1 CAR[0,+1]) following earnings announcements on day 0, for the hedge portfolio that is long the top SUE quintile and short the bottom SUE quintile. We compare the announcement returns of stocks that exist in two consecutive 10-year periods. For each pair of consecutive 10-year periods between 1934 and 2022, we report the mean CARs across the sample of earnings announcements. We also report the proportion of the market's total eventual 62-day response to earnings announcements that are realized immediately on the first two days or over the sixty days following the announcement (CAR[2,61] / CAR[0,61] or CAR[0,1] / CAR[0,61]).

Period	CAR [2, 61]	t Value	CAR[2,61]/CAR[0,61]	CAR [0, 1]	t Value	CAR[0,1]/CAR[0,61]
1934-43	1.88%	3.21	40.52%	2.84%	16.6	59.48%
1944-53	2.54%	6.48	51.51%	2.45%	21.19	48.49%
1944-53	2.29%	6.49	48.82%	2.46%	23.49	51.18%
1954-63	3.50%	10.46	61.03%	2.29%	23.73	38.97%
1954-63	3.43%	11.56	60.56%	2.28%	26.1	39.44%
1964-73	3.38%	10.47	62.53%	2.07%	23.29	37.47%
1964-73	3.87%	13.12	61.26%	2.51%	28.1	38.74%
1974-83	3.79%	15.54	68.33%	1.79%	26.15	31.67%
1974-83	4.24%	18.1	72.31%	1.65%	27.31	27.69%
1984-93	3.30%	13.27	73.45%	1.21%	17.28	26.55%
1984-93	3.97%	18.84	71.15%	1.64%	24.62	28.85%
1994-03	2.37%	8.72	59.01%	1.68%	19.06	40.99%
1994-03	2.04%	6.53	54.52%	1.73%	18.66	45.48%
2004-13	1.77%	6.75	37.24%	3.08%	30.94	62.76%
2004-13	1.36%	5.39	30.91%	3.13%	30.88	69.09%
2014-22	0.70%	2.23	15.83%	3.86%	28.34	84.17%

Table A.4. Nonlinear Trend in PEAD and CAR[0,+1]: Alternative Return Definitions and Portfolio Formations

This table presents the results from estimating Equation (2) using alternative return definitions and portfolio formations. The dependent variable is either PEAD (CAR[+2,+61]) or the initial response to earnings news (CAR[0,+1]), measured by the cumulative abnormal return for stock *i* compounded over days a to b, following the earnings announcement of firm *i* during quarter *t*. The coefficient of Adj_SUE (β_1) is analogous to the return on a hedge portfolio that is long the top SUE quintile and short the bottom SUE quintile. The variable, Trend, indexes the period of the observation each quarter. For ease of exposition of the coefficients, we divide "Trend" and "Trend^2" by 1000. All models include the control variables in Table VIII, and add the interactions of all the control variables with Trend and Trend^2 (the coefficients of the controls and the interaction terms are omitted for brevity). In models (1) and (5), we use portfolios sorted based on SUE deciles. In models (2) and (6), we use market-adjusted cumulative abnormal returns. In models (3) and (7), we use portfolios sorted based on SUE deciles and market-adjusted cumulative abnormal returns. In models (4) and (8), we use a larger window to measure announcement returns from day -1 to +1, i.e., CAR[-1,1]. Table II provides variable definitions. All models include fixed effects for day of the week. Standard errors are adjusted for heteroscedasticity and clustered by the day of the announcement. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Variables		CAR [2, 61]				CAR [0, 1]			
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Adj_SUE	(β_1)	0.011**	0.012**	0.018***	0.009**	0.040***	0.035***	0.040***	0.047***
		(2.02)	(2.36)	(3.06)	(2.01)	(24.70)	(24.16)	(24.31)	(27.67)
Trend	(β_2)	0.595***	0.404*	0.246	0.615***	0.078	0.069	0.081	0.073
		(3.14)	(1.90)	(1.03)	(3.24)	(1.33)	(1.11)	(1.30)	(1.04)
Adj_SUE x Trend	(β_3)	0.449***	0.373***	0.312***	0.398***	-0.224***	-0.200***	-0.224***	-0.129***
		(7.84)	(6.89)	(4.86)	(7.76)	(-12.10)	(-11.88)	(-11.87)	(-6.94)
Trend^2	(β_4)	-0.001**	-0.001	-0.000	-0.001***	-0.000	-0.000	-0.000	-0.000
		(-2.53)	(-1.19)	(-0.55)	(-2.64)	(-0.61)	(-0.41)	(-0.63)	(-0.02)
Adj_SUE x Trend^2 (β_5)		**-0.0013*** **	**-0.0011*** **	**-0.0009*** **	**-0.0011*** **	**0.0007*** **	**0.0007*** **	**0.0008*** **	**0.0003*** **
		(-9.13)	**(-7.92)**	**(-5.80)**	**(-9.05)**	**(15.78)**	**(15.51)**	**(15.50)**	**(7.51)**
Constant		-0.045***	-0.022	-0.012	-0.045***	-0.020***	-0.019***	-0.021***	-0.021***
		(-2.97)	(-1.27)	(-0.68)	(-2.94)	(-4.33)	(-3.77)	(-4.16)	(-3.76)
Other Controls		Y	Y	Y	Y	Y	Y	Y	Y
Trend x Controls		Y	Y	Y	Y	Y	Y	Y	Y
Trend^2 x Controls		Y	Y	Y	Y	Y	Y	Y	Y
Observations		462,855	462,855	462,855	462,855	462,855	462,855	462,855	462,855
Adjusted R^2		0.005	0.007	0.010	0.005	0.026	0.025	0.026	0.034

Table A.5. Nonlinear Trend in PEAD and CAR[0,+1] for Subsamples of Stocks Based on Firm Attributes over Each 10-Year Period since 1934

Panels A and B present results from estimating Equation (1) applied to CAR[+2,+61] and CAR[0,+1], respectively, over each 10-year period beginning in 1934 for subsets of stocks based on firm size (small versus large based on NYSE terciles), book-to-market (i.e., value versus growth), ivol, illiquidity, and turnover. These subsets of stocks are selected each quarter as the tercile portfolios with low versus high values of each firm attribute. The dependent variable is the cumulative abnormal return compounded over days +2 and +61, or days 0 to +1, following the earnings announcement of firm *i* in quarter *t*. For brevity, we only report the coefficient of Adj_SUE (β_1), which is analogous to the return on a hedge portfolio that is long the highest SUE quintile and short the lowest SUE quintile each quarter. For each model, we include all control variables from the model in Table V, as well as fixed effects for day of the week. Table I provides variable definitions. Standard errors are adjusted for heteroscedasticity and clustered by the day of the announcement. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Nonlinear Trend in PEAD, CAR[2,+61]

Decades	CAR[2,61]									
	Small	Large	Value	Growth	High Ivol	Low Ivol	Illiquid	Liquid	High Turnover	Low Turnover
1934-43	0.013 (0.88)	0.011* (1.77)	0.022* (1.81)	0.026*** (3.35)	0.008 (0.61)	0.024*** (4.28)	0.011 (0.92)	0.014** (2.02)	0.019** (2.14)	0.020* (1.71)
1944-53	0.022*** (2.89)	0.024*** (5.53)	0.022*** (3.62)	0.027*** (5.43)	0.025*** (3.39)	0.023*** (5.55)	0.022*** (3.11)	0.027*** (5.86)	0.030*** (5.40)	0.021*** (3.89)
1954-63	0.026*** (4.00)	0.037*** (9.74)	0.022*** (4.46)	0.044*** (9.44)	0.041*** (7.45)	0.031*** (8.96)	0.027*** (5.10)	0.040*** (9.37)	0.036*** (7.50)	0.027*** (5.92)
1964-73	0.052*** (8.88)	0.034*** (9.15)	0.035*** (7.12)	0.056*** (11.82)	0.058*** (8.49)	0.038*** (11.49)	0.048*** (8.64)	0.037*** (8.72)	0.050*** (9.74)	0.041*** (8.71)
1974-83	0.065*** (15.09)	0.027*** (7.95)	0.045*** (10.92)	0.046*** (11.87)	0.062*** (11.02)	0.039*** (14.87)	0.059*** (9.33)	0.031*** (9.82)	0.061*** (14.48)	0.036*** (9.78)
1984-93	0.058*** (17.53)	0.022*** (7.11)	0.039*** (10.65)	0.048*** (13.82)	0.056*** (10.22)	0.036*** (15.94)	0.065*** (11.52)	0.030*** (11.81)	0.059*** (15.11)	0.036*** (11.09)
1994-03	0.049*** (12.19)	0.011*** (2.99)	0.039*** (8.47)	0.035*** (7.80)	0.052*** (6.82)	0.028*** (11.45)	0.050*** (8.40)	0.020*** (6.34)	0.042*** (8.94)	0.028*** (7.30)
2004-13	0.031*** (9.43)	0.006* (1.93)	0.028*** (5.93)	0.017*** (5.53)	0.029*** (5.02)	0.012*** (5.59)	0.050*** (7.06)	0.011*** (4.31)	0.020*** (5.59)	0.019*** (4.96)
2014-22	0.029*** (5.31)	0.011*** (3.58)	0.030*** (5.91)	0.013*** (3.21)	0.028*** (3.39)	0.012*** (4.93)	0.046*** (5.69)	0.007** (2.33)	0.017*** (4.06)	0.015*** (2.84)

Table A.5., continued

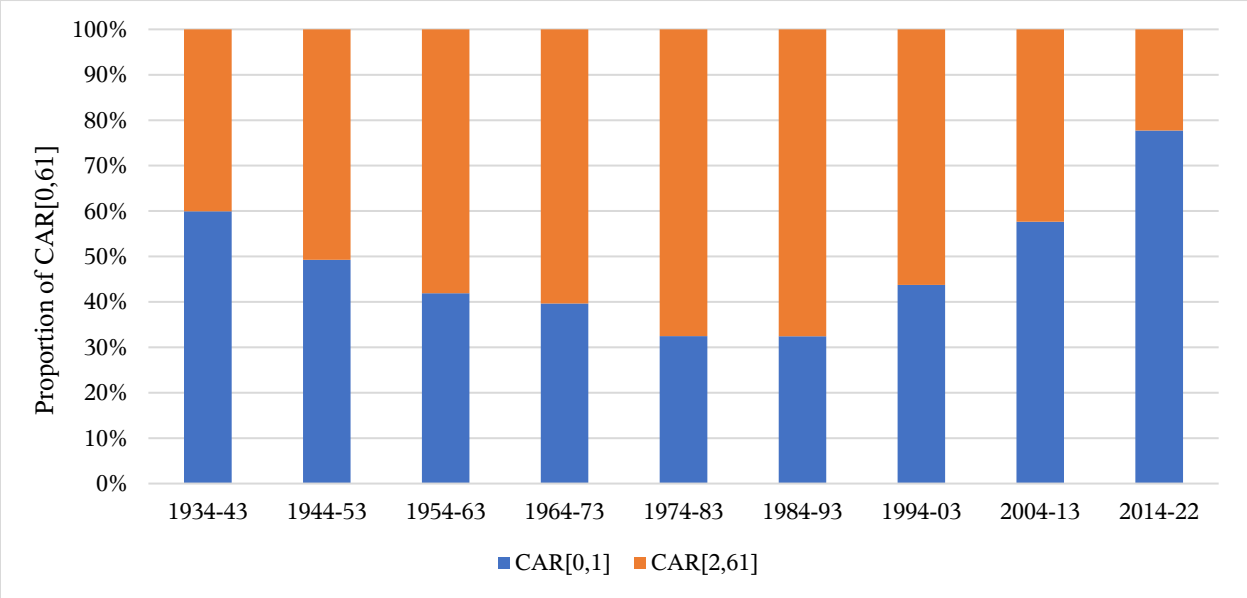
Panel B. Nonlinear Trend in the Initial Response to Earnings News, CAR[0,+1]

Decades	CAR[0,1]									
	Small	Large	Value	Growth	High Ivol	Low Ivol	Illiquid	Liquid	High Turnover	Low Turnover
1934-43	0.038***	0.017***	0.038***	0.020***	0.031***	0.020***	0.036***	0.019***	0.041***	0.011***
	(8.94)	(9.70)	(10.81)	(10.07)	(8.37)	(11.49)	(9.24)	(10.73)	(11.99)	(5.31)
1944-53	0.034***	0.017***	0.027***	0.020***	0.027***	0.021***	0.032***	0.017***	0.039***	0.011***
	(14.46)	(14.80)	(14.42)	(13.46)	(12.76)	(17.49)	(15.89)	(14.28)	(19.22)	(9.64)
1954-63	0.032***	0.018***	0.028***	0.019***	0.025***	0.021***	0.026***	0.019***	0.038***	0.010***
	(17.27)	(17.20)	(19.19)	(15.41)	(16.47)	(18.21)	(16.26)	(15.74)	(22.38)	(11.28)
1964-73	0.037***	0.020***	0.032***	0.026***	0.034***	0.023***	0.033***	0.024***	0.050***	0.010***
	(20.89)	(18.63)	(21.94)	(18.78)	(17.04)	(24.04)	(18.80)	(21.02)	(27.80)	(11.15)
1974-83	0.027***	0.013***	0.024***	0.021***	0.026***	0.017***	0.033***	0.015***	0.039***	0.008***
	(22.26)	(16.28)	(21.11)	(21.25)	(16.22)	(25.87)	(16.49)	(20.18)	(26.85)	(11.76)
1984-93	0.033***	0.010***	0.022***	0.024***	0.033***	0.016***	0.037***	0.013***	0.046***	0.007***
	(30.42)	(11.10)	(20.62)	(22.65)	(19.51)	(24.39)	(20.34)	(16.98)	(28.47)	(9.90)
1994-03	0.035***	0.010***	0.029***	0.024***	0.034***	0.018***	0.032***	0.016***	0.053***	0.008***
	(28.82)	(7.93)	(20.61)	(18.31)	(16.63)	(22.82)	(17.13)	(15.94)	(25.35)	(10.86)
2004-13	0.053***	0.019***	0.042***	0.039***	0.051***	0.030***	0.042***	0.027***	0.075***	0.016***
	(38.25)	(14.73)	(23.94)	(28.44)	(23.09)	(29.69)	(15.47)	(24.26)	(34.59)	(17.76)
2014-22	0.060***	0.021***	0.047***	0.041***	0.056***	0.033***	0.053***	0.027***	0.081***	0.018***
	(32.52)	(14.38)	(23.46)	(23.46)	(21.12)	(26.03)	(18.75)	(21.20)	(31.89)	(17.93)

Figure A.1. Price Response to Earnings Surprises over 10-Year Periods from 1934 to 2022: Alternative Samples

These figures present the proportion of the market's total eventual 62-day response to earnings announcements that are realized immediately on the first two days around the earnings release (CAR[0,1] / CAR[0,61]), using alternative return definitions and portfolio formations. We compute the mean initial 2-day announcement return (CAR[0,1]) and total 62-day announcement return (CAR[0,61]) for the top and bottom deciles of firms by SUE each quarter, averaged across all quarters during each 10-year period. We then plot the ratio of these average values for every 10-year period from 1934 to 2022. In Panel A, we do not exclude smaller stocks with prices less than $1, or firms below the fifth percentile NYSE size cutoff. In Panel B, we only keep NYSE firms in our sample. In Panel C, we only keep NYSE firms and do not exclude firms with prices less than $1, or firms below the fifth percentile NYSE size cutoff. In Panel D, we only keep NYSE and AMEX firms.

Panel A. All Stocks



Panel B. NYSE Stocks

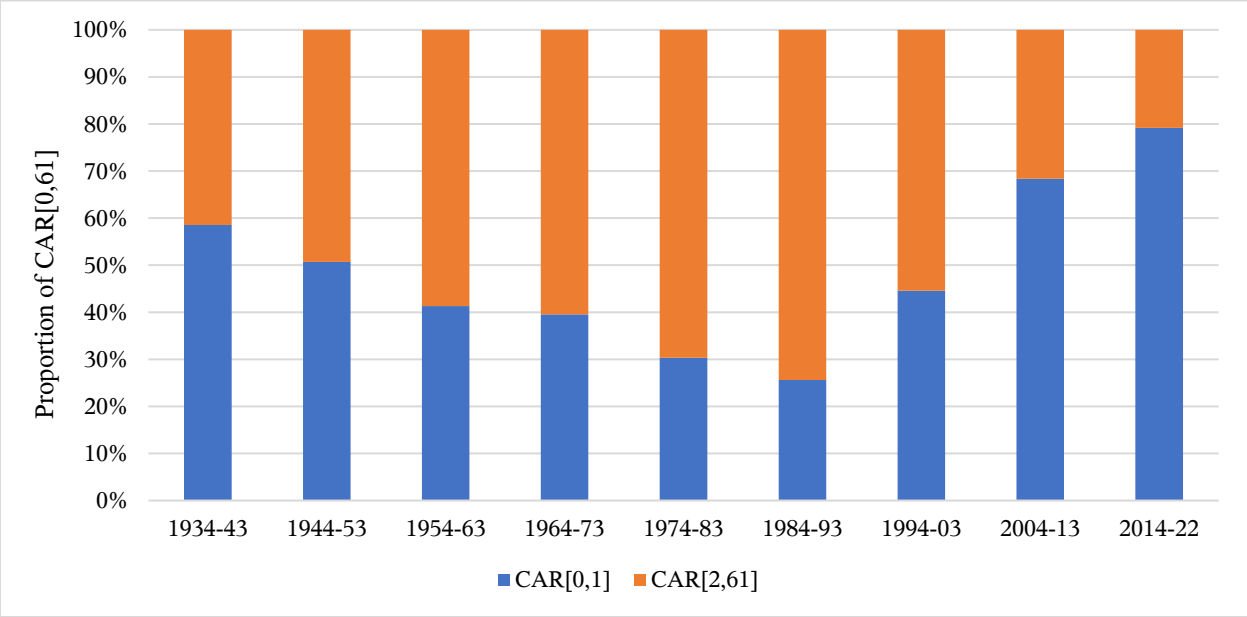
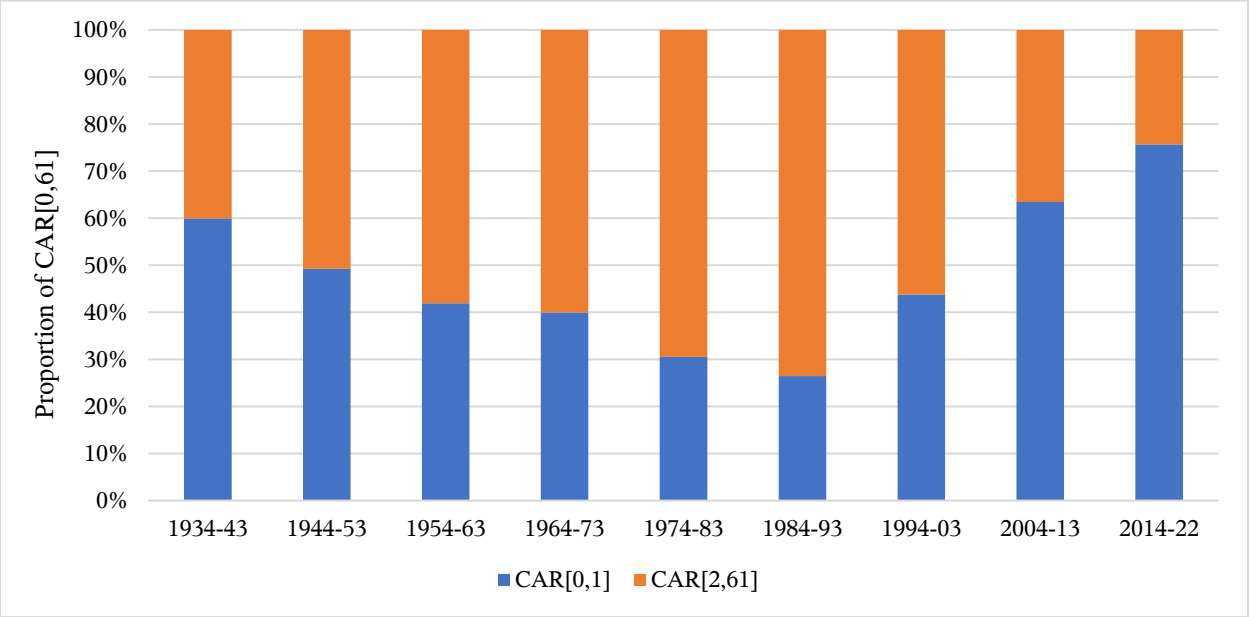


Figure A.1., continued

Panel C. All NYSE Stocks



Panel D. NYSE + AMEX Stocks

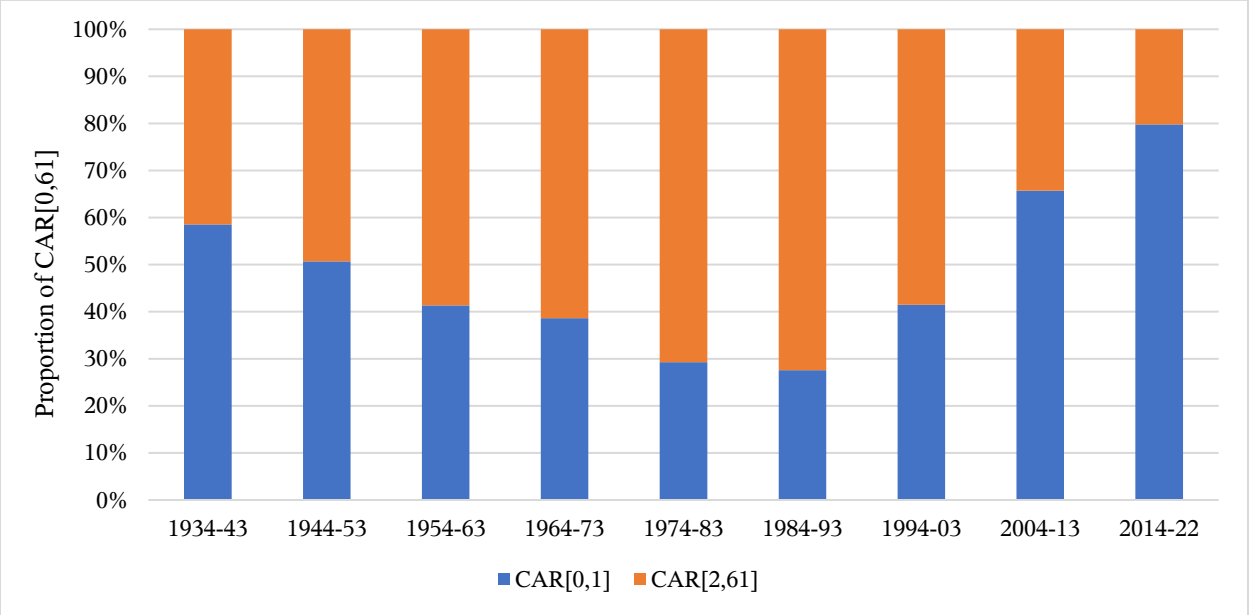


Figure A.2. Proportion of Total Announcement Return Realized from Day +2 to +61: Holding Stocks that Exist over Each of the First Four or Five 10-Year Periods

This figure plots evolution in the proportion of total announcement returns realized immediately around the earnings release on days 0 and 1 (CAR[2,61] / CAR[0,61]) for firms that exist during each of the first four and five 10-year periods. The **blue** (**orange**) bars reflect the firms that continuously exist during each of the first four (five) 10-year periods. For each 10-year period, we report the mean CARs across the sample of earnings announcements.

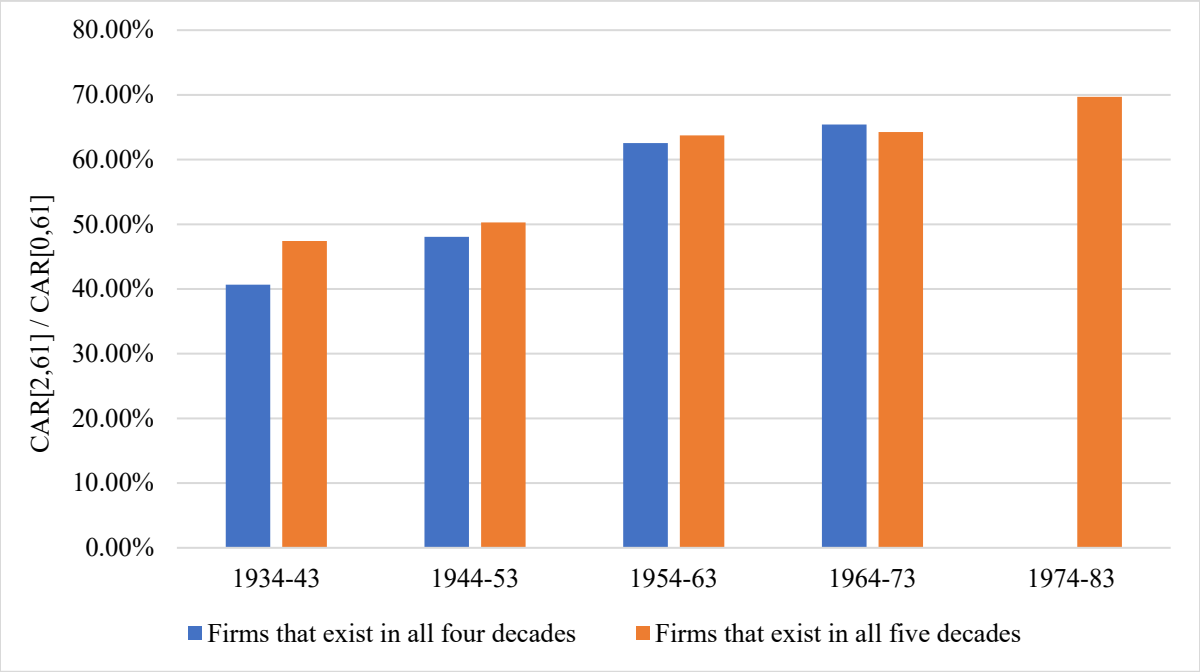


Figure A.3. Proportion of Total Announcement Return Realized from Days +2 to +61: Holding Same Stocks over Successive 10-Year Periods

This figure plots evolution in the proportion of total announcement returns realized after the earnings release from days +2 to +61 (CAR[2,61] / CAR[0,61]) for firms that exist during each consecutive 10-year period. The **blue** bars reflect the earlier 10-year period, while **orange** bars reflect the later period. For each 10-year period, we report the mean CARs across the sample of earnings announcements.

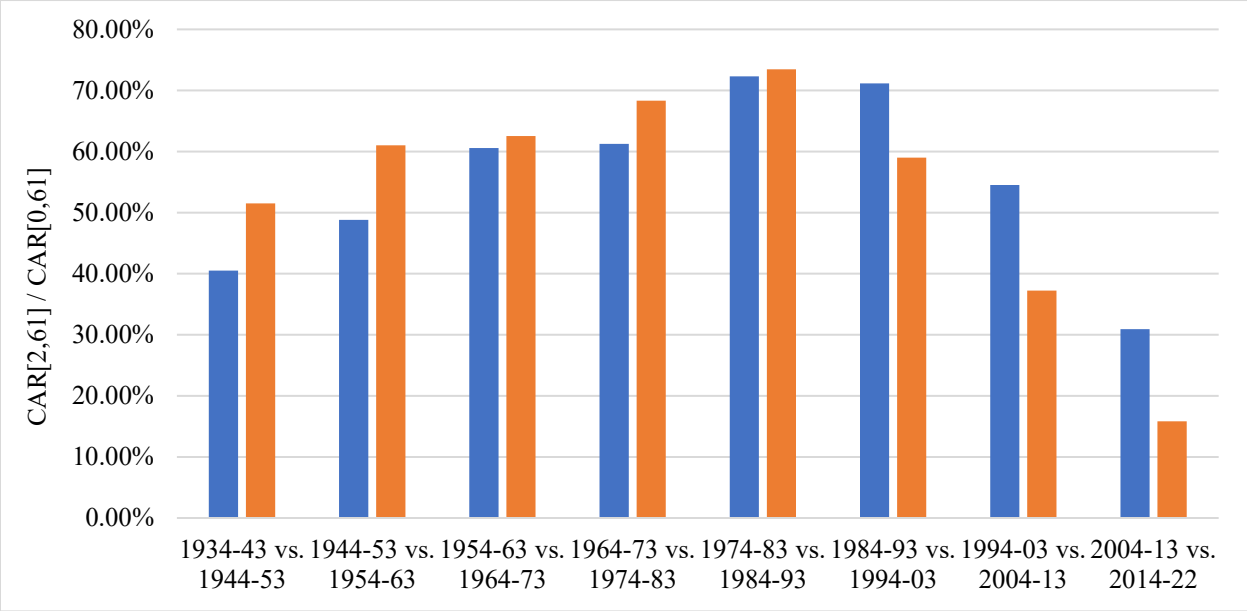
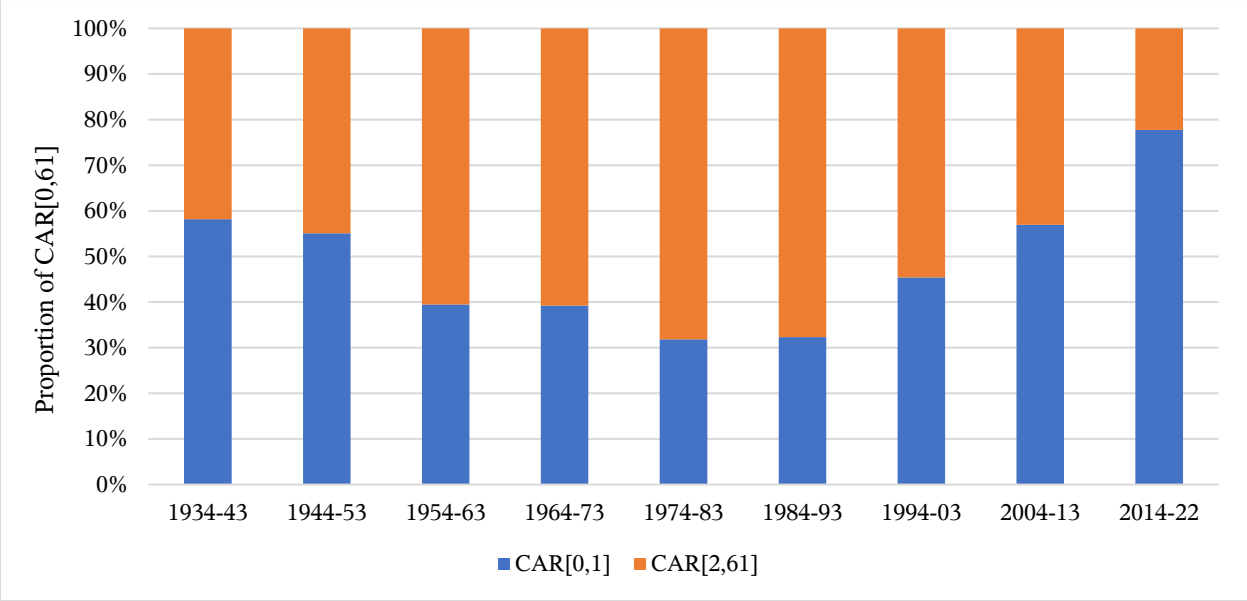


Figure A.4. Price Response to Earnings Surprises over 10-Year Periods from 1934 to 2022: Alternative Return Definitions and Portfolio Formations

These figures present the proportion of the market's total 62-day response to earnings announcements that is realized on the first two days around the earnings release (CAR[0,1] / CAR[0,61]), using alternative return definitions and portfolio formations. We compute the mean initial 2-day announcement return (CAR[0,1]) and total 62-day announcement return (CAR[0,61]) for the top and bottom deciles of firms by SUE each quarter, averaged across all quarters during each 10-year period. We then plot the ratio of these average values for every 10-year period from 1934 to 2022. In Panel A, we use portfolios sorted based on SUE deciles. In Panel B, we use market-adjusted cumulative abnormal returns. In Panel C, we use portfolio sorts based on SUE deciles and market-adjusted cumulative abnormal returns. In Panel D, we use a larger window from day -1 to +1, to measure CAR[-1,1] / CAR[-1,61].

Panel A. SUE Decile Portfolios



Panel B. Market-Adjusted Cumulative Abnormal Returns

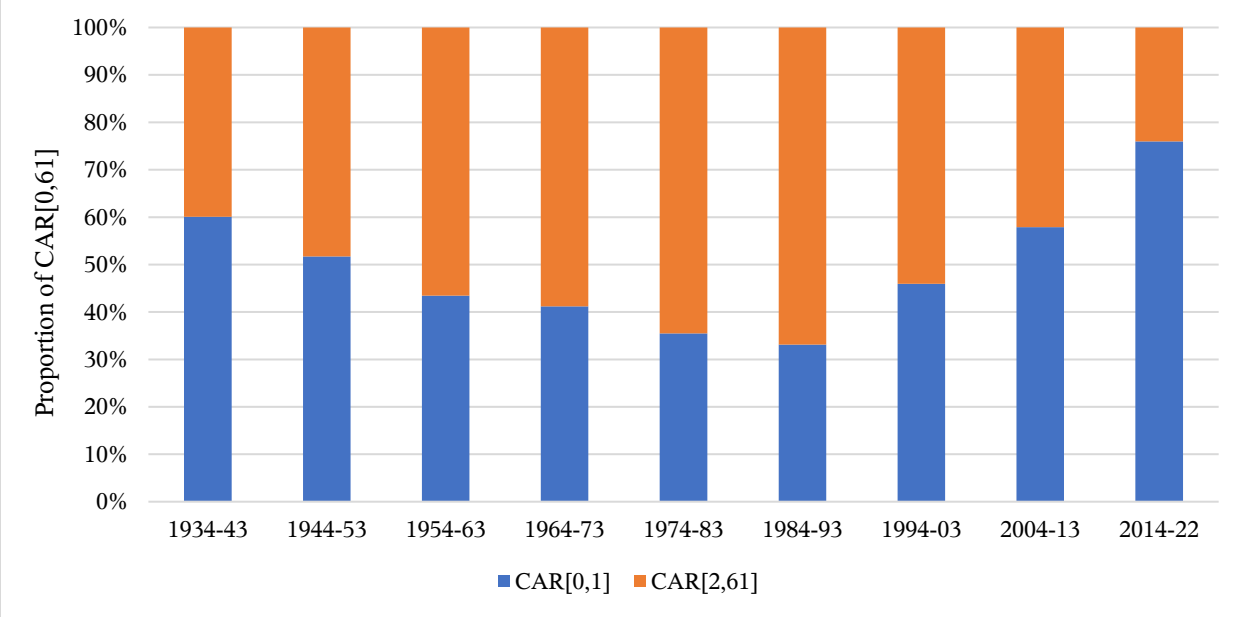
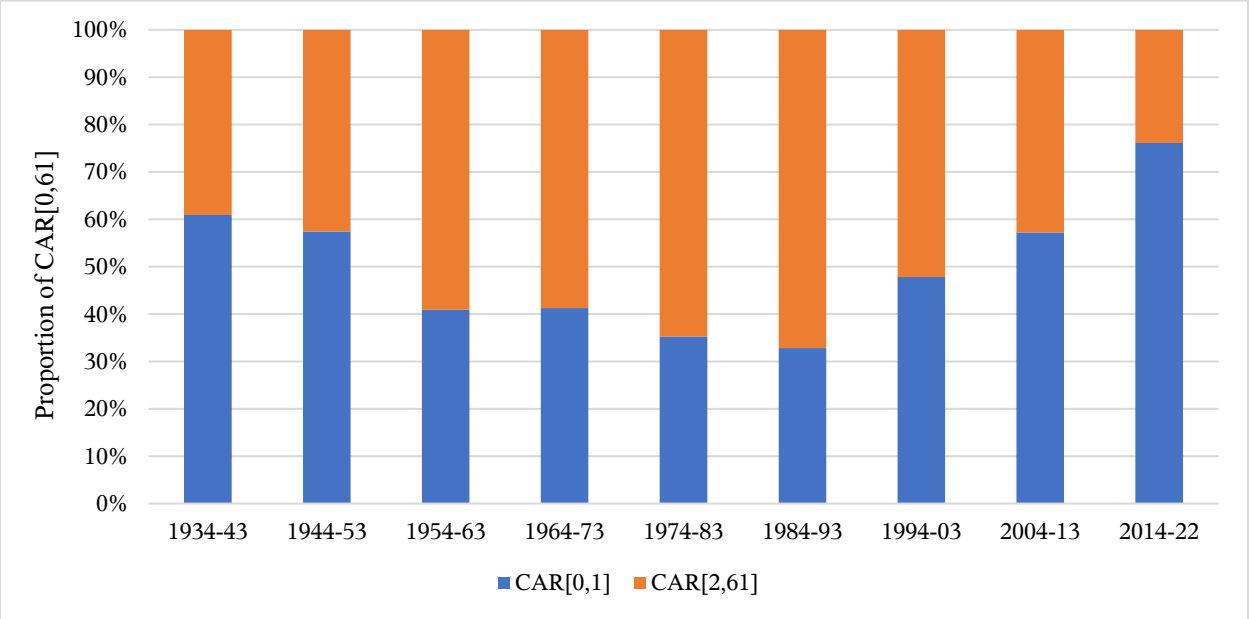


Figure A.4., continued

Panel C. SUE Decile Portfolios and Market-Adjusted Cumulative Abnormal Returns



Panel D. Expanded Announcement Window beginning on Day -1: CAR[-1,+1]

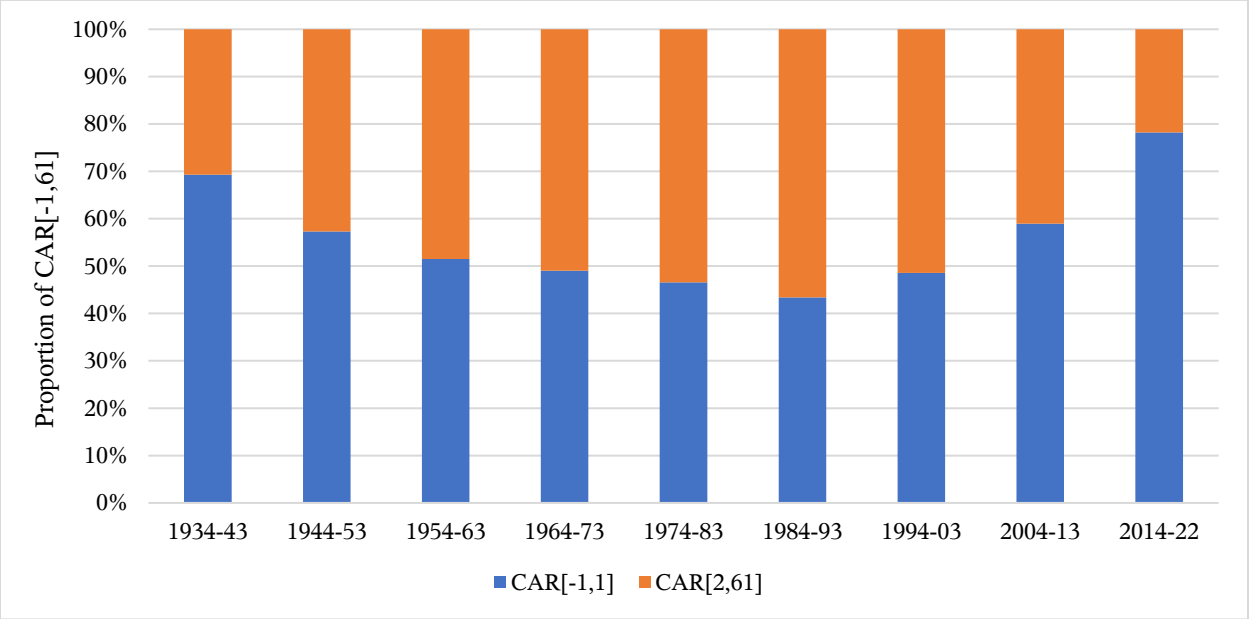
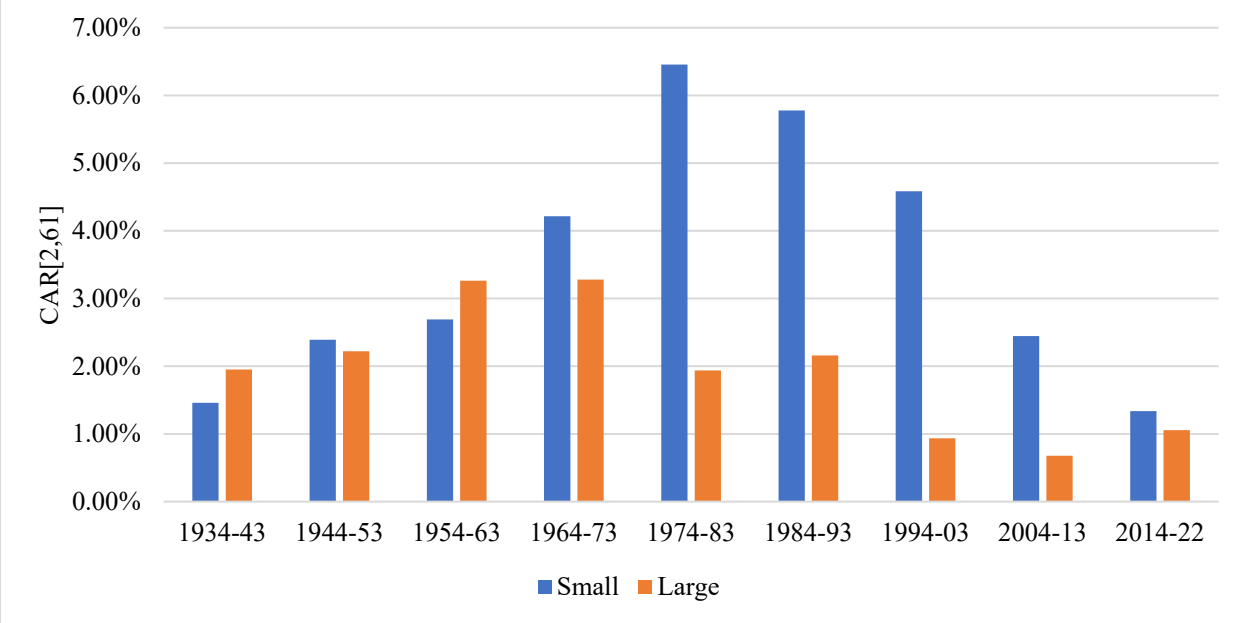


Figure A.5. Evolution in PEAD: Small vs. Large and Value vs. Growth Stocks

Panels A and B plot evolution in the magnitude of PEAD (CAR [2,61]) and the immediate response (CAR[0,1]), respectively. Panel C plots evolution in the proportion of total announcement returns realized from days +2 to +61 (CAR[2,61] / CAR[0,61]) for small versus large stocks separately. Panels D through F plot the analogous evolution for value versus growth stocks. Each quarter, we first group stocks into size terciles based on NYSE size cutoffs or book-to-market. We then compute the mean for the top or bottom tercile of stocks sorted by each firm attribute, averaged across all quarters in each 10-year period.

Panel A. Evolution in CAR [2,61]: Small versus Large Stocks



Panel B. Evolution in CAR [0,1]: Small versus Large Stocks

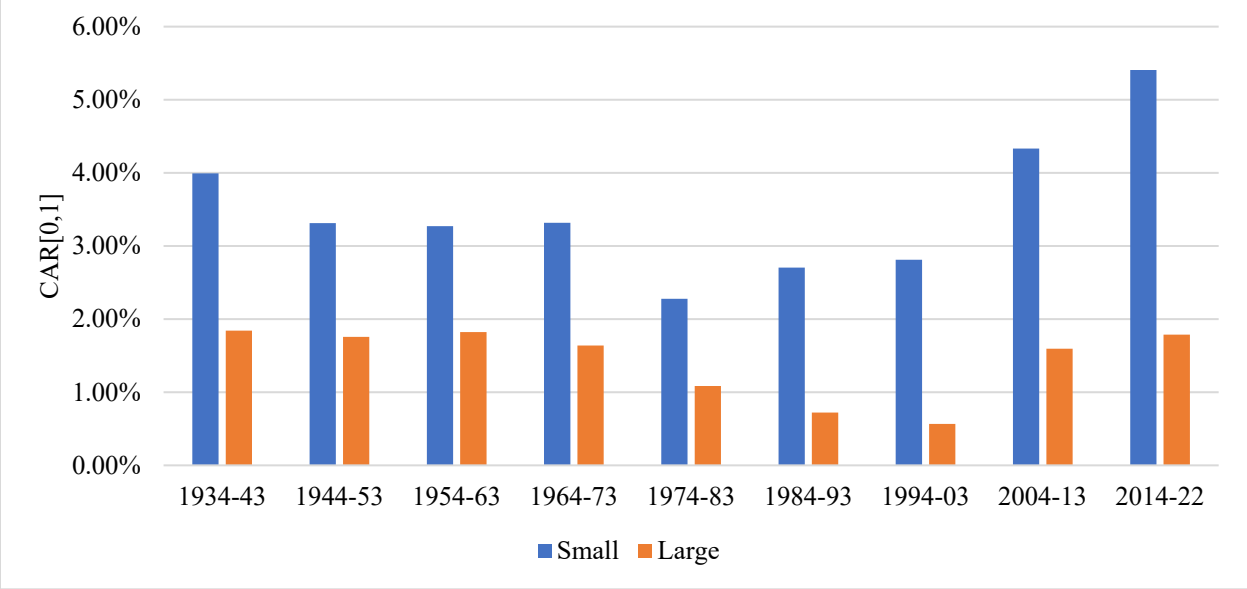
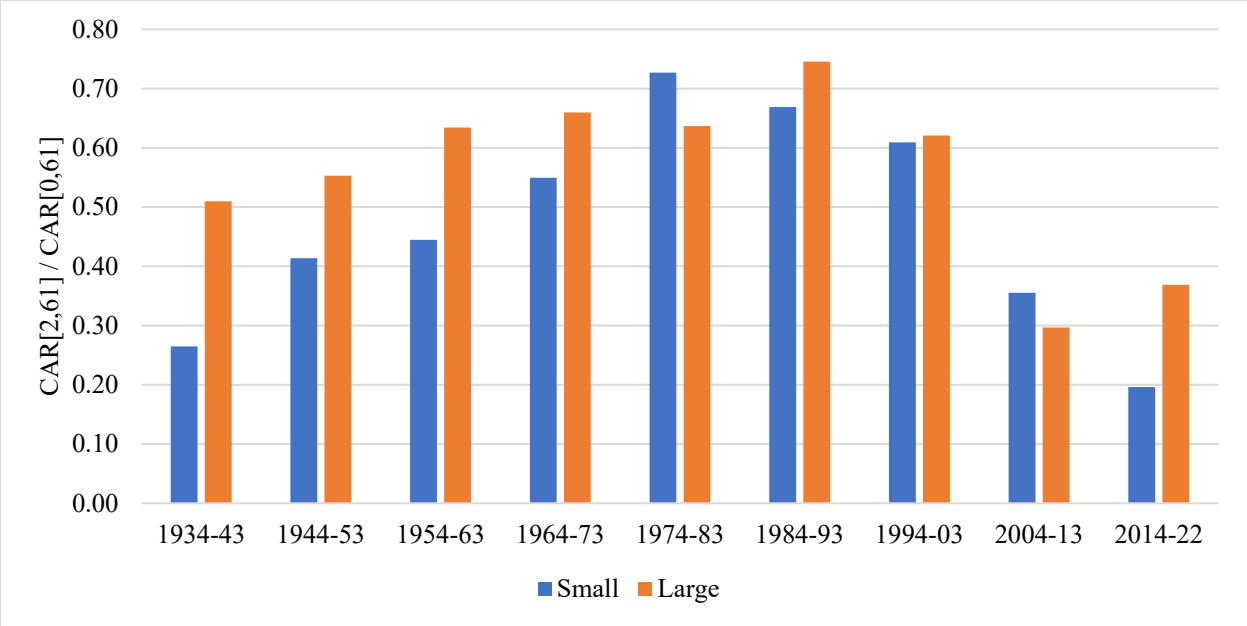


Figure A.5., continued

Panel C. Proportion of the Total Announcement Return Realized from Days +2 to +61, CAR[2,61] / CAR[0,61], for Each 10-Year Period: Small versus Large Stocks



Panel D. Evolution in CAR [2,61] : Value versus Growth Stocks

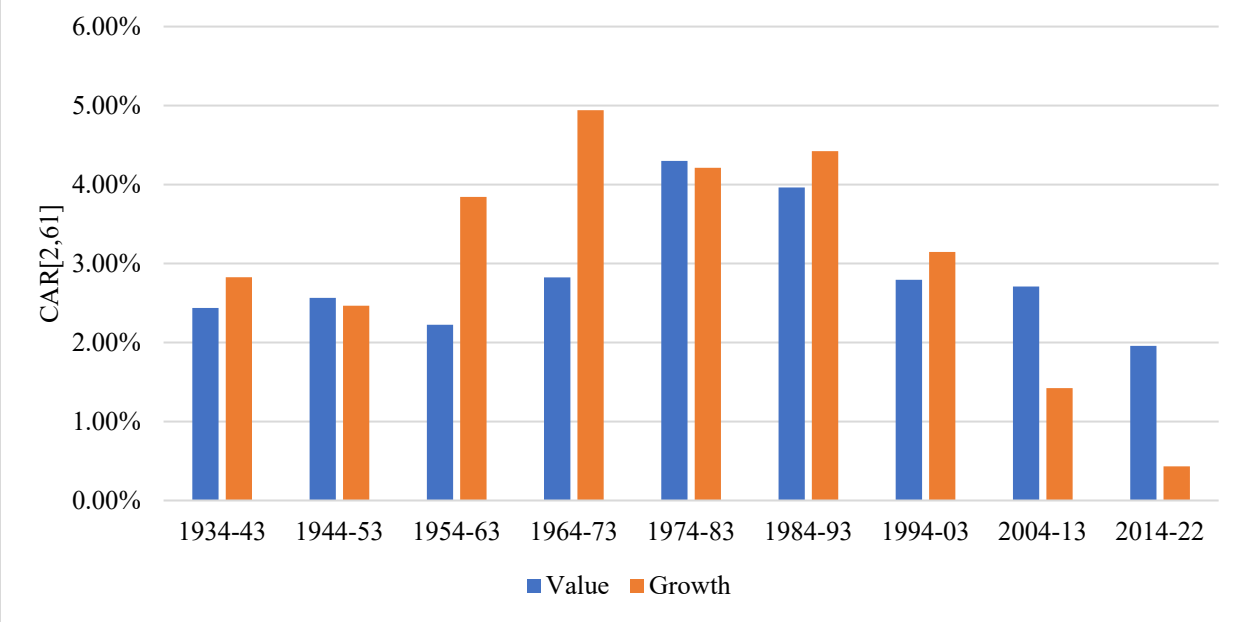
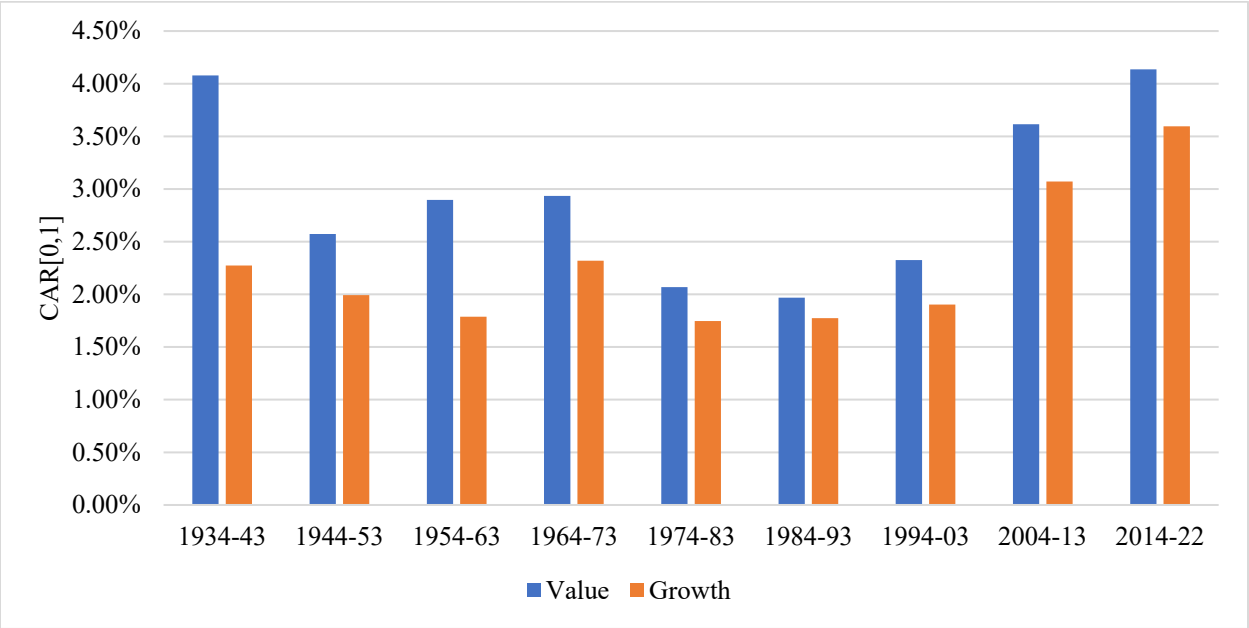


Figure A.5., continued

Panel E. Evolution in CAR [0,1] : Value versus Growth Stocks



Panel F. Proportion of the Total Announcement Return Realized from Days +2 to +61 CAR[2,61] / CAR[0,61], for Each 10-Year Period: Value versus Growth Stocks

